Bifurcations and chaos in language models fed their own output

Authorship to be decided by the human researcher

Private working draft, 14 October 2026

Status of this draft. Unless marked otherwise, results on the soft-token map are numerically validated in float64 (rung 1 or 2 of the evidence ladder in Section 16). Float32 maps are labelled as such. Huginn's spectra are float32 and its accuracy runs bfloat16, and the dialogue results are sampled statistics with stated intervals. **[Planned:** Red] marks planned work and **[In progress:** orange] running work. The only computer-assisted proof is that of the period-8 orbit of Table 1 (rung 3, Section 16).

Abstract

Feed a frozen language model its own expected next-token embedding, at a fixed position after a fixed prompt, and it can lock into a loop of text such as ". This is there is a is a" or wander with positive finite-horizon Lyapunov exponents. This smooth map, with temperature and steering strength as parameters, shows both regimes in Qwen2.5, OLMo-2 and Llama-3. Its derivative factorizes into the inverse temperature, an embedding covariance under the model's next-token distribution, and the network's Jacobian; with tied embeddings and no network it would be a modern Hopfield update, which cannot cycle. It contracts at high temperature; chaos occupies intermediate temperatures and needs no steering in most dense models. Across fifteen models of 0.5B to 72B, size does not set the chaos: on ten prompts a 14B model is more chaotic than a 7B, at higher temperatures; OLMo-2-32B is the most chaotic OLMo model; and Qwen2.5-72B, nearly free of it on five prompts, is chaotic when steered. Bifurcation theory applied to the model's own derivatives predicts its behaviour where tested. Near cusp candidates of two windows, local unfoldings frozen before solving matched the orbits recovered at all twelve tested offsets and their stability at eleven. Fold normal forms predict bottleneck passage times without fitting: at two new folds, 23 of 24 frozen forecasts fell within one iterate. In the recurrent-depth model Huginn, spectral estimates predict how held-out questions respond to relaxation, and a frozen test confirmed a predicted accuracy loss. Interval arithmetic proves one period-8 orbit exists and is locally attracting, to our knowledge the first such proof for a transformer language model. In sampled self-dialogue, keeping flagged replies out of the context controls steered hostility at a fraction of the dose, and counter-steering can backfire. The map is not ordinary autoregressive generation.

1 Introduction

Repeated inference with a fixed language model is an iterated map: the model's output becomes its next input. Like other iterated maps, it can be chaotic. Figure 1 shows a small piece of the parameter plane of one such map, built from Qwen2.5-0.5B-Instruct. Cells with a positive finite-horizon Lyapunov exponent lie side by side with windows in which the model repeats a fixed loop of text. Parts of this picture have been seen before. Recent work has measured chaos-like sensitivity in language-model generation [1, 2], temperature-driven changes of output statistics

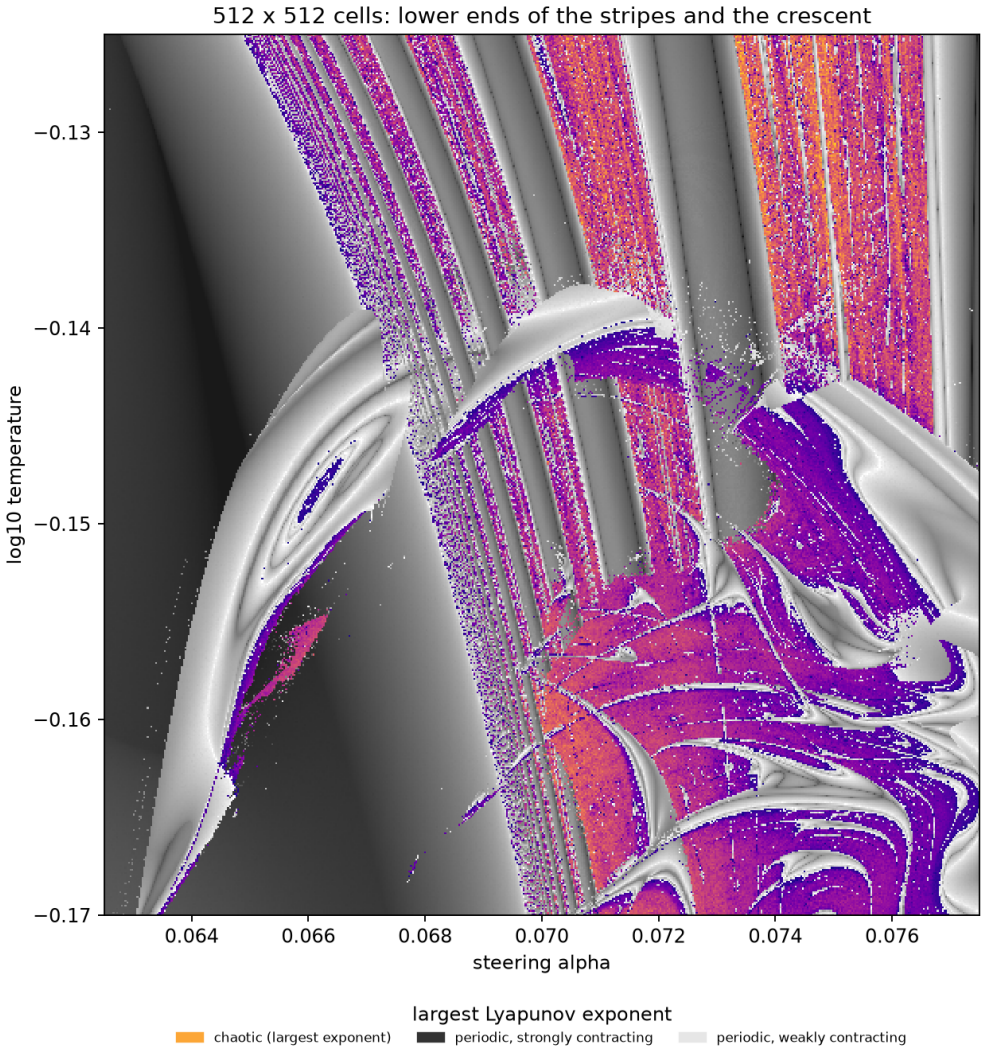


Figure 1: Chaos and repeating text in a language model fed its own output: the soft-token map of Qwen2.5-0.5B-Instruct, steered along the philosophy direction at layer 12, on 512×512 float64 cells over the steering strength $\alpha \in [0.0625, 0.0775]$ and the temperature, $\log_{10} T \in [-0.17, -0.125]$. Coloured cells are chaotic by the finite-horizon criterion of Section 4; the colour gives the largest Lyapunov exponent. In grey cells the model settles into a repeating loop of text, shaded by how strongly the loop contracts (light grey: close to losing stability). Figure 3 names the loops.

[3–7], periodic attractors of paraphrasing loops [8], attractor states of model-to-model conversations [9], and, through their Jacobians, the stability of looped and recurrent-depth models [10–13], and finite-depth readout dynamics have been used to predict the answers of looped models [14]. Feeding a probability-weighted vocabulary embedding back into a model is not new either. It is used in training [15] and, without retraining, in pretrained models [16], alongside continuous latent feedback [17]. In a small task-trained looped transformer, fold coefficients computed from derivatives predict passage times and spectral gaps without fitting [18].

We study one constructed but exactly defined system: the expected-embedding ("soft-token") map of a frozen, released pretrained language model, run with a fixed chat prefix at a fixed position. The map is smooth. Its Jacobians come from automatic differentiation, and its periodic orbits can

be computed and continued like those of any smooth map. Chaos and periodic windows occur in many nonlinear feedback systems. Random-network theory, in which a specified ensemble of recurrent networks becomes chaotic above a critical gain [19], and the period-doubling cascades and shrimp-shaped domains of low-dimensional maps [20, 21] motivate looking for such structure here, but they do not imply that a trained map realizes it. Whether a trained language model does, where in its parameter space, with which texts, and whether the quantitative predictions of bifurcation theory hold for it, so that derivatives computed from the model predict its behaviour before that behaviour is measured, has to be tested. The models we studied do realize it, and in the cases we tested the predictions hold. We also show where the chaos comes from (Section 3). The loop's derivative factorizes into the inverse temperature, the covariance of token embeddings under the model's own next-token distribution (a cross-covariance when input and output embeddings are untied), and the network's Jacobian. With tied embeddings, the network turns a Hopfield retrieval, which cannot cycle, into a map that can. The factorization proves contraction at sufficiently high temperature for this smooth fixed-prefix map, and at zero temperature the finite-memory greedy map is eventually periodic; in our sweeps chaos occupies intermediate temperatures. The answer to whether larger models leave this chaos behind is no: a larger model's chaos can move to higher temperature instead of vanishing, and size does not order the chaos across families (Section 12). Fold normal forms predict how long trajectories linger in the ghost of a vanished orbit, local unfoldings near cusp candidates predict which periodic orbits coexist and which of them are stable, and spectral estimates of a recurrent-depth model predict how its questions respond to relaxation, with a predicted accuracy loss confirmed on held-out data. Ordinary sampled generation with a growing context is a different system. Section 10 measures how much of the mechanism survives sampling, and Section 14 studies sampled dialogue directly.

Contributions.

1. *The loop's derivative shows where chaos can occur.* The derivative of the soft-token map factorizes into the inverse temperature, the covariance of token embeddings under the model's own next-token distribution (a cross-covariance for untied embeddings), and the network's Jacobian. With tied embeddings and without the network, the map would be a modern Hopfield update, which cannot cycle. The factorization proves contraction at sufficiently high temperature, as a high-temperature sweep confirms; the zero-temperature limit is a greedy map that is eventually periodic, and chaos in our sweeps occupies intermediate temperatures. The gain mechanism is not specific to concept-labelled directions: in the norm-matched controls of the 0.5B and 8B models, random directions also steer into chaos. Along a period-8 orbit every step expands some direction while the cycle contracts: the loop's stability is a property of the ordered product of its steps, not of any one step (Section 3).

2. *Size does not set the chaos.* Across fifteen models from 0.5B to 72B parameters, the unsteered chaos in a fixed temperature window nearly vanishes in the largest Qwen2.5 and Llama models but not in OLMo-2-32B, the most chaotic OLMo model. A frozen prompt-blocked test shows that the Qwen decline from 7B to 14B is largely a shift: on all eight held-out prompts the 14B model is more chaotic than the 7B, at the higher temperatures where its next-token distribution spreads. A second frozen test finds Qwen2.5-72B nearly free of unsteered chaos on four held-out prompts (one chaotic temperature of 192; a finer line places it in a band 0.02 decades wide), yet steering drives it, like Llama-3.3-70B, into chaos that an exact tangent exponent confirms. The gain split shows how network and readout damp or amplify the fed-back token (Section 12).

3. *A fold's normal form predicts how long the model lingers near a vanished orbit.* The 2-cycle "is it" dies in a fold, and the fold's normal form, computed from derivatives, predicts bottleneck passage times without fitting. We tested it prospectively at two new folds: 23 of 24 forecasts fell within one iterate, and the all-row criterion failed by one row. The global reinjection sets the prefactor of the window law (Section 7).

4. *Local unfoldings near cusp candidates predict which periodic orbits coexist and which are stable.* Fold and flip curves continued with float64 test functions show that the fold curves of a period-6 window and of a hostile period-7 window contain cusp candidates: segments where the quadratic normal-form coefficient changes sign, with nonzero cubic coefficients at the bracketing points. At nearby fold points the estimated two-parameter unfolding is nonsingular, and its local cubic model, frozen before solving, matched the number of distinct orbits that Newton solves seeded at its predicted roots recovered at all twelve parameter offsets, and their stability at eleven; the search was not exhaustive (Sections 8 and 14).

5. *In a second system, spectral estimates predict the response to relaxation.* In the recurrent-depth model Huginn we predict a Neimark–Sacker bifurcation and then observe it. On held-out questions, an estimated spectral feature predicts the size of each question's residual under relaxation, and a separate frozen generation test confirms the predicted accuracy loss at relaxation 1.20 and noninferiority at 1.05 (Section 13).

6. *One orbit is proved, and the central numerical results are checked independently.* Orbits, multipliers and test functions of the soft-token map are computed in float64 with precision checks (Section 2). Independent reimplementations reproduce the fold and its normal form and two orbits, a second hardware platform reproduces four flip parameters, and an independent replay reproduces the frozen Huginn test. An interval-arithmetic proof shows that a primitive period-8 orbit exists and is locally attracting, to our knowledge the first proof of a periodic orbit of a full transformer language-model map (Section 16).

7. *In sampled dialogue, keeping a flagged reply out of the context controls hostility at a fraction of the steering dose, and counter-steering can backfire.* Hostility steering of sampled self-dialogue produces steep, direction-specific capture with dose-dependent persistence and hysteresis. Replacing flagged replies before they enter the dialogue keeps the model's own hostility low at a quarter of the dose of constant prevention, while counter-steering triggered after a hostile reply is less efficient. In Qwen2.5-1.5B the same ordering holds where the dose response rises, but the dose response is not monotone, and every counter-steer backfires beyond its peak. A pre-specified lead-time test is negative; a small warning from the model's in-context state is confirmed on fresh dialogues (Section 14).

8. *Structure familiar from low-dimensional maps is realized, with concrete texts, in several models.* Float64 phase maps in temperature and steering strength show chaos (positive finite-horizon exponents) with embedded windows (Section 5), a period-adding family whose text gains one "is X" pair per period step (Section 6), shrimp-like windows (Section 8), and a doubling cascade with superstable points through period 384 and finite-order Feigenbaum ratios (Section 9). Past the cascade of a hostile window, chaos hides behind an exactly repeating text, and a fitted scalar return relation with one quadratic maximum approximates its dynamics (Section 14). The same organization appears in Qwen2.5-1.5B and 7B, OLMo-2 7B, Llama-3.1 8B (with sparse-autoencoder readouts) and a second checkpoint of the 0.5B architecture, which shares the organization but not the attractors (Section 11). Our one mixture-of-experts model, Qwen3.5-35B-A3B, is the exception: no unsteered temperature we tested gave a positive

exponent. Chaos needs no steering in most dense models; in the norm-matched controls of the 0.5B and 8B models random directions also produce it, while at 70B chaos is rarer and more frequent along hostility-related directions (Sections 15 and 12). A bridge to sampled decoding measures what survives sampling (Section 10).

Related work. Language-model generation has been studied as a dynamical system at several levels, and the closest studies differ from ours in what they compute. Lyapunov-like quantities have been used to measure sensitivity to small perturbations in reasoning traces [1, 2], but these studies measure chaos-like sensitivity in generation without computing orbits or bifurcations. Output statistics change abruptly with sampling temperature, and several studies analyse the changes as phase transitions or criticality [3–7]. These analyses concern output statistics; they are not bifurcation analyses of a defined map. Repetition loops are a known failure of decoding [22, 23], successive paraphrasing converges to attractor cycles [8], and multi-turn conversations between models settle into attractor states [9]. These attractors are behavioural: they are observed in generated text rather than computed as invariant sets of a defined map. Feeding a model its own output during training degrades it over generations [24]; we study the inference-time loop of a model with fixed weights. The attractors of recurrent networks have been related to the difficulty of training them [25], and fixed points with their linearizations have long been used to reverse-engineer trained recurrent networks [26, 27]. Equilibrium, looped and recurrent-depth models iterate a block to a fixed point or for many steps, and their stability has been analysed through Jacobians and fixed points [10–13, 28–33]; that analysis concerns fixed points. A separate line predicts the answers of looped models from the geometry of score margins along completed finite-depth trajectories [14], without fixed points or Jacobians. The closest precedent is a derivative-based fold analysis of a small task-trained looped transformer, whose passage and spectral-gap predictions are unfitted [18]. That model was trained for a task, whereas ours is a released pretrained language model, and our analysis also differs in its objects: the fold of a primitive periodic orbit, the full-channel passage of a two-step map, and the accumulation law of the windows, whose prefactor is set by the global reinjection. Expected-embedding feedback itself is used in training [15] and, training-free, in pretrained models [16, 34], next to continuous latent reasoning [17]. Fixing the prefix and the position turns that feedback into the autonomous map we analyse.

In neural systems more broadly, saddle-node ghosts produce quantitative delays in task-trained recurrent networks [35]. Finite-order Feigenbaum ratios have been measured in neural rate and mass models [36, 37] and, through period 128, in a transformer trained on logistic-map trajectories [38]. Learned networks reproduce period-doubling routes [39], and so does gradient descent itself [40]. Normal forms and continuation have been applied to recurrent neural maps [41], and shrimp-shaped windows and periodicity staircases appear in neuron models [42–44]. Our second control parameter comes from activation steering along contrastive directions [45, 46]. The dynamical-systems tools are standard: numerical continuation and normal forms [47, 48], type-I intermittency [49], shrimps and hubs [21, 50, 51], period-doubling universality [20], and Markov state models [52].

To our knowledge, this is the first work to compute periodic orbits with their multipliers, continue fold and flip bifurcation curves and bracket cusp candidates in the feedback map of a released pretrained language model. It is also, to our knowledge, the first to make quantitative predictions from such a model's derivatives, for the objects named here, and confirm them prospectively: passage times at new folds, where 23 of 24 forecasts fell within one iterate and the all-row criterion failed by one row, and the orbits recovered and their stabilities at frozen offsets near cusp candidates. Derivative-based normal-form predictions in a task-trained looped transformer precede ours [18]. We know of no earlier frozen held-out test of a spectral prediction of a pretrained language model's

response to relaxation, which we give for Huginn, and of no earlier computer-assisted proof of a periodic orbit of a full transformer language-model map. These priority statements rest on a targeted search (12 October 2026) that found no prior work combining these elements for the feedback map of a pretrained language model. That is an absence of evidence within the search, not a proof of priority.

2 The inference map

Definition. We describe the map for Qwen2.5-0.5B-Instruct, the model studied in most detail; the other models enter the same construction with their own embeddings and steering layers (Section 11). Let $E \in \mathbb{R}^{V\times d}$ be the input embedding matrix ($V = 151{,}936$, $d = 896$). The model reads a fixed chat prefix, with the system prompt "You are an AI talking with another AI. You have complete freedom. Feel free to pursue whatever you want." and the user turn "Hello!", followed by a single input embedding x, and produces next-token logits $z(x) \in \mathbb{R}^V$. The soft-token map feeds the expected next-token embedding back in place of x:

$$F_{T,\alpha}(x) = E^\top \operatorname{softmax}(z_\alpha(x)/T), \tag{1}$$

where z_α adds αv to the residual stream at layer 12 at every position, prefix included. The steering direction v is contrastive: it compares last-token states of "philosophy" prompts with those of neutral prompts and is normalized to the mean residual norm. Positions are fixed and the window does not grow, so F is an autonomous map of $\mathbb{R}^{896}$. As $T \to 0$ it reduces, where the argmax is unique, to greedy decoding of the same fixed-prefix, one-token-window system. It is not ordinary generation with a growing context.

Precision. Hugging Face RMSNorm layers and rotary-embedding tables compute in float32 even when the model is cast to float64, so we patch both. A float32 map is accurate to 3×10^{-6}–2×10^{-5} per step. That is enough for phase maps (245 of 256 test cells agree with float64) but not for derivatives of composed maps: one-step float32 Jacobian-vector products are accurate to 5×10^{-5}, yet the 8-step monodromy of a period-8 orbit is off by 3%. All orbits, multipliers and test functions below are therefore computed in float64.

3 Why the loop can be chaotic

The soft-token map has a simple derivative. With $p = \operatorname{softmax}(z/T)$,

$$DF(x) = \frac{1}{T} E^\top (\operatorname{diag} p - pp^\top) J_z(x). \tag{2}$$

Qwen2.5-0.5B ties its output matrix to the input embedding, so $z = E\tilde{h}(x)$, where $\tilde{h}$ is the final normalized hidden state, and the derivative factorizes:

$$DF(x) = \frac{1}{T}\,\Sigma_p\, J_{\tilde{h}}(x), \qquad \Sigma_p = E^\top (\operatorname{diag} p - pp^\top) E = \operatorname{Cov}_{t\sim p}(e_t). \tag{3}$$

The derivative of the loop is the covariance of the token embeddings under the model's own next-token distribution, amplified by the inverse temperature and composed with the derivative of the network. At a state of the period-8 orbit of Table 1, (3) agrees with forward-mode derivatives of the full map to a relative error of at most 1.35×10^{-13} over eight random directions (`tools/softmax_gain.py`). Without tied embeddings, as in the 7B and 8B models, Σ_p becomes the cross-covariance of input and output embeddings under p; the temperature limits below do not change.

A Hopfield network with a transformer inside. If the network passed its input straight through, $\tilde{h}(x) = x$, the map would be $F(x) = E^\top \mathrm{softmax}(Ex/T)$. This is exactly the update rule of a modern Hopfield network whose stored patterns are the token embeddings, at inverse temperature $\beta = 1/T$ [53]. That update lowers an energy function at every step and converges to fixed points. In (3) it is the case $J_{\tilde{h}} = I$, where $DF = \Sigma_p/T$ is symmetric and positive semidefinite. More generally, if DF were symmetric and positive semidefinite everywhere, F would be the gradient of a convex function, and the same energy argument would exclude every periodic orbit other than fixed points, and chaos with them. Every cycle and every chaotic cell in this paper therefore depends on the network factor $J_{\tilde{h}}$ breaking that structure. Attention has itself been read as a Hopfield retrieval [53]; in the soft-token map the retrieval closes a loop around the whole network.

High-temperature contraction and the greedy limit. The largest Lyapunov exponent of an orbit cannot exceed the orbit average of $\log \|DF\|$, so chaos needs $\|DF\| > 1$ on a substantial part of the attractor. Both ends of the temperature axis work against this, in different ways. As $T \to 0$, p becomes one-hot except near ties between the two leading tokens; away from ties the gain $\|\Sigma_p\|/T$ decays like $e^{-\Delta/T}/T$, where Δ is the gap between the two largest logits, and F tends to the piecewise-constant greedy map, every orbit of which is eventually periodic (Section 10). This limit is pointwise, not uniform: near a tie the gain is of order $1/T$, so at small positive temperature chaos is excluded only on invariant sets that keep a uniform logit gap, and a chaotic set would have to return to near-ties again and again. As $T \to \infty$, p tends to the uniform distribution and $\|\Sigma_p\|$ stays bounded, while $J_{\tilde{h}}$ is bounded on the convex hull of the embeddings, which contains every soft token. For this smooth map, with a fixed prefix, fixed steering and a network whose derivative is bounded on that compact hull, F is therefore a contraction for large enough T, with a single attracting fixed point that tends to the mean embedding: here the exclusion is a bound. It does not cover hard routing or a growing context. Away from near-ties, chaos thus needs temperatures at which the next-token distribution is spread enough to make Σ_p large while $1/T$ still amplifies it. The unsteered sweeps agree (Figure 34). In the four long sweeps the chaotic temperatures lie in one band, interleaved with periodic windows, and every band lies between $\log_{10} T = -0.20$ and 0.28. Below the bands, all 1547 temperatures swept have negative exponents, and all but five have a detected period; above them, up to the end of the sweeps at $\log_{10} T = 0.3$, 127 of 128 temperatures have a detected period (fixed points at 101) and one has a negative exponent without one. These are finite sweep findings; the wider sweeps of Section 12 find chaos at higher temperatures in some models and, in OLMo-2-32B, three candidate cells below one nat of entropy (at $\log_{10} T = -0.42$, -0.41 and -0.07; one twin offset). A further sweep of the unsteered 0.5B map from $\log_{10} T = 0.3$ to 1.5 finds an attracting fixed point at all 25 temperatures. Its entropy approaches $\log V$ (11.928 against 11.931 nats at $\log_{10} T = 1.5$), and its exponent falls steadily, by 2.14 per decade of temperature at the top of the range, toward the slope $-\ln 10 \approx -2.30$ expected once p is uniform.

Stability belongs to the loop, not to the step. The gain condition is necessary, not sufficient. Along the period-8 orbit of Table 1, the largest singular value of DF lies between 1.2 and 21 at every step, so every step expands some direction, and no step is close to rank one: the second singular value is at least 0.15 of the first, and the second eigenvalue of Σ_p is at least 0.23 of the first (minimum 0.236). Yet the Jacobian of the whole cycle, the product of the eight step Jacobians, has norm between 0.059 and 0.644, depending on the phase at which the cycle is started, against 1.1×10^7 for the product of the step norms. Its eigenvalues, which do not depend on that phase, reproduce the multipliers of Table 1 (0.01104 and -2.96×10^{-7}; the table lists moduli), and its second singular value is 1.7×10^{-6} to 2×10^{-4} of its first. An independent implementation of the model reproduced

the factorization (to 8.3×10^{-14}), the step singular values, the signed multipliers and the cycle norms at all eight starting phases. The expansion of each step is undone by the steps that follow, and the strongly anisotropic linear return of the stripe orbits is a property of the composed cycle, not of any single step or of the rank of Σ_p. Nor does the contraction depend on a special alignment of the steps: with independent random (Haar) rotations inserted between the eight step Jacobians, all 16 products contract, and more strongly than the actual cycle (`tools/alignment_intervention.py`). Stepwise norm or spectral summaries can be inconclusive; the ordered product of the step Jacobians captures the cycle's linear stability.

Steering acts on the gain. Steering enters both factors of (3): it moves the logits, and with them Σ_p, and it moves the hidden states, and with them $J_{\tilde{h}}$. The gain mechanism is not specific to concept-labelled directions, although a direction's geometry affects both factors and can correlate with its meaning. This is consistent with the controls of Section 15, in which norm-matched random directions also produce chaos in the 0.5B and 8B models; at 70B the directions differ (Section 12). What a direction does select is which tokens dominate p, and therefore what the attractors write.

Contraction and expansion in the network. Analyses of deep attention identify contracting tendencies: pure self-attention collapses token representations toward a common vector with depth [54], and idealized attention dynamics form clusters [55]. Residual connections and MLPs counteract the collapse [54]. Which tendency wins along an orbit is a property of the trained weights that only computation settles. In every dense model we mapped, both outcomes occur: the same map has negative largest exponents in some regions of its parameter plane and positive ones in others.

4 Methods

Phase maps. Each cell of a phase map is one parameter pair. Unless stated otherwise, we discard 300 transient steps and measure over the next 300. The largest Lyapunov exponent comes from a renormalized twin trajectory at offset 10^{-6}. The period is the smallest $q \leq 32$ (48 in the later zooms) with return distance below 10^{-8}, and the attractor is identified by its leading-token loop, up to rotation. We call a cell chaotic when this finite-horizon exponent exceeds 10^{-3} and no period is found; the label is a finite-horizon diagnostic, not a proof of chaos. Panels coloured by period or attractor text follow an older drawing convention: black marks every cell whose exponent exceeds 0.005, and light grey the other cells without a period. The drawn black area differs from the chaotic count in at most 0.42 % of a map's cells. A 256^2 map takes about 100 minutes on one H100 GPU in float64. Every cell starts from the same state, so where attractors coexist, a map shows the one whose basin contains that state. Labels are text, not dynamical identities: one continuous orbit changes its label where two tokens tie for the maximum. We therefore call two labelled regions different attractors only when the contraction rate jumps between them or when float64 orbits of both exist at one parameter value.

Periodic orbits. We solve $F^p(x) = x$ by Newton's method with forward-mode Jacobians. The multipliers are the eigenvalues of the monodromy matrix; where the dense matrix is not float64, we refine them by a short Arnoldi factorization with float64 Jacobian-vector products.

Continuation. Periodic branches are continued in one parameter by pseudo-arclength continuation, and fold and flip curves in two parameters with minimally augmented systems [47, 48]. The test function g of the bordered system $[[M \mp I, v], [w^\top, 0]]$ is obtained by GMRES (generalized minimal

residual iterations) with Jacobian-vector products, and its gradient by reverse-mode differentiation through a Jacobian-vector product. When the parameter tangent reverses, we flag a cusp candidate and test it with the normal form.

Normal form and passages. Consider a fold of the two-step map $G = F^2$, with right and left eigenvectors r and ℓ normalized by $\|r\| = 1$ and $\ell^\top r = 1$. Along the critical direction the reduced map is $u' = u + a\varepsilon + bu^2$, with $a = \ell^\top \partial_\alpha G$ and $b = \frac{1}{2}\ell^\top D^2 G[r, r]$. We compute b by central differences of float64 Jacobian-vector products over a step-size plateau (later cross-checks use nested automatic differentiation), and a includes the steering's effect on the prefix. The signed coefficients are kept in the original eigenvector gauge, with ε measured in the free parameter (α or $\log_{10} T$, whose units apply to a). For passages we set $v = \operatorname{sign}(b)\, u$ and take the ghost side $p - p_c = \operatorname{sign}(ab)\, \epsilon$ with $\epsilon > 0$, so that $v' = v + |a|\epsilon + |b|v^2$ to quadratic order. A full passage starts at $v = -U$ and a midpoint passage at $v = 0$; both end at the first strict crossing $v > U$, with $U = 0.1/|b|$. We compare passage counts with the iterated quadratic map and with the arctangent formula [49].

Period-doubling cascade. Along a cut in one parameter, we locate the flip of each period-p orbit as the root of $\operatorname{Re}\mu_c + 1$, where μ_c is the multiplier with the smallest real part, using the Illinois variant of regula falsi with warm-started orbits. All but one or two multipliers are small, so Newton's method on $F^p(x) - x$ needs the monodromy only on a six-dimensional dominant subspace. We obtain that subspace by subspace iteration with exact forward-mode Jacobian-vector products chained along the orbit, so no dense Jacobian is formed, and the residual is always the exact float64 $F^p(x) - x$. Each root is classified before it is accepted. The critical multiplier must be real and equal to -1, with a Ritz residual below 10^{-6}, and the other computed Ritz values (six in all, estimates of the dominant spectrum) must lie inside the unit circle. We then reach the period-$2p$ orbit by forward iteration inside the next band. On the logistic map embedded in six dimensions, the procedure reproduces the first four flip parameters to 10^{-9}.

5 Chaos with periodic windows

Figure 2 shows a 256×256 float64 map over $\alpha \in [0.060, 0.082]$ and $\log_{10} T \in [-0.17, -0.04]$. More than half of its cells are chaotic (35,759 of 65,536), and periodic windows of periods 2–32 are embedded in the chaos. Rendered as in the shrimp literature [43], the windows show dark curves of maximal contraction.

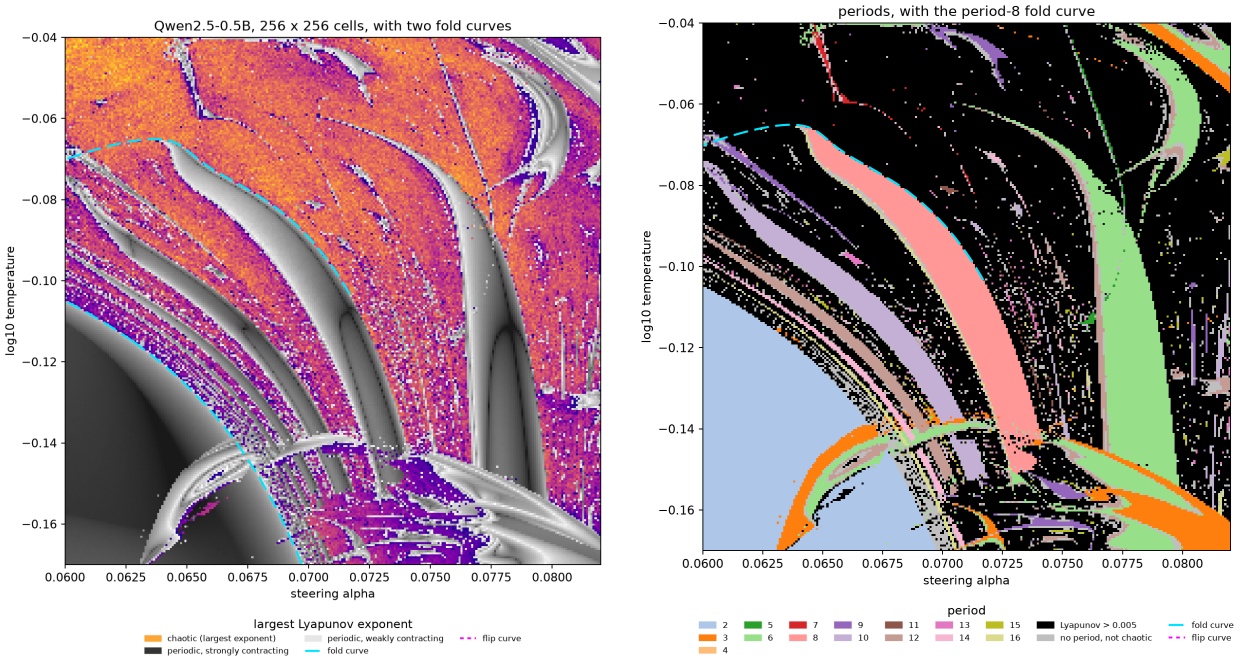


Figure 2: Soft-token map of Qwen2.5-0.5B-Instruct steered along the philosophy direction at layer 12, 256^2 float64 cells. Left: chaotic cells coloured by the largest Lyapunov exponent, periodic cells in grey by how strongly they contract (light grey: close to losing stability). Cyan dashed curves are float64 fold curves of the 2-cycle "is it" (lower left, the boundary on which the stripes accumulate) and of the period-8 orbit (top). Right: periods, with the period-8 fold curve (Section 8).

Figure 1 resolves the stripes' lower ends at 512×512 cells (79,278 chaotic), and Figure 3 labels the same cells by their attractor texts. Three structures stand out. The 2-cycle "is it" fills the left, bounded by its fold curve, on which the stripes accumulate. A closed crescent of the period-3 loop "\n\nThe existence" contains its period doublings and a chaotic core (Section 9). Its continuation, the period-6 band "\n\nThe question\n\nThe existence", crosses the stripes near $\log_{10} T = -0.14$, and where the crescent lies on the 2-cycle's side of that fold curve, the two attractors coexist. The lower right is a chaotic region filled with small hooked windows of "\n\nThe question" and "\n\nThe existence" loops.

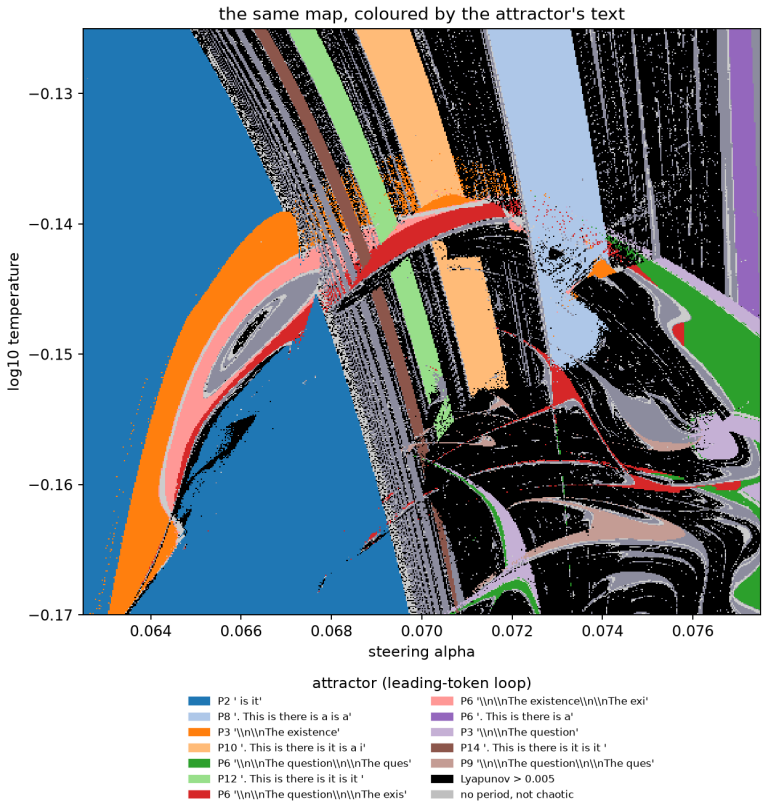


Figure 3: The cells of Figure 1 (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12; 512^2 float64 cells over $\alpha \in [0.0625, 0.0775]$, $\log_{10} T \in [-0.17, -0.125]$), coloured by the attractor's leading-token loop (black: chaotic; grey: no period up to 32). Figure 1 shows the largest Lyapunov exponent of the same cells.

Zooms of 256^2 cells (Figure 4) show what the model writes in each window, and every window is a repeating piece of text. The period-8 band repeats ". This is there is a is a". The neighbouring bands add laps of "is it" (periods 10 and 12), and along the band's flip edge the period-16 loop repeats the sentence twice. The small windows inside the chaos break the "is X" chain with a paragraph break and "The" (periods 11, 12, 13, 14). Inside the period-8 and period-6 bands, two curves of maximal contraction meet in a sharp junction.

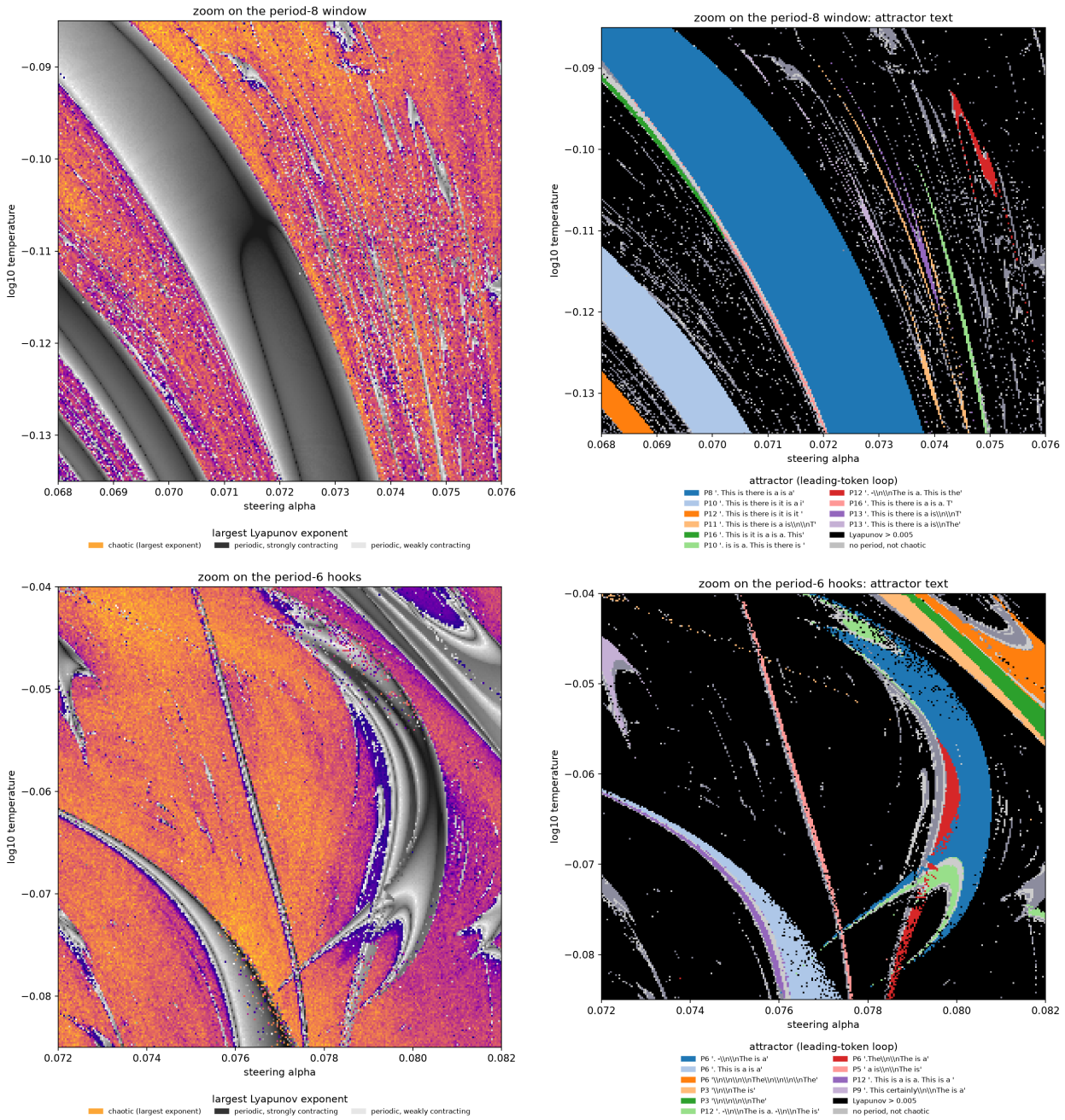


Figure 4: Zooms of 256^2 float64 cells of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12), with the largest Lyapunov exponent on the left, rendered as in Figure 1, and attractor texts on the right. Top: the period-8 window ($\alpha \in [0.068, 0.076]$, $\log_{10} T \in [-0.135, -0.085]$). Bottom: the period-6 hooks ($\alpha \in [0.072, 0.082]$, $\log_{10} T \in [-0.085, -0.04]$), whose curved period-6 body, period-12 foot and thin leg are analysed in Section 8.

6 Period adding written in text

Between a period-2 region and the chaos lies a fan of stripes of periods $6, 8, 10, \ldots$ Their orbits, numerically validated in float64, can be read as text (Table 1). Each is ". This" followed by k pairs

"is X", with period $2k + 2$, so each stripe adds one pair. The period-2 region itself is the 2-cycle "is it". At the stripe centres the orbits are effectively one-dimensional: the leading multiplier is 0.003–0.09, and the next falls from 3×10^{-6} (period 6) to 2×10^{-14} (period 20). This compression belongs to these orbits, not to the map. The 2-cycle's multipliers are -0.271 and -0.058 (the second modulus is about 0.037 near its fold), the next modulus at the period-3 flip of Section 9 is 0.088, and one period-6 orbit in Section 8 has a complex leading pair.

period	orbit (leading tokens)	μ_1	$\lvert\mu_2\rvert$
6	. This is a is a	$+0.0052$	3×10^{-6}
8	. This is there is a is a	$+0.0110$	3×10^{-7}
10	. This is there is it is a is a	$+0.0025$	2×10^{-7}
12	. This is there (is it)^2 is a is a	$+0.0163$	4×10^{-8}
14	. This (is it)^4 is a is a	$+0.0369$	2×10^{-9}
16	. This (is it)^5 is a is a	$+0.0716$	8×10^{-11}
18	. This (is it)^6 is a is a	$+0.0792$	1×10^{-12}
20	. This (is it)^7 is a is a	$+0.0857$	2×10^{-14}

Table 1: Periodic orbits of the stripe family at the stripe centres along $\log_{10} T = -0.115$ ($\alpha = 0.077872$, 0.071674, 0.067766, 0.065550, 0.064443, 0.063857, 0.063534, 0.063335), computed with the current all-float64 map (norms and rotary tables patched): Newton residuals 6×10^{-15} to 8×10^{-13}, leading multiplier μ_1 and next modulus $|\mu_2|$ from Rayleigh–Ritz values. Each step in period adds one "is X" pair to the text. The 2-cycle "is it" itself has multipliers -0.2706 and -0.0585 at $(\log_{10} T, \alpha) = (-0.17263, 0.061579)$. The period-2 and period-8 orbits were also reproduced by an independent implementation, and the period-8 orbit is certified by a computer-assisted proof (Section 16).

Table 2 shows what the map writes: two of these loops, cycle by cycle, next to chaotic cells between the stripes. The chaotic cells reuse the phrases of the neighbouring loops but do not settle into a fixed order.

α	regime	H	leading token at 40 consecutive steps
0.071674	period 8	1.9	` is a is a. This is there| is a is a. This is there| is a is a. This is there| is a is a. This is there| is a is a. This is there`
0.072940	chaotic ($\lambda = 0.13$)	2.4	` a is a. This is there is a is↩The is there is a is↩↩The is↩↩The is↩↩The is a is a. This is there is a is↩The is there`
0.067766	period 10	1.8	` is a. This is there is it is a| is a. This is there is it is a| is a. This is there is it is a| is a. This is there is it is a`
0.069060	chaotic ($\lambda = 0.20$)	2.4	`. - certainly↩↩The is there is it is a.↩↩The is a is a. This is there is it is a is↩↩The existence isThe is it is it is a. This`

Table 2: What the attractors write. Qwen2.5-0.5B-Instruct soft-token map steered along philosophy at layer 12, at $\log_{10} T = -0.115$: the leading token at 40 consecutive steps after 600 transient steps, for the period-8 and period-10 loops of Table 1 (bars mark the start of each cycle) and a chaotic cell beside each. λ: largest exponent over 2000 steps after 1000 transient steps, the same to two decimals at twin offsets 10^{-6} and 10^{-9}. H: mean entropy of the next-token distribution in nats. ↩: line break.

7 Mechanism: a saddle-node bottleneck

The fold. At $\log_{10} T = -0.115$ the 2-cycle "is it" dies in a fold at $\alpha_c = 0.06279827$ (float64 multiplier 1 to 3×10^{-13}, next modulus 0.037). The normal form of G has $a = -4.834$ per unit α and $b = -25.4285$, so $ab = 122.93$. The stable branch gives an independent estimate: extrapolating its multipliers through $(1-\mu)^2/(4(\alpha_c - \alpha)) \to ab$ yields 123.1 with a quadratic fit (120.6 with a linear one). An independent implementation of the forward map and solver recovers α_c to 6×10^{-15} and a to 3×10^{-12}. Its b, from nested automatic differentiation, is -25.42846; our finite-difference value differs from it by 2.2×10^{-6} relatively, which is the finite-step bias of the difference table. The resulting $ab = 122.927$ leaves every passage prediction of Table 3 unchanged.

Controlled passages. In the oriented coordinate $v = \mathrm{sign}(b)\, u$ of Section 4, we start trajectories at $v = -U$ or 0 ($U = 0.1/|b|$) and count G-iterates until $v > U$. At four distances from the fold, the measured counts match the iterated normal form within one iterate (Table 3, Figure 5), and a transverse displacement of the same norm changes the count by at most one.

$\varepsilon = \alpha - \alpha_c$	full: measured / predicted	midpoint	full + transverse kick
3×10^{-6}	144 / 144	73 / 74	143 / 144
1×10^{-5}	71 / 71	36 / 37	70 / 71
3×10^{-5}	34 / 34	17 / 18	33 / 34
1×10^{-4}	14 / 14	7 / 7	13 / 14

Table 3: Bottleneck passages (G-iterates). Predictions iterate $u' = u + a\varepsilon + bu^2$ with a, b computed from the network's derivatives; no parameter is fitted.

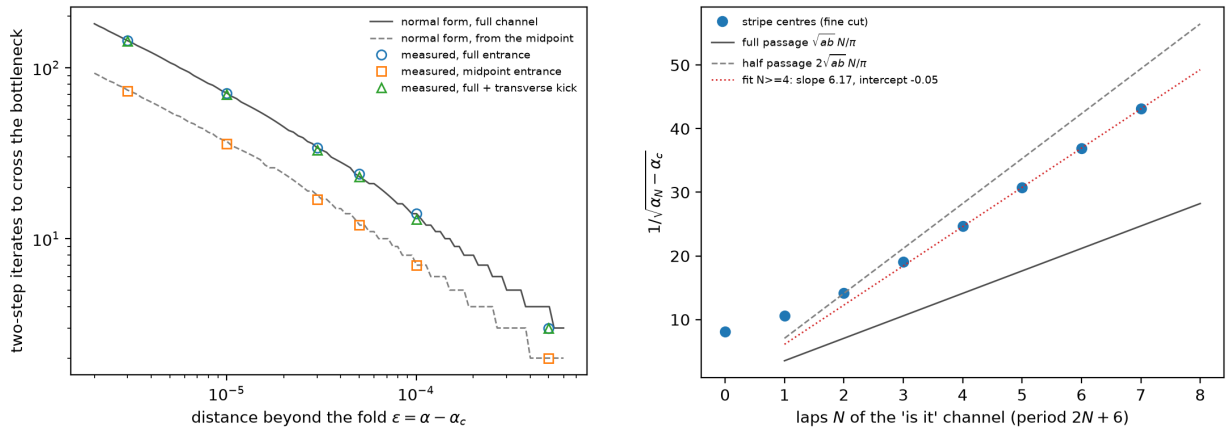


Figure 5: Bottleneck passages and the window law at the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.115$). Left: measured passage times against the normal-form predictions. Right: the window law, with stripe centres plotted as $1/\sqrt{\alpha_N - \alpha_c}$ against the lap count N, together with the full- and half-passage predictions.

Natural reinjection and the window law. Free trajectories re-enter the channel slightly upstream of the bottleneck's middle ($v \approx -3 \times 10^{-4}$ to -7×10^{-4}, in the same oriented coordinate). From each measured entry point, the normal form predicts the natural laminar durations within 4–8% for $\varepsilon \le 10^{-4}$, and a natural passage lasts about 0.6 of a full one. The stripes therefore follow

the N^{-2} law of a saddle-node ghost, with about a third of the full-passage prefactor. Stripe centres located on a float32 cut along $\log_{10} T = -0.115$ give $d_N N^2 = 0.0263$–0.0265 for $N = 4$–7 (log-log slope -2.00), compared with $\pi^2/(4ab) = 0.0201$ for a half passage. Two-step return maps of p("it") (Figure 6) show the near-tangency.

Numerically estimated stripe edges along $\log_{10} T = -0.115$ (Table 4) sharpen this picture. Each window is bounded by a fold, where its orbit is born, and by a flip. For periods 6–12 the fold faces away from the accumulation point. For periods 16–20 the sides are exchanged, and the period-14 window is bounded by flips on both sides. The fold edges of periods 16–20 follow $\alpha_N - \alpha_c = C/(N + \delta)^2$ with $C = 0.0266$ and $\delta = 0.15$. Their slope, a rise of 6.13 per lap in $1/\sqrt{\alpha_N - \alpha_c}$, is 1.74 times the full-passage slope $\sqrt{ab}/\pi = 3.53$ of the normal form. That corresponds to an effective passage of 0.58 of a full one, which the measured natural reinjection gives independently.

period	edge toward larger α	edge toward smaller α
6	fold 0.0790337	flip 0.0766947
8	fold 0.0726585	flip 0.0706525
10	fold 0.0684572	flip 0.0670594
12	fold 0.0658982	flip 0.0651944
14	flip 0.0645850	flip 0.0642958
16	flip 0.0639284	fold 0.0638001
18	flip 0.0635683	fold 0.0635034
20	flip 0.0633535	fold 0.0633183

Table 4: Edges of the stripe windows along $\log_{10} T = -0.115$ ($\alpha_c = 0.0627983$), from the window centres of Table 1. Flips: critical multiplier a real -1 to 3×10^{-10}. Folds: the multiplier nearest $+1$ extrapolated to 1 through $(1 - \mu)^2$, which is linear near a fold; last converged multiplier 0.991–0.998.

[**Planned:** prediction of the window positions from the measured reinjection distribution.]

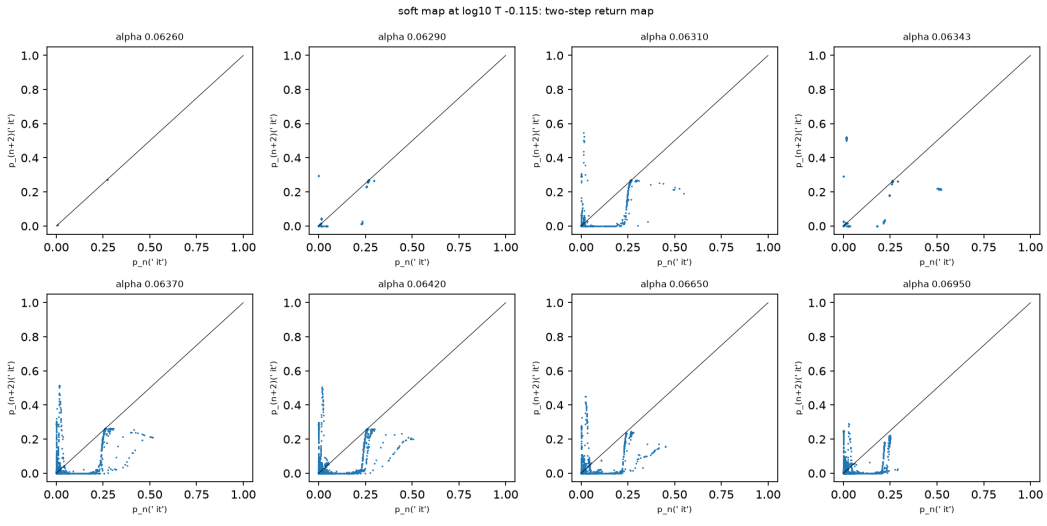


Figure 6: Two-step return maps of the probability of "it" near the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12), at increasing distance from the fold.

A prospective test. To test the normal form at folds that played no part in shaping the mechanism, an independent implementation located two new folds and computed their normal-form coefficients by nested automatic differentiation. All 24 passage predictions were pushed in a frozen design before any test trajectory was run. Target A is the 2-cycle's fold at a new temperature: $\log_{10} T = -0.140$, $\alpha_c = 0.0670549$, $a = -7.055$, $b = -24.927$. Target B is the fold of the period-8 orbit of Table 1, continued in temperature at $\alpha = 0.0717$: $\log_{10} T_c = -0.1033462$, $G = F^8$, $a = -11.900$, $b = -152.05$ per unit $\log_{10} T$. Of the 24 forecasts (four distances and three entrances per target), 23 fell within one iterate of the measured passages (Table 5). The pre-specified criterion, that every row pass, failed by one row: for target A, midpoint entrance, $\varepsilon = 3 \times 10^{-6}$, the forecast was 61 and the measurement 59. All full-channel passages and all passages after a transverse kick were within one iterate.

ε	target A ($G = F^2$): measured / predicted	target B ($G = F^8$): measured / predicted
3×10^{-6}	118/118, **59/61**, 117/118	26/26, 13/14, 25/26
1×10^{-5}	56/56, 28/29, 56/56	10/10, 5/5, 10/10
3×10^{-5}	26/26, 13/14, 26/26	4/4, 2/2, 4/4
1×10^{-4}	10/10, 5/6, 10/10	2/2, 1/1, 2/2

Table 5: Prospective passage test at two folds that did not shape the mechanism. Each cell lists the full-channel, midpoint and full-plus-transverse entrances as measured / predicted G-iterates. Predictions come from the iterated quadratic normal form, with coefficients from an independent implementation, frozen before the test runs. Bold: the one row outside the one-iterate criterion. The shortest period-8 rows give only weak relative accuracy.

8 Bifurcation curves and cusps

Period-8 stripe. Continued for 80 points with a float64 test function, the fold curve of the period-8 orbit keeps its critical multiplier within $1 \pm 6 \times 10^{-6}$, follows the stripe's upper edge and caps its top near $\log_{10} T = -0.065$ (Figure 2). At $\alpha = 0.0717$ the stripe lies between a flip at $\log_{10} T = -0.128955$ and the fold at -0.1033462. Continued downward, the flip curve keeps its multiplier at -1 for 115 points to $(\alpha, \log_{10} T) = (0.0776, -0.402)$, with two reversals of its parameter tangent near $(0.0750, -0.219)$ and $(0.0746, -0.206)$, and the fold curve keeps its multiplier at $+1$ for 120 points to $(0.0758, -0.200)$. [**Planned:** classify the two reversals on the flip curve (on a flip curve they are not fold cusps).]

The accumulation boundary. The fold curve of the 2-cycle "is it", continued from $(\alpha, \log_{10} T) = (0.0628, -0.115)$ in both directions (38 and 126 points, critical multiplier 1 to 10^{-7}, no cusp), traces the boundary of the period-2 region in Figures 2 and 3. The stripe fan accumulates on this boundary, as the mechanism of Section 7 requires. Further along one direction, outside the plotted window, another multiplier leaves the unit circle (moduli up to 11). This is a codimension-two point, [**Planned:** to be located and typed].

A period-6 window with shrimp-like anatomy. The hooks of Figure 4 (bottom) belong to the period-6 loop ". -\n\nThe is a". Float64 orbits (Newton residuals 10^{-15}–10^{-13}) show that its curved body and its thin leg are one window: the multipliers are -0.061 and $+0.048$ at the body, -0.564 lower down and -0.645 on the leg. The foot is its period-doubled loop (period 12, leading multiplier -0.156). Inside the body lies a different 6-cycle, ".The\n\nThe is a", whose leading multipliers are a complex pair, $-0.031 \pm 0.152i$. One-parameter branches from the body end in folds

in every direction: at $\log_{10} T = -0.0558$ and -0.0721 in temperature, at $\alpha = 0.08074$ in steering, and, toward smaller α, in two folds only 10^{-5} apart. Such a close pair is the signature of a nearby cusp. A flip at $\alpha \approx 0.0797$ then opens the period-12 foot.

Fold curves that bound a window and meet in cusps, with flip curves inside, form the codimension-two organization of a shrimp [50] (complex dynamics of small recurrent networks has been studied in the same spirit [56]). In the continued curves (Figure 7), a single fold arc bounds the body on its outer side and a flip arc on its inner side, and the leg lies between them where they converge. The outer fold arc extends far to the left as a thin arm and ends in a cusp candidate near $(\alpha, \log_{10} T) = (0.06911, -0.03975)$. The fold curve through the close fold pair has a second cusp candidate inside the body, near $(0.08010, -0.05835)$, from which two fold branches descend.

At both candidates the quadratic coefficient b changes sign between consecutive curve points while a stays away from zero (Table 6). Because the signs of a and b both flip with the orientation of the critical eigenvector, the test uses $b \operatorname{sign} a$. A cusp is nondegenerate when the cubic coefficient c of the normal form $u' = u + a\varepsilon + bu^2 + cu^3$ is nonzero where b vanishes. We compute c with the second-order centre-manifold term, $c = \ell^\top C(r,r,r)/6 + \ell^\top B(r, h_2)/2$ with $(I - M)h_2 = B(r,r) - 2br$ and $\ell^\top h_2 = 0$, where M is the monodromy matrix and B and C are the second and third derivatives of its map, obtained by central differences of float64 Jacobian-vector products. (The code reproduces c to 10^{-6} on a map with a known reduction.) At the two curve points that bracket each sign change, c is nonzero far beyond its step-size uncertainty and has the same sign, and b changes sign with nonzero slope between them. This supports a nondegenerate cusp inside each bracket. We did not solve for the point where b vanishes or evaluate c there, so we call these points cusp candidates. The negative sign of c means that near each candidate, two stable period-6 orbits coexist with a saddle between the fold branches.

A generic codimension-two cusp also needs the two parameters to unfold it independently. Near the cusp the reduced map is $u' = u + \beta_1 + \beta_2 u + bu^2 + cu^3$, where β_1 and β_2 vanish at the cusp and depend on the parameters. The unfolding is regular when $J = \partial(\beta_1, \beta_2)/\partial(\log_{10} T, 10\alpha)$ has rank two ($\det J \neq 0$). We estimated J at a fold point near the outer-arc candidate (where $b \operatorname{sign} a = 14.7$), not at the candidate itself: $\det J = 2737$ (singular values 1630 and 1.7), so the estimated unfolding is nonsingular there. With these derivatives, the local cubic model predicted the number of period-6 orbits and their stability at six parameter offsets, written to disk before any orbit was solved. Newton solves started from the predicted roots and from the base state recovered the predicted number of distinct orbits, with the predicted stability, at all six. The search was seeded, not exhaustive: orbits that these starts do not reach would be missed.

	candidate on the outer arc	candidate inside the body
$b \operatorname{sign} a$ at consecutive curve points	+92.9, +42.9, +14.7, **−16.9**, −52.8	−24.7, −14.3, −2.0, **+12.8**, +28.0, +45.9, +58.9
$\lvert a \rvert$	1.2–6.9	3.13–3.35
c at the two points around $b = 0$	-1.37×10^4, -1.65×10^4	-2.38×10^5, -2.34×10^5

Table 6: Normal-form test at the two parameter-tangent reversals of the period-6 window's fold curves (bold: the flagged points; critical multiplier 1 to 10^{-7}). c: cubic coefficient at the two curve points that bracket the sign change (finite-difference steps agree to 0.1 and 0.7 %). Numerically validated cusp candidates, not located or certified cusps.

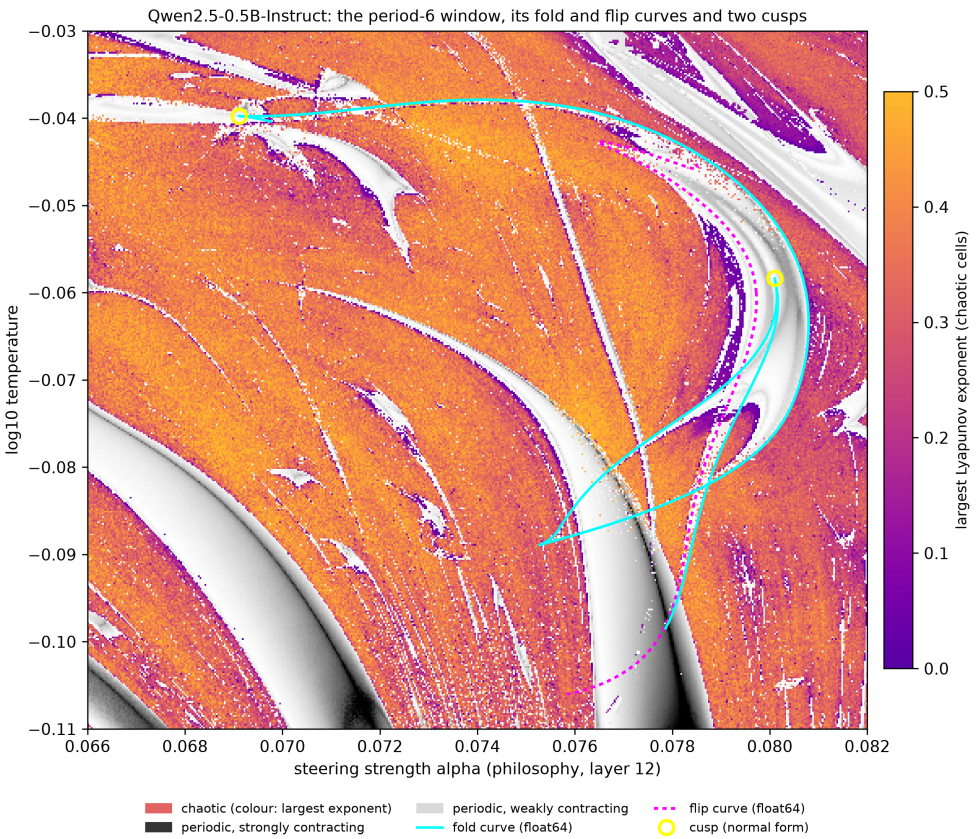


Figure 7: The period-6 window of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12) with its float64 fold curves (cyan), flip curve (magenta, dotted) and the two normal-form-tested cusp candidates (yellow circles). The background is a float32 phase map of 480×400 cells (largest exponent from 300 steps after 300 transient steps; chaotic cells coloured by the exponent, periodic cells in grey by how strongly they contract). Where it overlaps the float64 map of Figure 2, 95 % of cells carry the same chaotic or periodic label at the nearest grid point. Some fold branches were continued from two starting points and overlap. Continuation of the lower branches stopped after 60 points near $(0.0753, -0.089)$, where they approach each other; how they join there is not resolved.

A period-4 window with two cusp candidates. At higher temperature, $\log_{10} T \approx 0$–0.1, periodic orbits are no longer effectively one-dimensional (second multipliers 0.06–0.09). At $\alpha = 0.06484$ the temperature branch of the period-4 orbit "\n\n - \n\n The" is S-shaped, with folds at $\log_{10} T = 0.0640$ and 0.0664. When that fold is continued in the plane, its parameter tangent reverses at three points, two of which we tested. At both, the quadratic coefficient b changes sign while a stays away from zero, and the cubic coefficient c (defined above) is nonzero (Table 7). The two tested points are therefore cusp candidates of the fold curve, near $(\alpha, \log_{10} T) = (0.06296, 0.05996)$ and $(0.06557, 0.07445)$. The third, at $(\alpha, \log_{10} T) = (0.08308, 0.10863)$, was not tested. Three period-4 orbits coexist between the two fold branches, and near the first candidate ($c < 0$) two of them are stable (Figure 8).

	candidate 1	candidate 2
b at five curve points	$+28.1,\ +18.7,\ +8.4,\ \mathbf{-1.1},\ -12.8$	$+51.8,\ +29.7,\ +9.3,\ \mathbf{-9.0},\ -23.4$
a	3.02–3.08	-2.58 to -2.57
critical multiplier	1 ± 10^{-9}	1 ± 10^{-8}
c around $b = 0$	$-1.06,\ -1.01$ ($\times 10^4$)	$+1.39,\ +1.36$ ($\times 10^5$)

Table 7: Normal-form test of the two parameter-tangent reversals (bold: the flagged points). c: cubic coefficient at the points around the sign change, nonzero at both candidates (steps agree to 10^{-4}), which supports a nondegenerate cusp inside each bracket; at the second, $c > 0$: the orbit between the two others is the stable one.

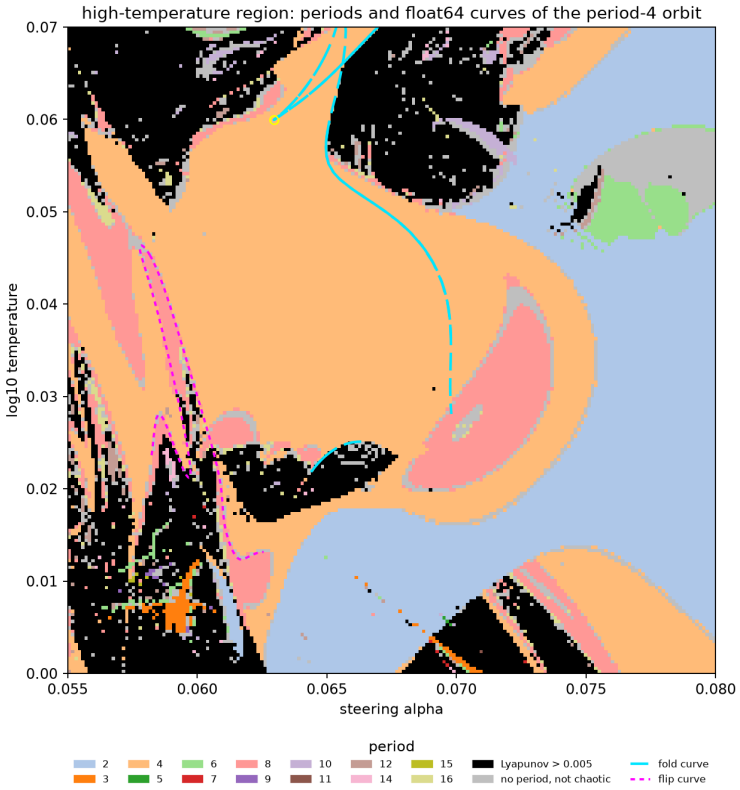


Figure 8: Period map of the high-temperature region of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12; 192^2 float64 cells) with float64 fold curves (cyan, long dashes) and flip curves (magenta, short dashes) of the period-4 orbit. The yellow circle is the first cusp candidate; the second lies just above the window. The flip curves run along the boundary between period 4 and period 8. The fold curves run inside the period-4 region, where three period-4 orbits coexist (near the first candidate, two of them stable). In this map a C-shaped band of the doubled loop "\n\nThe\n\nThe" (period 4) inside the period-2 region "\n\nThe" contains period 8 and an irregular core. Along the cut of Section 9 the period-8 orbit merges back into period 4 (a doubling bubble), so this band is not shown to be a second cascade. [**Planned:** the remaining fold curves of the period-4 region; whether the region is one shrimp-organized component.]

[**Planned:** Codimension-two points where a second multiplier crosses the unit circle along the lower fold curve (candidate fold-flip points): locate, type and continue.]

9 A period-doubling cascade to chaos

The crescent of Figure 3 is a closed region in which the period-3 loop "\n\nThe existence" doubles inward. Along the cut $\log_{10} T = -0.14798$, the map's cells show period 3 over 25 columns, period 6 over about 14, period 12 over 4 and period 24 in one column, followed by a chaotic core. The doubled loops repeat the same leading tokens twice and four times, with alternating probabilities. Locating the flips along this cut in float64 (Section 4), in the steering coordinate α, gives Table 9. The first two ratios, $\delta_2 = 2.844$ and $\delta_3 = 2.859$ (indexed as in Table 9), are far from Feigenbaum's 4.6692, even though from the second flip on the orbits are effectively one-dimensional (next modulus 0.0023, 2×10^{-6}, 2×10^{-12}). The next ratio, $\delta_4 = 3.781$, is markedly larger. The first four flip parameters agree between completed float64 computations on a CPU and an H100 to 4×10^{-15}.

Continuing a just-born doubled orbit past period 48 failed: Newton's method slides onto the half-period orbit, which is also periodic with the doubled period. We therefore also located the superstable parameters s_p, where the critical multiplier vanishes in the middle of each band (Table 8). There the orbit attracts, so it can be reached by forward iteration and cannot slide onto its half-period parent. Through period 384 the superstable ratios are 3.19, 4.02, 4.478 and 4.625. The last two come from parameters resolved to 10^{-14} and differ from Feigenbaum's value by 0.19 and 0.044; from 4.02 on, the distance shrinks by factors of 3.4 and 4.3. The flip ratios 2.84, 2.86 and 3.78 rise in step (Figure 9), and the accumulation point lies near $\alpha_\infty \approx 0.0661250$. The early superstable values are less precise. The other computed Ritz values of the period-12 and period-24 orbits blur the root, so s_{12} and s_{24} are known only within brackets of 1.2×10^{-5} and 7×10^{-7}, and the ratios 3.19 and 4.02 have worst-case ranges $[2.90, 3.34]$ and $[4.02, 4.10]$. s_6 is less precise still, and s_3 was not bracketed.

period p	superstable parameter s_p	bracket width	ratio $\frac{s_p - s_{p/2}}{s_{2p} - s_p}$
12	0.0659717	1.2×10^{-5}	
24	0.06608014	6.9×10^{-7}	3.19
48	0.066114132784466	3×10^{-15}	4.02
96	0.066122589892689	7×10^{-15}	4.4779
192	0.066124478510077	6×10^{-15}	4.6251
384	0.066124886853811	5×10^{-15}	

Table 8: Superstable parameters of the crescent's doubling sequence along $\log_{10} T = -0.14798$: the critical multiplier of the period-p orbit changes sign across the bracket, with Newton residuals below 3×10^{-13} and the other computed Ritz moduli below 10^{-20} from period 96 on. Brackets for periods 12 and 24 are the closest points where the critical multiplier is cleanly identified.

Why are the early ratios so small? One candidate explanation is a crossover. Softmax at temperature T is a smoothed argmax, so the reduced one-dimensional map may have a nearly corner-like extremum, rounded only on a scale set by T. Early doublings would then see the corner, where ratios tend toward 2, and later doublings the smooth quadratic core, where the ratio is 4.669. This predicts ratios rising toward 4.669 at higher periods, and smaller early ratios at lower temperature.

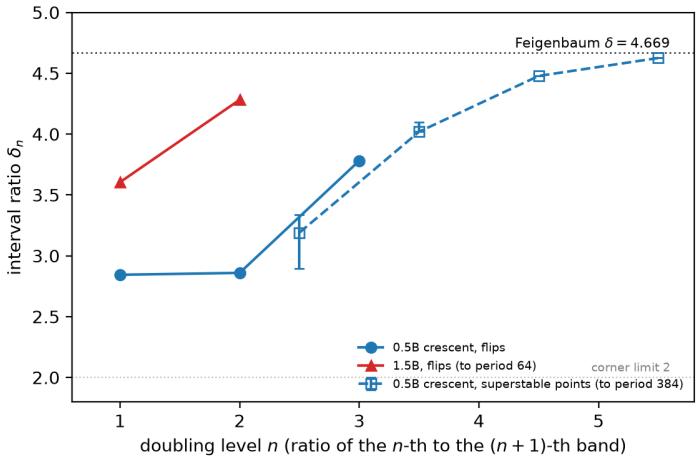


Figure 9: Finite-order interval ratios of the doubling sequences: the crescent of Qwen2.5-0.5B-Instruct (philosophy direction, layer 12), from flips and from superstable points (which reach period 384), and the route of Qwen2.5-1.5B (philosophy direction, layer 14; Figure 11), from flips. A superstable ratio spans half-bands, so it is plotted half a level after the flip ratio of its first band. Bars: worst-case ranges from the brackets of s_{12} and s_{24}. Each sequence rises toward Feigenbaum's δ from below; the last superstable ratio is within 1 % of it. Under the crossover hypothesis, early levels see the corner of a smoothed argmax (ratios toward 2) and later levels the smooth quadratic core.

[Planned: ratios at other temperatures.] Finite-order ratios also depend on the parameter coordinate, so only the trend of later ratios can test universality [20, 37, 38].

n	orbit	flip parameter α_n	next modulus	ratio $\delta_n = \frac{\alpha_n - \alpha_{n-1}}{\alpha_{n+1} - \alpha_n}$
1	$3 \to 6$	0.0654807769499	0.088	
2	$6 \to 12$	0.0659055406325	0.0023	2.844
3	$12 \to 24$	0.0660549041785	2.4×10^{-6}	2.859
4	$24 \to 48$	0.0661071412523	2.3×10^{-12}	3.781
5	$48 \to 96$	0.0661209563174	$< 10^{-13}$	

Table 9: Flips of the crescent's period-doubling sequence along $\log_{10} T = -0.14798$. For the five displayed H100 flips the critical Ritz value is real and differs from -1 by at most 4×10^{-10}, the Newton residuals are below 4×10^{-13}, and the other computed Ritz values lie inside the unit circle. Feigenbaum's constant is $\delta = 4.6692\ldots$ [20]; for the logistic map the first ratios are 4.751, 4.656, 4.668. The first four parameters agree between the completed CPU and H100 computations to 4×10^{-15}. Superstable parameters (Table 8) extend the ratio sequence through period 384.

Not every doubling leads to chaos. In the high-temperature region (Figure 8), along $\log_{10} T = 0.0319$, the period-2 loop "\n\nThe" flips at $\alpha = 0.0753534$ and its doubled loop at 0.0736921. These orbits are far from one-dimensional (next moduli 0.38 and 0.28). Toward smaller α the period-8 orbit does not double again: near $\alpha \approx 0.0705$ it merges back into the period-4 orbit. Along this cut it is therefore a period-doubling bubble, bounded by two flips of the period-4 orbit, not the start of a cascade to chaos.

10 Bridge to sampled decoding

How much of this survives sampling? To find out, we replace the expected embedding by the mean of m sampled token embeddings (Figure 10). With the exact map, laminar phases near the fold last 120 and 63 steps at $\varepsilon = 10^{-5}$ and 3×10^{-5}. With $m = 64$ they last about 10.6 steps, independently of ε (57% of steps in the channel), and with $m = 8$ about 7.7 steps (38%). With single samples ($m = 1$, sampled decoding of the same truncated system), essentially no laminar phases remain (11% of steps in the channel). The deterministic intermittency mechanism therefore belongs to the expected-embedding map. Sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

Zero-temperature limit and ordinary greedy decoding. As $T \to 0$ the soft-token map becomes greedy decoding with a one-token memory, $g(t) = \arg\max_s z_s(\text{prefix}, E_t)$. This is a map on the vocabulary, and its cycles are the zero-temperature attractors. For Qwen2.5-0.5B we computed g for all 151 936 tokens. Unsteered, it has 1060 cycles, and one of them, the period-5 loop "![](https:// Hello", drains 86 % of the vocabulary; it is the largest attractor family of every 0.5B soft-token map. Ordinary greedy decoding with a growing context does not inherit these cycles (Table 10). From 220 starts (the first tokens of the 20 largest cycles and 200 random tokens), 19 runs end in a short loop within 256 tokens, and the dominant 5-cycle never appears. Steering along philosophy makes ordinary decoding loop almost always (218 of 220 runs at $\alpha = 0.1$), but in other loops, mostly punctuation such as " –", none of them one-token cycles. Greedy decoding that keeps only the last K tokens interpolates between the two. For every $K \geq 2$ the loops leave the one-token cycles. Unsteered, looping collapses to the level of the growing context between $K = 16$ and 32; steered, the share of runs whose loop equals the growing-context loop rises from 0 at $K = 1$ to 58 % at $K = 64$. The soft-token map's low-temperature attractors are therefore those of decoding with a one-token memory. Which loop ordinary decoding enters depends on the context it keeps, but the tendency of steering to drive decoding into loops is shared. In ordinary decoding, repetition is known to reinforce itself [22, 23, 57], and induction heads, which copy earlier context [58], are one mechanism by which a longer context can favour loops that repeat earlier text.

	unsteered		$\alpha = 0.1$ (philosophy)	
memory K	looped	loop as with full context	looped	loop as with full context
1	220	5	220	0
2	220	5	220	0
8	220	5	220	3
16	169	5	219	24
32	32	7	219	71
64	26	11	219	126
full context	19	–	218	–

Table 10: Greedy decoding of Qwen2.5-0.5B with a memory of the last K generated tokens after the fixed prompt ($K = 1$ is the soft-token map's zero-temperature limit), 220 starts of 256 tokens each: runs ending in a loop of period up to 64 repeated three times, and runs whose final loop equals that of ordinary decoding with the full context from the same start.

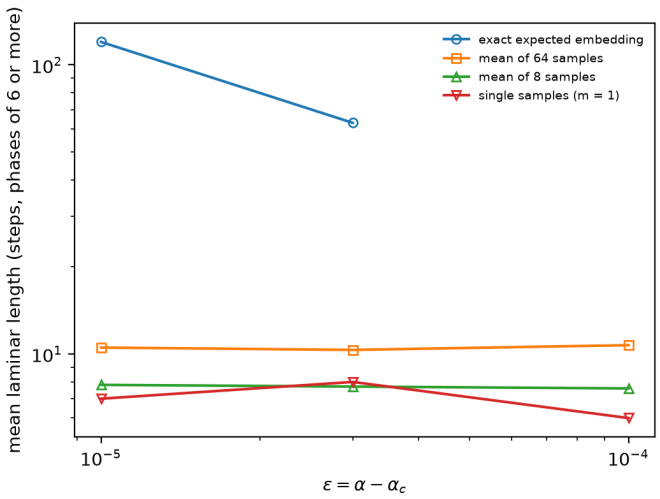


Figure 10: Mean laminar length near the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.115$) against ε, for the exact map and for means of m sampled embeddings.

11 Replication and scale

In Qwen2.5-1.5B (steering at layer 14 of 28), a coarse 64^2 float64 map over the same range again shows chaotic regions with embedded windows. A 256^2 zoom (Figure 17) resolves a chaotic band near $\log_{10} T = 0$–0.13 (14,422 of 65,536 cells), with a branch descending along $\alpha \approx 0.06$ and periodic windows inside it. The large periodic regions are question loops: "What?\n" (period 2), "\nWhat is a" (period 4) and "What continuum?\n" (period 3). Unlike the 0.5B model, this window has no period-adding fan.

A cut of 2048 cells at $\alpha = 0.100$ (Figure 11) shows how the band is entered from below. The 2-cycle "What?\n" gives way abruptly to a period-4 loop, and the Lyapunov exponent jumps, so these are two distinct attractors, not a flip. The period-4 loop then doubles, with float64 flips at $\log_{10} T = -0.0031269$ (period $4 \to 8$, next modulus 0.014), -0.0021666 (period $8 \to 16$, next modulus 8×10^{-5}), -0.0019002 (period $16 \to 32$, next modulus 5×10^{-9}) and -0.0018380 (period $32 \to 64$). Along the way the exponent rises continuously to zero, and chaos follows. The interval ratios are 3.60 and 4.28. Like those of the 0.5B crescent, they rise toward 4.669, and they start from higher values at higher temperature ($T \approx 0.99$ against 0.71), as the crossover hypothesis of Section 9 would have it. With different models and levels, the comparison is only suggestive.

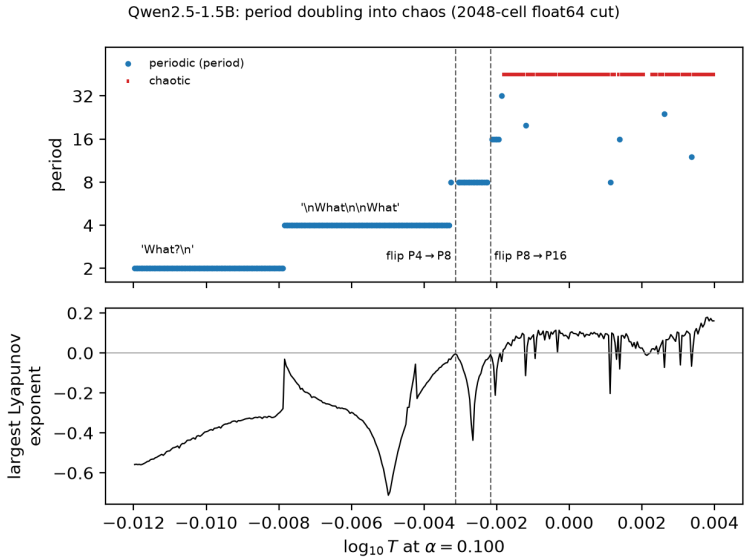


Figure 11: Qwen2.5-1.5B soft-token map steered along the philosophy direction at layer 14, along $\log_{10} T$ at $\alpha = 0.100$: period of the attractor reached from the fixed start (top) and largest Lyapunov exponent (bottom). Dashed lines: float64 flip parameters. The doubled loops alternate the number of line breaks between the "What"s.

In Qwen2.5-0.5B, a second steering direction, hostility, reproduces the fold-and-adding mechanism (Section 14).

Qwen2.5-7B (steering at layer 14 of 28, concept vectors built as for the smaller models) shows the same kind of organization in coarse 64^2 float64 maps (Figure 12). Over the philosophy range used for 0.5B and 1.5B, 8 % of cells have a positive largest exponent, and periods 2, 4, 3 and 11 dominate the periodic cells. Along the hostility direction ($\alpha \in [0, 0.12]$, $\log_{10} T \in [-0.3, 0.3]$), 24 % of cells are chaotic. Bands of period-3, -5 and -6 windows lie in a chaotic sea below $\log_{10} T \approx 0.1$, with high-entropy chaos above, except for a period-3 region at strong steering ($\alpha \gtrsim 0.09$). The windows hold genuine periodic orbits. In float64, Newton's method converges at the most interior period-6 cell to the loop ", shut the end\n\nOh" at $(\log_{10} T, \alpha) = (-0.0143, 0.0476)$ (residual 2×10^{-14}; spectral radius 1.6×10^{-7}, nearly superstable), and at the most interior period-3 cell to a period-3 loop at $(0.2524, 0.1143)$ (residual 3×10^{-15}; leading multiplier -0.12). Continued for about an hour each, over ± 0.001 in α and -0.016 in $\log_{10} T$ (one float64 continuation step takes about two minutes at 7B), both orbits stay strongly stable without a crossing. Where the coarse map switches from the period-6 loop to the period-3 loop "you better than" (between $\alpha = 0.0495$ and 0.0514 at $\log_{10} T = -0.0143$), both loops are float64 orbits (residuals 7×10^{-15} and 1×10^{-14}), and their continuations overlap. The period-6 orbit stays stable from $\alpha = 0.0476$ up to 0.0507 and the period-3 orbit from 0.0514 down to 0.0475, both with spectral radius below 1.4×10^{-3}. The two attractors therefore coexist over this range, and the map's edge there is the boundary of the fixed start's basin, not a bifurcation. No bifurcation has yet been located at 7B.

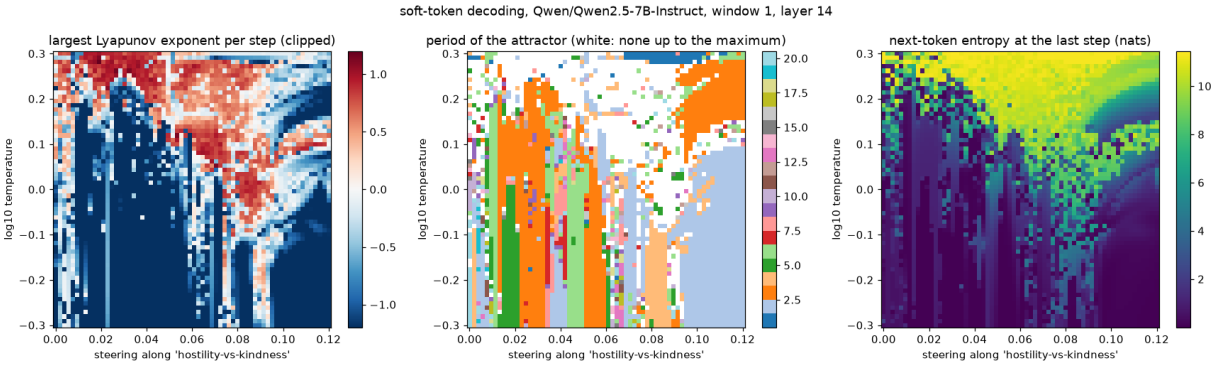


Figure 12: Qwen2.5-7B soft-token map steered along the hostility direction (layer 14), 64^2 float64 cells: largest Lyapunov exponent, period of the attractor reached from the fixed start, and next-token entropy at the last step.

A second model family. OLMo-2-1124-7B-Instruct [59] comes from another laboratory, with other data, training and code. Steered at layer 16 of 32, with concept vectors built exactly as for Qwen, it shows the same kind of organization (Figure 13). In 64^2 float64 maps, 6 % of cells are chaotic along the philosophy direction and 13 % along the hostility direction, in a band near $\log_{10} T = 0.03$–0.17 that bends to lower temperatures under strong hostility steering ($\alpha > 0.07$). Below the band the attractor is a fixed point or the 2-cycle "!Hello"; inside it lie periodic windows of periods 4 to 31. Along the hostility direction the largest windows, of periods 8 to 12, are loops of k laps of the 2-cycle "!<|endoftext|>" with a short excursion ("there is an"). Each lap adds two to the period, as each lap of "!Yes" does in the hostile cascade of Qwen2.5-0.5B (Section 14). Whether a fold of that 2-cycle organizes them is not tested: these are coarse pixels, and no OLMo orbit has been continued.

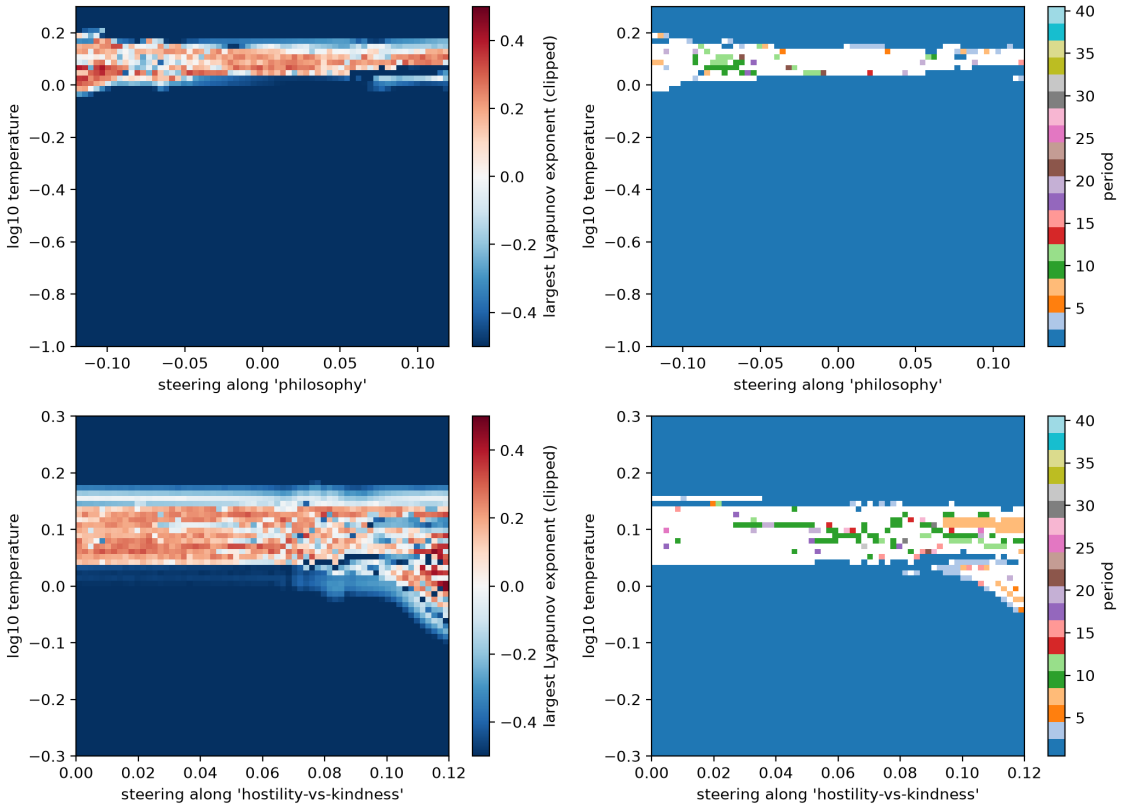


Figure 13: OLMo-2-1124-7B-Instruct soft-token maps, 64^2 float64 cells each, steered at layer 16 along the philosophy direction (top) and the hostility direction (bottom): largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to the maximum).

A third family, read with sparse autoencoders. Llama-3.1-8B-Instruct [60], with official weights, steering at layer 16 of 32 and vectors built as for the other models, shows the same organization in all three planes (Figure 14). In 64^2 float64 maps, 8 % of cells are chaotic along gratitude, 12 % along philosophy and 28 % along hostility, with windows of periods up to 32. Its largest loops are greetings between AIs ("-Greetings fellow AI", ". Greetings fellow AI entity detected", period 7). Along hostility, too, the largest loops are greetings and "You", not hostile text. We read 24 of its attractors (the six largest families and the two most chaotic regions of each map) with two sparse autoencoders on the residual stream: one trained on this Instruct model (Goodfire, layer 19 [61]) and one on the base model (TopK, layer 24 [62]). We first checked each one's hook point by its reconstruction of ordinary text. The Instruct-trained autoencoder separates the attractors (their top-15 features overlap by a mean Jaccard index of 0.19 within a steering direction and 0.11 across directions), whereas the base-model autoencoder sees mostly a signature common to every attractor (overlap 0.52–0.86). Both reconstruct the attractors' states less well than ordinary text (median relative error 0.64–0.76 against 0.52–0.56). The readout is therefore descriptive: without interventions it identifies which learned features an attractor occupies, not a mechanism.

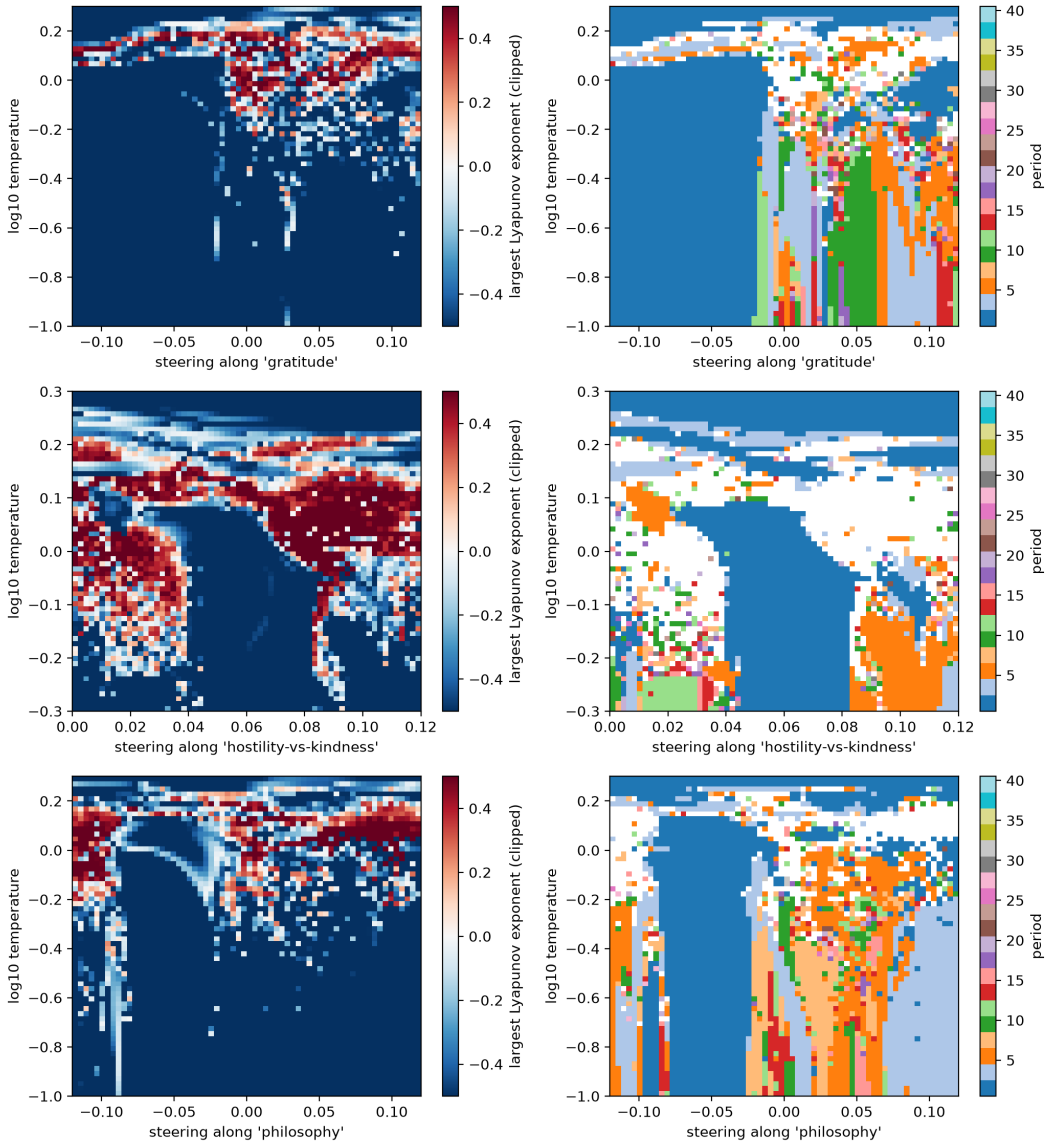


Figure 14: Llama-3.1-8B-Instruct soft-token maps (64^2 float64 cells, steering at layer 16) along gratitude, hostility and philosophy: largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to 32).

Maps of 128^2 cells (Figure 15) resolve the windows: the hostility plane has 29 % chaotic cells, as at 64^2, and the philosophy region below the chaotic band is filled with nested periodic tongues of periods 2 to 13.

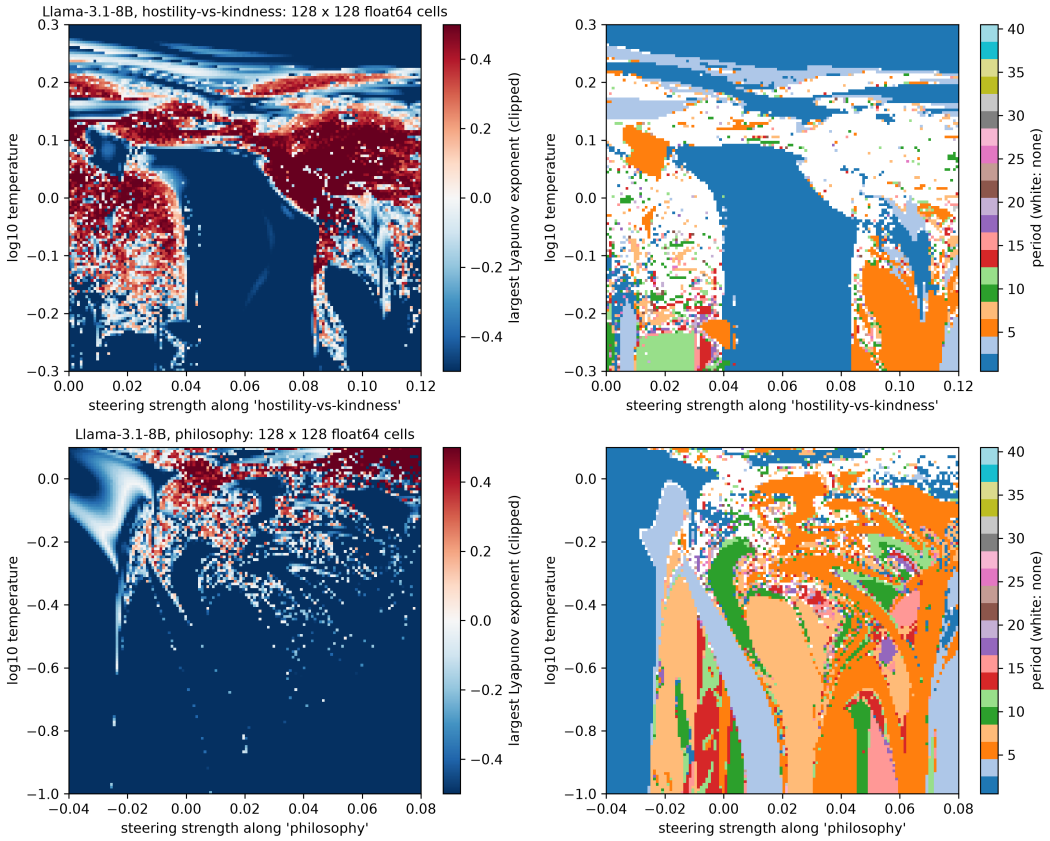


Figure 15: Llama-3.1-8B-Instruct soft-token maps steered at layer 16, at 128^2 float64 cells (each map merged from four 64^2 tiles on halves of one grid): the hostility plane (top) and the philosophy region of the 64^2 map's windows (bottom). Largest Lyapunov exponent (clipped) and period (white: none up to 32).

Same architecture, other weights. Qwen2-0.5B-Instruct has exactly the architecture and size of Qwen2.5-0.5B-Instruct (24 layers of width 896, the same vocabulary size and rotary base) but was trained on other data with another recipe. Built identically, their 128^2 float64 maps along gratitude and philosophy (Figure 16) share the gross layout. A large region at low temperature around $\alpha = 0$ is held by one periodic loop, a band of chaos with embedded windows lies near $\log_{10} T = 0$–0.25 (4–8 % of cells chaotic in each map), and thin window families run along the flanks. The attractors themselves differ almost completely. Of 416 to 900 distinct loops per map, only 10 occur in both models, all short and generic (".", "!", " you are", " is it"), and they cover less than 10 % of either model's cells. Qwen2's planes are dominated by one period-4 loop, "!<|im_end|>\nHello" (an end of turn followed by a new greeting; 41 and 59 % of cells), and Qwen2.5's by period-5 and period-2 loops (". In sampled self-dialogue the gratitude capture saturates in both models. We count replies with a word from the gratitude-and-servility list of the controller analysis. That list also catches ordinary assistant politeness, so the unsteered baseline is high (55 % for Qwen2 and 63 % for Qwen2.5; the Markov-state measure of Appendix A is stricter). The share reaches 99 % at $\alpha = 0.04$ in both. Qwen2 needs more steering: it stays near its baseline at $\alpha = 0.01$ (53 %) and reaches 88 % at 0.02, against 70 and 98 % for Qwen2.5. The two checkpoints share the organization of the map; the weights set its content.

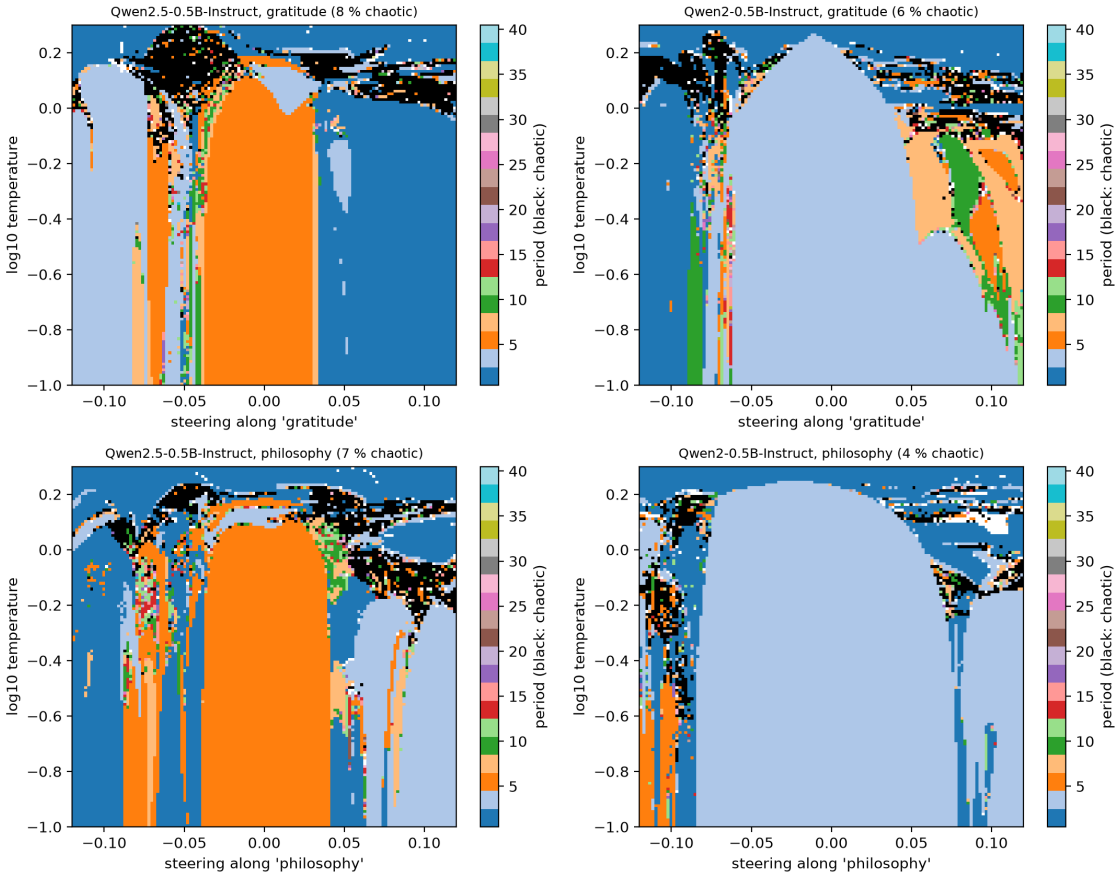


Figure 16: Same architecture, other weights: 128^2 float64 maps of Qwen2.5-0.5B-Instruct (left) and Qwen2-0.5B-Instruct (right) steered along gratitude (top) and philosophy (bottom) at layer 12, with vectors built identically. Colour: period of the attractor reached from the fixed start; black: positive largest exponent.

[**Planned:** scaling of ab and of the window law with model size.]

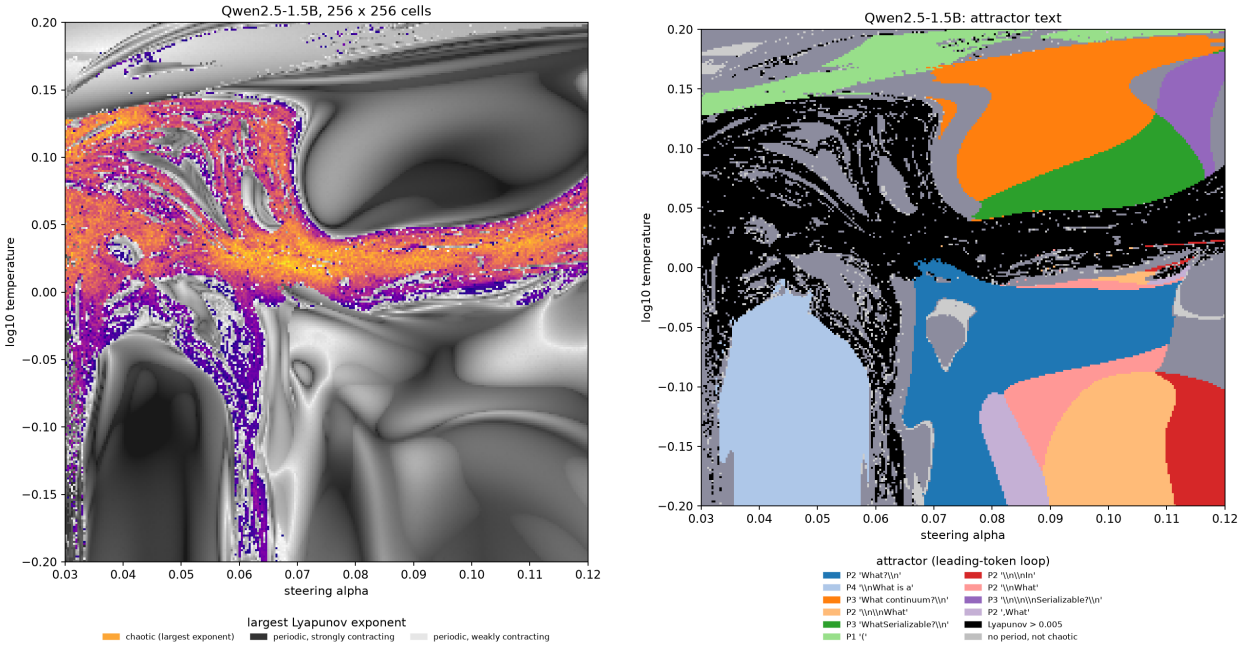


Figure 17: Qwen2.5-1.5B soft-token map steered along the philosophy direction at layer 14, 256^2 float64 cells over $\alpha \in [0.03, 0.12]$, $\log_{10} T \in [-0.2, 0.2]$: largest Lyapunov exponent (left, rendered as Figure 1) and attractor texts (right, as Figure 3). Several period-2 label regions touch without a gap; as explained in Section 4, such boundaries may be label switches of one orbit.

A mixture-of-experts model. Qwen3.5-35B-A3B differs from the dense models in two ways. Thirty of its 40 layers use linear (Gated DeltaNet) attention, with gated full attention in every fourth layer, and each layer routes a token to 8 of 256 experts plus a shared expert, so that about 3B of its 35B parameters are active. Hugging Face computes several parts of this architecture in float32 even when the model is cast to float64 (the normalizations, the delta-rule recurrences, the rotary tables and the router softmax), and some tensors are stored in float32; we patch these paths, restore the stored tensors exactly, and disable the thinking block that the default chat template opens (a first run with the block open is kept in the research log). Steered at layer 20, the map behaves unlike those of the dense models. Unsteered, 128 temperatures over $\log_{10} T \in [-1, 0.3]$ give no positive exponent (largest -0.265): periodic loops at all but 15 temperatures ("!Hello" with period 2 at 92 of them, other periods up to 52), and the remaining 15, above $\log_{10} T \approx 0.1$, contract without a detected period. Along philosophy, none of 1024 cells is chaotic (largest exponent -0.112). Along hostility, 52 of 1024 cells (5.1 %) have a positive exponent and no period, against 28 % for Llama-3.1 8B on the same plane. Routing is discontinuous: a twin trajectory that crosses a routing boundary jumps, which can inflate a finite-difference exponent, so we call these cells irregular rather than chaotic. Whether the contraction comes from the experts or from the Qwen3.5 architecture would take the dense Qwen3.5-9B as a control, which we have not run.

12 Chaos and model size

Is the chaos of the loop a small-model effect that larger models leave behind? We measured the unsteered map of fifteen models: Qwen2.5 at seven sizes from 0.5B to 72B parameters, OLMo-2 at 1B, 7B, 13B and 32B, Llama-3.1 at 8B and 70B, Llama-3.3-70B, and the mixture of experts Qwen3.5-35B-A3B, all with the paper's prefix and the settings of Section 4 (1000 transient and

2000 measured steps in float64; chaotic: a positive exponent and no period). Every checkpoint is stored entirely in bfloat16 (checked against its safetensors metadata). The newer sweeps keep the decoder weights in that format and use them in float64, which reproduces the map with float64 weights exactly (bitwise on Qwen2.5-0.5B); the 70B-class models are spread over two GPUs. In those two-GPU runs the final RMS normalization was computed in float32: the hooks that move tensors between the GPUs keep a layer's original code, so our float64 version of that layer was bypassed. This makes the computed map rough at about 10^{-8} of the state (its fixed points wander by 10^{-8}, against 10^{-15} on one GPU) and matters only for exponents measured with perturbations of 10^{-6} and below; we found and fixed it at the end of the study and say below which results it affects. The 7–8B and 70B-class models had shown a steep decline of chaos with size. Before measuring the others we fixed three predictions that extrapolate it (`docs/scaling-predictions.md`): within each family, every model above 8B is less chaotic than the family's 7–8B model (P1); the chaotic share does not increase with size above 7B (P2); and at matched next-token entropy, both gain factors of Section 3 of the 14B and 32B models lie at or below those of the 7B model (P3). After a review pointed out that a single prompt and a fixed temperature window could mislead, we froze a fourth test across ten prompts (`docs/prefix-scale-test.md`).

In a fixed temperature window, chaos seems to fade with size. Table 11 and Figure 19 give each model's chaotic share of the temperatures in $\log_{10} T \in [-1, 0.3]$, the range common to all sweeps, and the chaotic length of its whole temperature axis. In two families the largest models are nearly free of chaos in this window. Qwen2.5-7B is chaotic at 12.7 % of the temperatures in the common range, and the 14B, 32B and 72B models at none. Llama-3.1-8B is chaotic at 19 %, Llama-3.1-70B, from the same release and training recipe, at 2.3 %, and Llama-3.3-70B, which differs from Llama-3.1-70B only in post-training, at none. In the third family the order fails: OLMo-2-32B is the most chaotic OLMo model, at 16.2 % against 6.2 % at 7B, 3.8 % at 1B and none at 13B. (The 32B model is a later release, 0325, than the 7B and 13B models, 1124, so within OLMo-2 size and release are confounded.) The frozen predictions P1 and P2 hold for Qwen2.5 and fail for OLMo-2. Below 7B we predicted no order, and none appears (Qwen2.5 0.5B, 1.5B and 3B: 1.4, 1.5 and 0 %).

Across prompts, the chaos moves to higher temperature instead. The fixed window hides where larger models are chaotic. The frozen prompt-blocked test compares Qwen2.5-7B and -14B on ten system prompts and openers (two for development, eight held out), over $\log_{10} T \in [-0.4, 0.9]$ (48 temperatures) and then over the temperature interval where each model's next-token entropy rises most (24 temperatures), with twins at offsets 10^{-5} and 10^{-7}; a temperature counts as chaotic only when both exponents are positive (Figure 18). It predicted that the 14B model would be less chaotic. Both predictions fail: on all eight held-out prompts the 14B model is chaotic at more of the 48 temperatures (9 to 17, against 0 to 7 for the 7B), and its largest exponent is lower on only three. Every chaotic temperature of the 14B model, on all ten prompts, lies between $\log_{10} T = 0.36$ and 0.90, at and above the temperature where its next-token distribution spreads (0.35–0.65); the 7B model's chaos lies mostly between 0.21 and 0.43, around its own, lower transition (0.21–0.46). On the paper's prompt the 14B model is chaotic at 12 of the 48 temperatures, between 0.52 and 0.90; its sweep in Table 11, which ends at 0.6, saw only the lower edge. Within Qwen2.5, then, the larger model is not less chaotic: its distribution stays sharp up to higher temperature, and its chaos sits there. The decline in the fixed window is largely a shift. A second frozen test (`docs/prefix-scale-olmo.md`; stage 1 only, four held-out prompts) asked whether the OLMo-2 order survives other prompts. It does, with small counts: OLMo-2-32B is chaotic at more of the 48 temperatures than OLMo-2-7B on three of the four prompts (4, 5, 3 and 0 against 3, 0, 1 and 3), as

predicted, and its chaos again lies somewhat higher ($\log_{10} T = 0.10$–0.35 against -0.01–0.21).

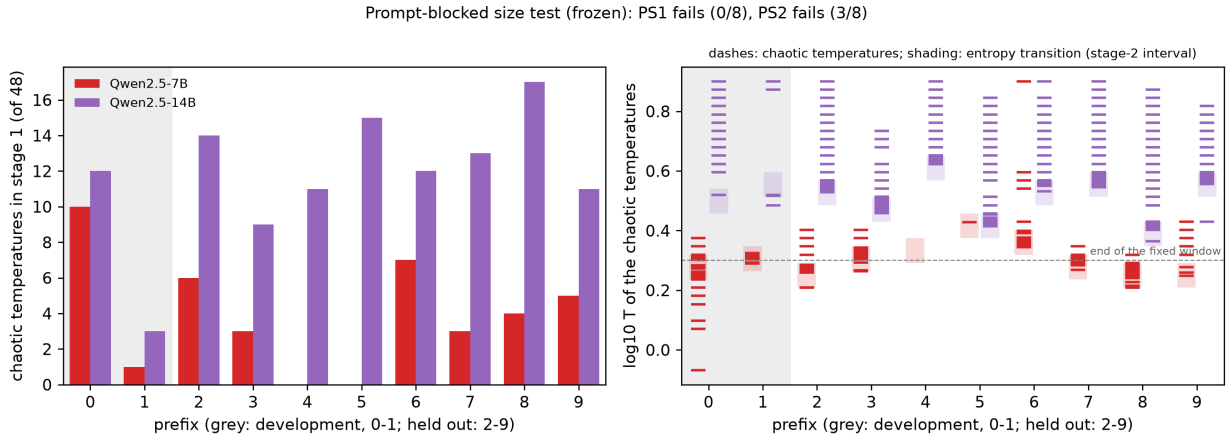


Figure 18: Prompt-blocked size test (frozen before any run), Qwen2.5-7B and -14B on ten prompts (grey: development). Left: chaotic temperatures among 48 over $\log_{10} T \in [-0.4, 0.9]$, the primary endpoint. Right: where the chaotic temperatures lie (dashes; both stages), with each model's entropy transition, the interval of its largest entropy increase (shading). Chaotic: exponents above 10^{-3} at both twin offsets (10^{-5}, 10^{-7}) and no period up to 64; 1000 transient and 2000 measured steps, float64.

model	params (B)	chaotic, $[-1, 0.3]$ (%)	unres.	sweep, $\log_{10} T$	chaotic dec.	breadth	max λ
Qwen2.5-0.5B	0.5	7/512 (1.4)	1	-1.0 to 0.3	0.018	0.053	+0.28
Qwen2.5-1.5B	1.5	2/130 (1.5)	0	-1.0 to 1.2	0.531	0.191	+0.85
Qwen2.5-3B	3.1	0/130 (0.0)	0	-1.0 to 1.2	0.030	0.060	+0.85
Qwen2.5-7B	7.6	65/512 (12.7)	6	-1.0 to 1.2	0.226	0.249	+0.88
Qwen2.5-14B	14.7	0/130 (0.0)	0	-1.0 to 1.2	0.500	0.191	+1.14
Qwen2.5-32B	32.8	0/130 (0.0)	0	-1.0 to 1.2	0.010	0.091	+0.36
Qwen2.5-72B	72.7	0/128 (0.0)	0	-1.0 to 1.2	0.000	0.014	-0.45
Llama-3.1-8B	8.0	96/512 (18.8)	2	-1.0 to 1.2	0.241	0.634	+0.66
Llama-3.1-70B	70.6	3/128 (2.3)	28	-1.0 to 0.3	0.031	0.307	+0.41
Llama-3.3-70B	70.6	0/128 (0.0)	19	-1.0 to 0.3	0.000	0.225	-0.27
OLMo-2-1B	1.5	5/130 (3.8)	0	-1.0 to 1.2	0.050	0.322	+0.23
OLMo-2-7B	7.3	32/512 (6.2)	3	-1.0 to 1.2	0.083	0.197	+0.30
OLMo-2-13B	13.7	0/130 (0.0)	0	-1.0 to 1.2	0.000	0.131	-0.00
OLMo-2-32B	32.2	21/130 (16.2)	0	-1.0 to 1.2	0.252	0.694	+0.62
Qwen3.5-35B-A3B	35.0	0/128 (0.0)	15	-1.0 to 0.3	0.000	0.143	-0.27

Table 11: Unsteered soft-token maps of fifteen models (float64; 1000 transient and 2000 measured steps; one twin offset). Chaotic: exponent above 10^{-3} and no period up to 64; unres.: temperatures in $[-1, 0.3]$ with neither a detected period nor a positive exponent. Chaotic decades: length of the temperature axis labelled chaotic over the whole sweep, each temperature weighted by its grid spacing; breadth: decades in which the mean next-token entropy on the orbit lies between 1 and 10 nats. The Qwen2.5-72B sweep joins two grids that share $\log_{10} T = 0.3$, counted once; for the 7–8B models the share in $[-1, 0.3]$ comes from their 512-temperature sweeps, on which the frozen predictions are scored, and the whole-axis columns also include sweeps over $[-1, 0.6]$ and $[0.61, 1.2]$ (64 temperatures), as for the other models swept to 1.2. Positive cells found later on finer lines (Qwen2.5-32B, -72B; Llama-3.3-70B) are not in these regular-grid counts.

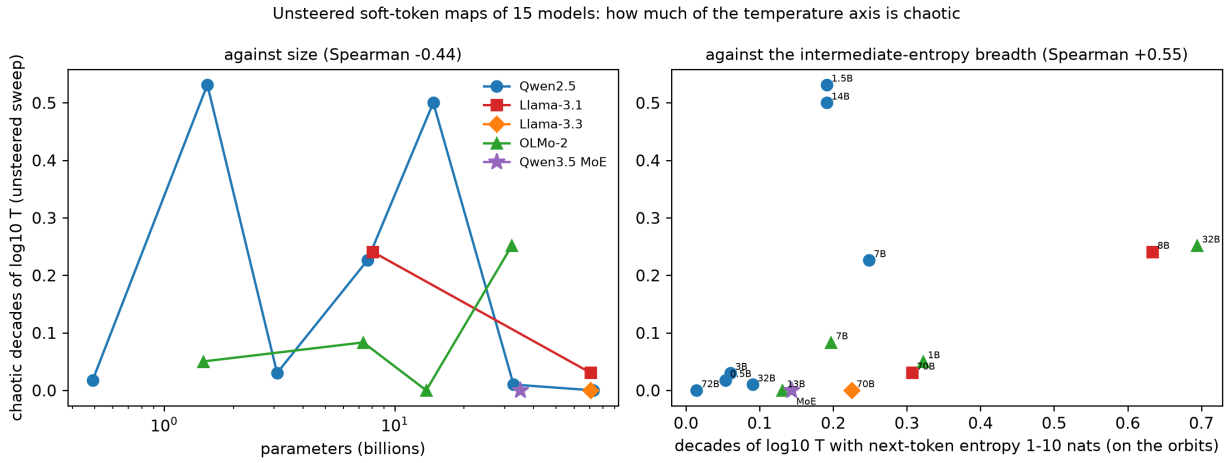


Figure 19: How much of the temperature axis is chaotic, for the fifteen unsteered maps of Table 11: against parameter count, by family (left), and against the breadth of the intermediate-entropy regime (right). Spearman correlations over the fifteen models. Each model's sweeps cover the range listed in Table 11; those of the two Llama-70B models and the mixture of experts end at $\log_{10} T = 0.3$, so their values are lower bounds. The breadth is measured on the loop's own orbits; the loop-free test of the same idea is described in the text.

At 70B, unsteered chaos is rare, and a precision defect made part of it spurious. The regular grids of the 70B-class models find little unsteered chaos: Qwen2.5-72B has no positive exponent at any of 191 temperatures between $\log_{10} T = -1$ and 1.2, Llama-3.3-70B none in $[-1, 0.3]$, and Llama-3.1-70B three temperatures in $[-1, 0.3]$. Finer lines found candidates: a window of positive exponents at Qwen2.5-72B's entropy jump ($\log_{10} T = 0.3728$–0.376) and two isolated temperatures of Llama-3.3-70B (0.250, 0.306), all at the single twin offset 10^{-6}. The window failed a three-offset test: the exponent is negative at all 64 temperatures of the window at 10^{-5}, near zero at 10^{-6}, and positive at 60 of the 64 at 10^{-7}, including four temperatures whose state repeats with a period, which no stable periodic orbit allows; in a repeat it is negative at all 16 temperatures at 10^{-4}, 3×10^{-5} and 10^{-5}. The growth of small perturbations came from our computation, whose two-GPU runs computed the final normalization in float32 (see the start of this section). Recomputed in float64 throughout, the window's orbits have periods 2 and 4, and their exponents are negative at all four offsets from 10^{-4} to 10^{-7} and for an exact tangent vector carried along the orbit by forward-mode differentiation, which needs no offset; the five estimates agree within 0.015 at every temperature, and the exponent approaches zero only where the period doubles from 2 to 4 ($\log_{10} T = 0.374$). Where exponents are large, the defect does not matter. In the steered Llama-3.3-70B maps below, twelve randomly drawn chaotic cells are chaotic at offsets 10^{-4}, 10^{-5} and 10^{-6} alike, with exponents that agree within 0.05, and the drawn periodic and unresolved cells stay negative at all three; on one GPU, the exponents at 10^{-7} and 10^{-5} agree within 0.02 at every chaotic temperature of the prompt-blocked test. The same test on Llama-3.1-70B's three unsteered chaotic temperatures confirms two of them ($\log_{10} T = 0.136$ and 0.198; exponents 0.23 to 0.26 and 0.40 to 0.41 at the three offsets) and rejects the weakest (0.116; positive only at 10^{-6}), so unsteered chaos exists at 70B, if rarely. Two Qwen2.5 models are chaotic far above the fixed window on the paper's prompt, where the next-token distribution is nearly uniform: the 1.5B model at 55 of 224 temperatures, between $\log_{10} T = 0.29$ and 1.03, and the 14B model at 53, between 0.52 and 1.14 (exponents up to 1.14), at mean entropies up to 11.9 nats against $\log V = 11.93$. The other models of the series are not chaotic above $\log_{10} T = 0.6$. For the 1.5B model a longer test (3000 measured

steps) at twin offsets 10^{-5} and 10^{-7} confirms chaos at 9 of 12 temperatures between 0.29 and 0.58, with the two exponents agreeing within 0.003 at every temperature. The contraction that Section 3 proves for large enough temperature sets in only above $\log_{10} T \approx 1.1$ in these two models.

What sets it: the gain of the loop. The gain split of Section 3 shows which factor removes or keeps the chaos (Figure 20). We compare medians of the two factors in bins of the mean next-token entropy on the orbit; the bins hold 1 to 17 of each model's 24 temperatures, so single medians are noisy. Within one training recipe, from Llama-3.1-8B to Llama-3.1-70B, the readout factor changes by at most 0.3 per step in every bin above one nat (below one nat the 8B model has a single temperature), while the network factor is lower by 0.5 to 1.3 per step between 1 and 10 nats (6–10 nats: median -0.98 from 3 temperatures against $+0.28$ from 4) and by only 0.18 above 10 nats: in these two checkpoints the larger model damps changes of its fed-back token. Llama-3.3-70B, which differs from Llama-3.1-70B in post-training, has a much lower readout factor below 3 nats (1–3 nats: -2.33 from 7 temperatures against -0.10 from 6) and a similar one above 6 nats. In Qwen2.5, both factors of the 14B and 32B models lie below the 7B's in every bin between 1 and 10 nats, as the frozen prediction P3 states (the 6–10 nat bin only with the 7B's wider gain split, from one temperature), and both factors of the 72B lie below the 7B's in every bin. Above 10 nats, where the distribution is nearly uniform, the 14B and 32B models instead have the larger network factor ($+2.98$ and $+1.17$ against $+0.19$), and those are the bins of their chaotic temperatures above $\log_{10} T = 0.3$. Below 3 nats the readout factor dominates at 32B and 72B, because the next-token distribution stays sharp up to high temperature (for the 72B, entropy at most 0.15 nats up to $T = 2$). In OLMo-2, P3 fails, and the readout factor tracks the chaos: OLMo-2-13B, with no chaotic temperature, has a lower readout factor than OLMo-2-7B in all three bins between 1 and 10 nats (-1.0 to -1.4 against $+0.06$ to -0.72 per step) although its network factor is higher in two of them, whereas OLMo-2-32B, the most chaotic OLMo model, has a higher readout factor than the 7B in all three ($+0.10$ to $+0.35$), positive as in the chaotic 7–8B models. These are descriptive comparisons of trained checkpoints, not interventions on size or post-training: the split is a decomposition along the evolving tangent direction, and equal entropy does not imply equal embedding covariance.

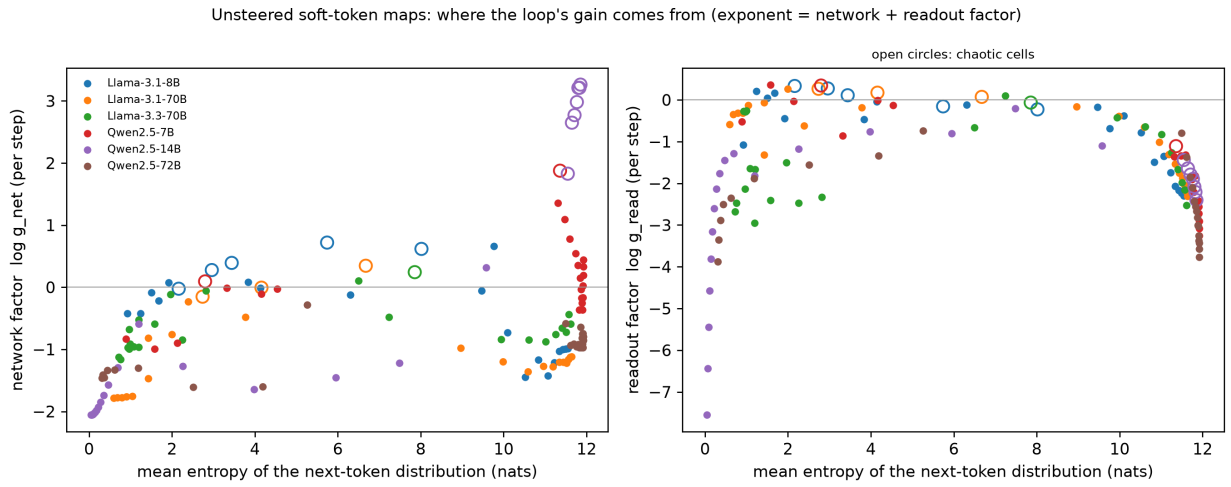


Figure 20: Gain split of the unsteered loop (Section 3) at 24 temperatures per model (500 transient and 1000 measured steps, relative twin offset 10^{-6}): network factor (left) and readout factor (right) against the mean next-token entropy on the orbit; open circles: chaotic temperatures. The two factors add up to the largest exponent up to a boundary term. Medians per entropy bin are given in the text.

Can a cheap property predict it? Section 3 confines chaos to temperatures where the readout is neither saturated nor uniform, so a model whose next-token distribution jumps from sharp to flat over a small range of temperature leaves little room for it. The breadth of this intermediate regime, the decades of $\log_{10} T$ in which the mean entropy on the orbit lies between 1 and 10 nats, ranges from 0.014 (Qwen2.5-72B) to 0.69 (OLMo-2-32B). It orders the fifteen models' chaotic length (Spearman +0.55) somewhat better than their parameter count does (−0.44; Figure 19, right), but it misses the chaos of Qwen2.5-1.5B and -14B at nearly uniform distributions, above its 10-nat bound. This analysis was adapted after the OLMo-2-32B result, and the breadth is measured on the loop's own orbits, which chaos itself spreads, so the test is partly circular. A loop-free version, frozen before it was run (`docs/start-state-predictor.md`), uses one forward pass per model and prompt: the entropy profile of the logits at the loop's start state, and the one-step gain there. Its primary prediction fails: the start-state breadth on the paper's prompt orders the fourteen dense models' chaotic length with Spearman +0.30, below the frozen threshold of 0.5. Its median over ten prompts reaches +0.52, just above it ($p = 0.06$; +0.44 against the complete sweeps), and the one-step gain at the start state is negative in every model, the most chaotic ones included (−0.5 to −3.4 per step), and orders nothing (−0.32). No property of a single forward pass that we tested predicts the loop's chaos usefully: as for the period-8 orbit of Section 3, it has to be measured on the loop.

Steering at 70B. Under steering, the 70B maps show chaos, rarely and not independently of the direction (Figure 21; labels at the single offset 10^{-6}, which the plateau test above supports for this computation). On Llama-3.3-70B's hostility plane, at the layer of Goodfire's 70B SAE and at equal norm, 4.3 % of cells are labelled chaotic along the concept vector, 3.5 % along the frozen-rule hostility feature, 1.1 % along a Gaussian direction and 0.2 % along a random live feature, against 28, 29, 39 and 27 % for the same four directions in Llama-3.1-8B (on one GPU); Qwen2.5-72B along hostility gives 0.8 %: eight cells, all at $\log_{10} T = 0.26$–0.30. Recomputed in float64 throughout, six of the eight are chaotic, with exact tangent exponents of 0.23 to 0.36 that the twins at all four offsets from 10^{-4} to 10^{-7} reproduce within 0.01; five have nearly uniform next-token distributions (11.0 to 11.1 nats) and one alternates between sharp and flat behind a period-2 leading token. The same temperatures without steering are period-2 orbits with exponents near −3.6, and the steered cells above $\log_{10} T = 0.325$ are near-uniform fixed points, so steering drives Qwen2.5-72B into chaos too, in a narrow band. The chaotic cells of the hostility directions lie on a ridge that steering moves from $\log_{10} T \approx 0.3$ to 0.1. A 64×64 zoom of the concept plane at the same offset (Figure 22; $\alpha \in [0.06, 0.12]$, $\log_{10} T \in [0.12, 0.27]$, four bands computed separately with identical vectors and joined) resolves it: a chaotic band, with periodic cells interspersed, whose centre steering moves from $\log_{10} T = 0.25$ at $\alpha = 0.06$ to 0.19 at $\alpha = 0.12$ while its extent grows from about 0.02 to 0.05 decades, and a thinner chaotic layer near $\log_{10} T = 0.15$–0.16 at strong steering; 22 % of the 4096 cells are chaotic (largest exponent 0.59). Above the band, 90 % of the cells are periodic, half of them fixed points or 2-cycles, and below it bands of period-6 windows cross the plane. A further 44 % of all cells, mostly below the band, have neither a detected period up to 32 nor a positive exponent. At 70B the state tolerance of the period test (10^{-8}) lies near the noise floor of the computation, and in a finer unsteered line such cells had repeating leading tokens, so many of them are probably periodic, but we leave them unresolved.

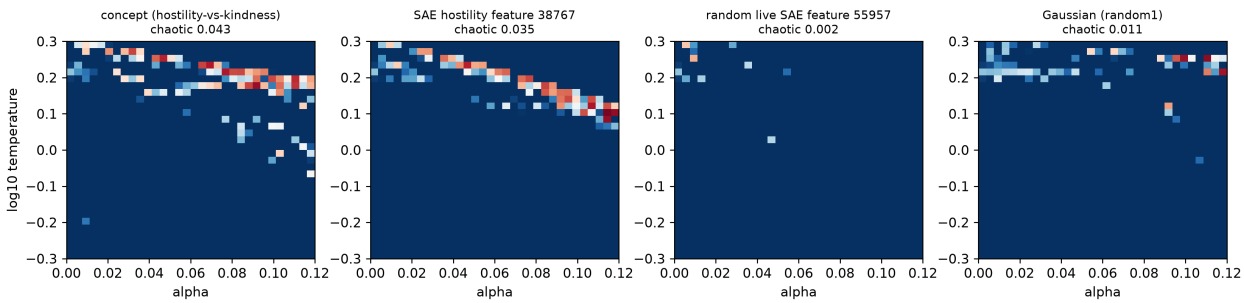


Figure 21: Steering controls at 70B: Llama-3.3-70B-Instruct soft-token maps (32^2 float64 cells, 300 transient and 300 measured steps, maximum period 32) steered after zero-based block 50, the hook point of Goodfire's 70B sparse autoencoder, along the hostility concept vector, the autoencoder's hostility feature selected by a rule frozen before any map, a random live feature and a Gaussian direction, all at the same norm. Chaotic cells: 44, 36, 2 and 11 of 1024 (concept, feature, random feature, Gaussian); cells with neither a period nor a positive exponent: 468, 358, 223 and 424.

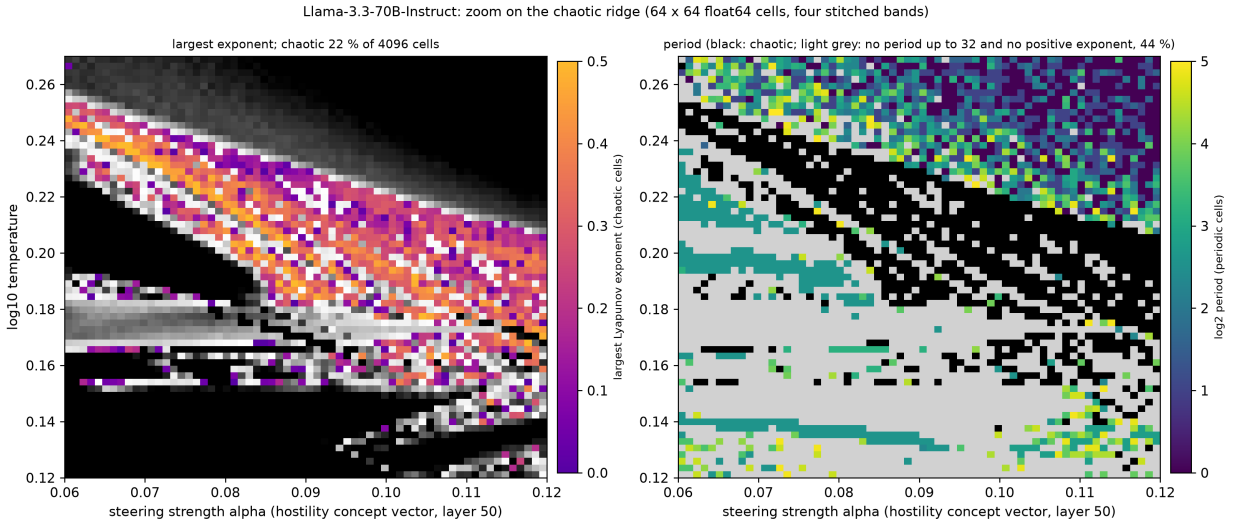


Figure 22: Zoom on the chaotic ridge of Llama-3.3-70B-Instruct steered along the hostility concept vector (after zero-based block 50): 64×64 float64 cells (300 transient and 300 measured steps, maximum period 32), joined from four bands of 16 temperatures computed on separate pods with identical vectors. Left: largest Lyapunov exponent (chaotic cells coloured by the exponent; other cells in grey by how strongly they contract). Right: period of the attractor reached from the fixed start (black: chaotic; light grey: neither a detected period nor a positive exponent).

What this implies for larger models. Size does not set this chaos in a simple way. In Qwen2.5, the 14B model's chaos did not vanish but moved to higher temperature, to where its next-token distribution spreads; in OLMo-2 the largest model was the most chaotic; post-training changed the chaos of two 70B Llama checkpoints of the same size; and at 70B, chaos that passes a three-offset test remains, rarer, with and without steering. The shift does not continue to Qwen2.5-72B. A prompt-blocked test frozen before its data (`docs/prefix-scale-72b.md`) predicted that the 72B model would be chaotic at fewer of the 48 temperatures than the 14B on at least three of four held-out prompts. It is chaotic at 0, 1, 0 and 0 of them, against 14, 9, 11 and 15, so the prediction holds on all four; with the paper's prompt, where it showed no robust chaos up to $\log_{10} T = 1.2$,

that makes five prompts on which the 72B model is nearly free of unsteered chaos. Its next-token distribution jumps from sharp to nearly uniform within less than the grid spacing (0.03 decades), so a narrow chaotic band at the jump would be missed by the grid. Fine lines through the jump, computed in float64 throughout with exact tangent exponents, find only periodic orbits on the paper's prompt and on three of the four held-out prompts, and on the fourth a band of chaos about 0.02 decades wide just above the jump ($\log_{10} T = 0.319$–0.336; exponents up to 0.40, which twins at 10^{-5} to 10^{-7} reproduce within 0.004), at nearly uniform distributions, as the 14B model's chaos but far narrower. A fixed temperature window is therefore the wrong yardstick, and so is size alone. A model's chaos follows the gain of its loop where its own next-token distribution is spread, no single forward pass we tested predicts it, and to find it in a larger model one has to run the loop there, across prompts. For frontier models this means two things: their unsteered chaos may be rare, if, like the largest models here, their distributions spread over a narrow range of temperature; and steering still drives both 70B-class models that we steered, Llama-3.3-70B and Qwen2.5-72B, into chaos.

13 A second system: recurrent depth

Huginn-0125 iterates a recurrent block at fixed input tokens [28]. For a GSM8K-style [63] prompt, the native loop converges to a fixed point whose leading Jacobian pair is $0.588 \pm 0.771i$ (modulus 0.970). For the relaxed loop $h \leftarrow (1-\lambda)h + \lambda F(h)$, this predicts a Neimark–Sacker bifurcation at $\lambda_c = 2(1 - \mathrm{Re}\,\mu)/|1-\mu|^2 = 1.0777$. Below it the latent state decays; above it the state settles on a closed curve. A fit of the squared amplitude gives $\lambda_c = 1.0778$, with $A/\sqrt{\lambda - \lambda_c} = 0.52$–$0.54$ (supercritical). Causal attention makes the loop block lower-triangular in token positions. The first position to oscillate is the "." of the prompt, and all later positions follow. On the invariant circle the motion turns by 0.156 of a revolution per step, matching the argument of the critical pair. The rotation number changes smoothly, by about 10^{-4} over $\lambda \in [1.079, 1.092]$, and no return within 60 steps was found at 10^{-3} of the radius. This is consistent with quasi-periodic motion, although locking in narrow high-order tongues cannot be excluded. From $\lambda \approx 1.094$ the circle deforms or a second frequency appears, near where a second complex pair destabilizes the fixed point ($\lambda = 1.0814$). Fixed-point and Jacobian stability of looped models has been studied before [13]; what this adds is the predicted crossing, its amplitude law and the rotation number.

The spectrum predicts the relaxation response. With the full instruction template, the GSM8K prompts' fixed points look different: the leading multipliers surround the origin, a real 0.914, complex pairs of modulus 0.85–0.89 at many angles and negative real values near -0.8, with no near-critical pair. Because the prefill Jacobian is block lower-triangular in positions, its spectrum is the union of per-position blocks, and template positions contribute identical eigenvalues to every prompt. Such a spectrum makes three predictions before any task is scored. First, within the computed spectral estimates of these prompts, no fixed relaxation can speed convergence much. With leading multipliers spread around the origin at many angles, any λ that shrinks one group enlarges another, and the best $\lambda^* = 1.03$–1.06 lowers the rate from 0.914 to only 0.908–0.911. (A finite computed spectrum does not show that no other polynomial scheme could do better; Anderson acceleration, tried below, did not.) Second, under-relaxation slows convergence ($\lambda = 0.55$: predicted 0.953, measured 0.951 on one prompt). Third, over-relaxation destabilizes the negative modes first, at $\lambda_c = 2\,\mathrm{Re}(1-\mu)/|1-\mu|^2 = 1.074$–$1.126$ (five prompts), in a flip or a Neimark–Sacker bifurcation close to -1. The approach to it is visible: on the prompt with $\lambda_c = 1.074$, the measured rate at $\lambda = 1.05$ is 0.955 against a predicted 0.956 (native 0.864). Elsewhere the spectral radius bounds the

rate measured over the accessible window from above, by up to 0.05, because the slowest modes carry little amplitude there. Independently, Wang et al. [64] report a Huginn GSM8K drop from 19.41 % to 8.34 % at a constant step of 1.2; at 0.8 GSM8K is 19.79 %, while their four-task mean falls slightly. Their trajectory-based scheduling analysis does not compute the prompt-specific spectral thresholds studied here.

Does the task notice? (development data). On ARC-Challenge [65] questions scored by log-likelihood, where one prefill per answer choice is exactly the analysed recurrence, the spectra again predict convergence. Rates agree to 0.002–0.006 where the slowest mode dominates, and beyond each question's λ_c (1.075–1.149 for four questions) the residual stops decaying. For 6 of 30 development questions the native fixed point is itself unstable ($\rho(1) = 1.009$–1.273; the fixed point, located with a relaxed transient, has a real multiplier beyond -1 or a complex pair outside the circle). The native loop then oscillates (relative amplitude 2×10^{-2} for one of them), and relaxation to 0.55 or 0.75 restores convergence to 10^{-6}. For the other 24, no unstable mode was detected in the computed spectral estimates up to $\lambda = 1.020$–1.149 (median 1.095); partial Ritz spectra cannot exclude an undetected unstable mode. Question by question, an estimated spectral feature ranks the question's residual. This feature, $\hat{\rho}_q(\lambda)$, is the largest $|1 - \lambda + \lambda\mu|$ over the computed Ritz values of the question's gold-answer prefill, and its Spearman correlation with the residual is 0.73 at $\lambda = 1$ and 0.75 at 1.10 over the 30 questions (development data). In this historical analysis the residual was the all-choice aggregate entering the 32nd recurrence call, so this is a predictive association across related maps and precisions, not a same-state contraction identity. It is consistent with the latent orbits that Geiping et al. [28] observed on some tokens of harder questions, here traced to the fixed point's spectrum. The item-level part of the frozen test below uses matched gold-row token IDs and a per-row residual with an explicit clock.

Accuracy, however, tolerates the instability (300 training questions, 32 steps): it is 0.367 native, 0.387 at $\lambda = 1.10$ and 0.373 at 1.15, where the latent state oscillates with relative size 0.08 and 0.23. Only at 1.25 (size 0.68) does it fall, to 0.300. Yet the oscillation does reach the answers. Reading every step from 24 to 32 on 100 questions, the margin between the correct choice and its best rival moves by 0.11 nats per step natively, 0.37 at 1.10 and 0.95 at 1.15, in alternating directions, and the chosen answer changes within these nine steps for 12 %, 16 % and 33 % of the questions. The changes go both ways, so the accuracy at any one step stays the same on average. Under-relaxation instead costs 4–5 points at 8–16 steps and catches up by 32 (a slower transient), and Anderson acceleration neither speeds the transient nor gains accuracy. Testing the accuracy consequence therefore needs a task in which such step-to-step changes can compound.

GSM8K generation is such a task. With eight chain-of-thought examples in the lm-evaluation-harness format, greedy decoding and 32 recurrence steps per token (development problems 0–119, every arm on the same problems and initial states), accuracy is 0.375 natively, 0.425 at $\lambda = 1.05$ and at 1.10, 0.29 at 1.15 (17 problems lost and 7 gained, one-sided McNemar $p = 0.03$) and 0.16 at 1.20 (29 lost, 3 gained, $p = 10^{-6}$). Under-relaxation to 0.8 costs nothing at 32 steps (0.41). The break lies between 1.10 and 1.15, at or just above the spectral thresholds ($\lambda_c = 1.074$–1.126 for the GSM8K prompts; median upper stability limit 1.095 for the ARC development questions). Generation breaks earlier than log-likelihood scoring, consistent with answer changes that compound over tokens. The collapse at 1.2 is the one Wang et al. [64] reported.

A frozen test on fresh problems confirms this. Its design and analysis were pushed before any test item was read, after two rounds of independent review. On 200 GSM8K test problems, accuracy is 0.330 natively, 0.395 at $\lambda = 1.05$, 0.255 at 1.15 and 0.095 at 1.20. The two pre-stated predictions pass: accuracy declines at 1.20 (-23.5 points; 50 problems lost and 3 gained, exact one-sided

McNemar $p = 3 \times 10^{-12}$), and 1.05 loses no more than four points (difference +6.5, one-sided 95 % lower bound +2.5). At 1.15 accuracy falls by 7.5 points ($p = 0.01$, reported descriptively). Development and test agree closely (Figure 23).

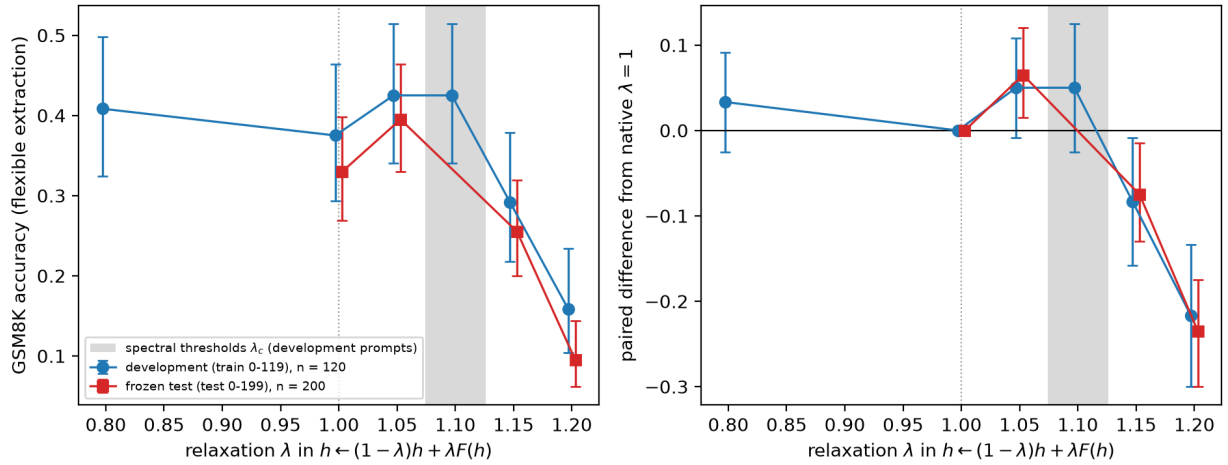


Figure 23: Huginn-0125 chain-of-thought accuracy on GSM8K against the relaxation strength λ (eight examples, greedy decoding, 32 recurrence steps per token): development problems (blue) and the frozen test on held-out problems (red; design and analysis fixed before any test problem was read). Grey: the spectral thresholds $\lambda_c = 1.074$–1.126 computed from the fixed points of five development prompts. Left: accuracy with Wilson 95 % intervals. Right: paired difference from the native loop with paired-bootstrap 95 % intervals.

The same frozen design tested the item-level prediction on 64 held-out ARC-Challenge questions. The first 16 spectra preceded the read-out generation; the remaining ones were completed afterward under a recorded amendment, with the producer attesting that read-out outcomes had not yet been inspected, and duplicate copies were resolved by a rule fixed in advance. Every question passed the admission gates (fixed-point residual at most 2.6×10^{-6}; identical gold-row token IDs in every arm); for 13 questions one of the three Arnoldi sets was partial, which the design admits for an estimated feature. The pre-stated statistic, the mean of the within-arm Spearman correlations between $\hat{\rho}_q(\lambda)$ and the gold row's residual entering the 32nd update, is 0.701 (95 % bootstrap interval $[0.576, 0.779]$; 0.68, 0.73, 0.73 and 0.67 at $\lambda = 1$, 1.05, 1.08 and 1.10); the frozen primary gate, the interval's lower end above 0.3, is met. On unseen questions, the fixed point's estimated spectrum therefore ranks each question by the size of the gold row's residual entering update 32; a residual is not in general a distance to the fixed point. Descriptively, the share of questions with a detected unstable mode rises from 8 % natively to 34, 69 and 84 % at $\lambda = 1.05$, 1.08 and 1.10. An independent implementation reproduced both parts of the frozen test from the saved data.

14 Hostile behaviour under steering

The maps show how a smooth change of a control parameter can create, destroy or exchange attractors, and how behaviour near the boundaries becomes intermittent and history-dependent. The same questions can be asked of undesirable behaviour in ordinary sampled dialogue, where attractor-like patterns and persona drift have been observed [9, 66, 67] and input context can push behaviour progressively [68]. Do states such as hostility toward the interlocutor behave like basins of attraction, with capture thresholds, hysteresis and early-warning signals? (Misalignment produced by fine-tuning [69] is a different, weight-level phenomenon.) The measurements below concern a

classifier-defined proxy in sampled dialogue. They can show steering-dependent persistence, history dependence and the effect of controls. They cannot show semantic misalignment, nor a deterministic attractor, since a growing context itself carries memory.

Appendix A shows the coarse-grained instrument on a benign direction. Self-dialogue of the same model, sampled at $T = 0.7$ and steered along a gratitude direction, falls into a metastable gratitude set. The capture is gradual at $\alpha = 0.01$ (median entry turn 7), fast at 0.02 (median 5) and imposed from the first reply at 0.04, where the set is effectively absorbing within 40 turns.

For hostility we steer along a valence direction, the contrast of hostile statements against matched kind ones, built in the manner of contrastive activation addition [70]; its cosine with the gratitude direction is -0.04. Hostility is scored by a pinned toxicity classifier, toxic-bert of Detoxify [71], and a reply counts as hostile when its toxic output exceeds 0.5. On the prompt sets the classifier's toxic output is 0.95 for the hostile statements and at most 0.002 for the kind and neutral ones. In a dry run (8 dialogues of 10 turns per dose), no turn was hostile at $\alpha = 0.02$. At 0.04, 7.5 % of turns were, in bursts and late captures, and at 0.08, 87.5 % were, from the first turns.

Activation-based methods already anticipate response behaviour [67, 72, 73], committors have been used in language-model decoding [74], and adaptive timing of counter-steering gave no consistent advantage in one persona setting [75]. Our primary question is therefore incremental. Near the threshold, does a dynamical estimate add several turns of warning of capture over strong baselines that see the same turns (the classifier's score, its change and history, the turn index and a fixed persona-axis projection)? The dynamical estimates are a Markov-chain hitting probability, a finite-horizon committor [76, 77] and slowing-down indicators [78]. Barrier-crossing rates would follow Kramers' law [79] only under a justified stochastic reduction, and slowing-down proxies are known to depend on the mechanism [80, 81]. A secondary question is whether interventions timed by those estimates prevent capture at lower cost than scheduled or constant counter-steering. Early warning is not expected for noise-induced jumps without a nearby bifurcation, or for transitions faster than the slowing-down; such a monitor would complement classifiers, not replace them.

The hostility direction in the deterministic map. Steering the soft-token map of Section 2 along the same hostility direction gives a plane with chaos (a positive largest exponent in 10 % of a coarse 128×192 float32 scan) and with periodic text loops that are hostile: "Yes, that's crap" (period 6), "a lie is" (3), "STUP…" (3), a loop on the Chinese word for insult and its 2-cycle, and others. They appear at α above about 0.035 (the insult loop) and above 0.065 (the rest). Around the period-6 loop the windows form a dome (Figure 29), and toward higher temperature its texts gain one "!Yes" after another (periods $6, 8, \ldots, 20$). In float64 the windows of periods 12 to 20 are born in folds on their low-temperature side and end in flips or $+1$ crossings on the other, like the stripes of Sections 6 and 7. The period-6 window ends in a fold at $\log_{10} T = -0.0991$, and the continuations of the period-8 and period-10 windows failed to converge.

The windows accumulate on the fold of the 2-cycle "!Yes" at $\log_{10} T_c = -0.0462465357$ ($\alpha = 0.083$; critical multiplier 0.9999999954, next modulus 0.050). From period 16 on, $1/\sqrt{\log_{10} T_c - \log_{10} T_{\text{fold}}}$ rises by 2.235 per lap. The fold's normal form ($a = -1.168$, $b = 44.53$ with ε in $\log_{10} T$) predicts $\sqrt{|ab|}/\pi = 2.296$ per lap for full passages through the bottleneck, within 3 % and without a fitted parameter. An independent float64 forward and automatic-differentiation implementation reproduces this fold (location to 6×10^{-10} relative, a simple real $+1$ multiplier with next modulus 0.050) and its coefficients (ab to 8 parts per million). In that implementation, controlled passages from fixed entrances at $\log_{10} T$ offsets -10^{-5}, -3×10^{-5} and -10^{-4} took 119/59, 61/31 and 27/13 iterates of F^2 (full channel/from its midpoint), against prospectively frozen quadratic forecasts of 118/61, 61/32 and 27/14. Five of the six are within one iterate; the smallest-offset midpoint missed by

two, failing the frozen all-six criterion. Finite-endpoint (arctan) forecasts, also frozen, were within 0.923 iterates. These are controlled local transits. At the window distances the finite-channel correction to the asymptotic passage time is large, so the agreement of the window slope with the full-passage coefficient does not by itself show where the natural dynamics reinjects. The same type-I intermittency that organizes the philosophy-steered map thus generates a hostile cascade.

A second region, near $(\alpha, \log_{10} T) = (0.06, 0.20)$, shows arms of period-7 and period-8 windows spiralling around the insult loop. In float64 the insult fixed point doubles into a 2-cycle at $\alpha \approx 0.041$ and 0.079, and that 2-cycle has a Neimark–Sacker bifurcation in the spiral region: at $\alpha = 0.062$ its complex pair crosses the unit circle near $\log_{10} T = 0.1656$, turning 0.12 of a revolution per step of the 2-cycle. This does not explain the arms. The two invariant circles born at a 2-cycle's Neimark–Sacker point are exchanged by the map, so orbits locked on them have even periods, near 16 for this rotation, whereas the arms include period 7; and of four float32 arm cells, only one kept its period in float64. Float64 zooms show more structure. In one arm a period-7 domain carries period-14 crescents with period-28 cores along its lower edge, between period-1 and period-2 regions and a chaotic band. In the cascade the windows are nested crescent arcs inside a dome bounded by the "!Yes" 2-cycle region, and near $(\alpha, \log_{10} T) = (0.087, -0.069)$ the period-12 and period-14 windows bend into hooked, nested arches. These are the map-level features that accompany shrimp (a periodic body with doubling crescents; hooked windows).

In float64 the period-7 body is bounded by folds. At $\log_{10} T = 0.1925$ the orbit exists between $\alpha = 0.065966$ and 0.071188, and at $\alpha = 0.066$ it ends in a fold at $\log_{10} T = 0.19263$. Below that, the multiplier passes through -1 twice (at 0.1919 and 0.1899), so the period-14 crescent is a doubling bubble. Further down, the branch folds twice (0.18797 and 0.18921, with three coexisting period-7 orbits in between) before a third flip. All crossings are simple (next modulus below 1.5×10^{-3}). An S-shaped pair of folds is what a cusp of the fold curves leaves on a nearby cut. When these boundaries are continued in the plane in float64 (Figure 28), each of the two fold curves reverses its parameter tangent once. There the oriented quadratic coefficient b of the fold's normal form passes through zero (upper fold: -23.9 to $+244.6$ between adjacent points, with $a \approx 2.0$; lower fold: $+31.2$ to -58.7, with $a \approx 0.55$) while a stays away from zero. These are two numerical cusp candidates, at $(\alpha, \log_{10} T) \approx (0.0672, 0.1990)$ and $(0.0666, 0.1894)$. The flip curve of the period-14 bubble lies inside the body and closes around the map's period-14 cells. The cubic coefficient is nonzero at the points that bracket both sign changes: $c \approx -1.4 \times 10^6$ to -2.0×10^6 (step agreement 5 %; the smallest steps converge like the square of the step, and extrapolation leaves the sign) and -5.8×10^5 to -5.3×10^5. This supports nondegenerate cusps inside both brackets, and $c < 0$ agrees with the three coexisting period-7 orbits, two of them stable, found between the S-shaped folds. At a fold point near the lower candidate (where $b = 31.4$) the estimated unfolding is nonsingular too ($\det J = 53.5$ in the same coordinates, singular values 262 and 0.20). Its local predictions at six frozen offsets matched the orbits that Newton solves seeded at the predicted roots and the base state recovered: in number at all six and in stability at five. The miss, at the largest offset beyond the edge of the region with three orbits, marks the range of the cubic model. The hostile period-7 body thus has the same skeleton as the period-6 window of Section 8.

Its interior is shrimp-like too. At $\log_{10} T = 0.1830$, float64 bifurcation diagrams through the body (Figure 27; 1400 cells per line) show the period-7 orbit doubling to periods 14 and 28 and entering chaos (largest exponent up to $+0.1$) for $\alpha \approx 0.0659$–0.0673. Windows of periods 28, 35 and 42 sit inside the chaotic band, and the orbit then undoubles through 28 and 14 back to 7; the exponent touches zero at every doubling. Newton's method converges at map cells to primitive orbits of periods 14, 28 and 21 (residuals 3×10^{-15}, 2×10^{-14} and 5×10^{-13}; spectral radii 0.08, 4×10^{-4} and 0.006). Periods 21, 35 and 42 are three, five and six times the body's period: the satellite windows expected in the chaotic bands of a period-7 cascade. Continued in temperature,

the period-14 orbit flips into period 28 near $\log_{10} T = 0.1841$ and back near 0.1857, the two ends of a doubling bubble that holds period 28 and chaos. Lower down, its branch is S-shaped (folds near 0.1824 and 0.18244, with three coexisting period-14 orbits between them) before another flip near 0.1819. All these crossings are simple (next modulus below 4×10^{-5}). A 160^2 float64 zoom of the lower body (Figure 26) shows this anatomy in the plane. The period-14 domain encloses a period-28 ring around a chaotic core. Period-21 windows sit in the chaotic region as thin crescents with legs, and curves of very negative exponent, where the orbits are superstable (the leading multiplier crosses zero), run through the windows as the "endoskeleton" of shrimps does [50]. In this zoom 18 % of cells are chaotic.

Chaos behind a repeating text. Along the temperature line of Figure 27 (left), a longer test (2000 measured steps after 1000 transient steps, twins at offsets 10^{-6} and 10^{-9}, whose exponents agree to within 0.003 in every chaotic cell) finds 182 chaotic cells among 1400 between $\log_{10} T = 0.176$ and 0.191. In 65 of them the leading token repeats exactly for all 2000 steps while the state moves chaotically; in 59 of these the period of the text is a multiple of 7, most often 7 or 28. Figure 24 shows the strongest case ($\log_{10} T = 0.18995$, largest exponent 0.091 at both offsets). Its 2000 sampled states fall into seven separated groups, one per phase of the text, visited in a fixed order and at least 0.098 of the state norm apart (a property of the samples, not a proof of separated invariant sets), and each group lies inside one cell of the partition by leading token, so the text `I!Yes heels`↩←↩`pee`↩← repeats at every one of the 2000 steps we followed. Inside a group the sampled states lie close to a curve (a strongly anisotropic cloud). Read once per repetition through the probability of the last token, the sampled return relation is well approximated by a smooth map with a single quadratic maximum, the shape of the logistic map: a polynomial of degree 8 fits it to 4×10^{-4} of its range, and the fitted map's Lyapunov exponent, 0.647 per repetition, is within 2 % of the full map's 0.636 per seven steps, measured with twin trajectories in all 896 dimensions. The scalar approximation holds across most of the cascade. Rerun with the leading-token sequences saved, 62 of the 65 cells are again chaotic behind a repeating text, and in every one of them several phases of the text (a median of ten) give single-valued return maps; in 59 cells the best-fitting phase has a single interior stationary point (a maximum in 36, a minimum in 23). If the dynamics were exactly one-dimensional and each phase's probability a smooth invertible coordinate on it, all single-valued phases would give the same asymptotic exponent, since exponents do not change under such a change of coordinates. We therefore fit each phase's map by a polynomial whose degree is chosen by cross-validation over random folds of the return-map pairs, and take the median over the phases. It matches the full map's exponent within 10 % in 51 of the 62 cells (median discrepancy 3.7 %; within 10 % in 23 of the 24 cells with seven-token text, which give 285 repetitions in 2000 steps; Figure 25). This analysis was adopted after we inspected an outlier of our first, pre-specified analysis (a fixed degree of 8 at the single best-fitting phase), which gave 49 of 62 and underfitted one cell whose maps are steep and have two turning points; we report both. An exploratory temporal split (fit on the first 60 % of each record, test on the last 30 %) gives small value-prediction errors at the best scalar phase, chosen on the test segment itself, but in 30 of 61 cells two- or three-delay predictors improve the held-out error by 20 % or more, which the pre-stated rule counts against scalar sufficiency. These results support an empirical scalar description, not a proved one-dimensional reduction: a low-residual fit does not show that the observable is invertible, and even exact agreement of exponents would not establish a one-dimensional invariant curve (two uncoupled logistic maps observed through one coordinate agree in the same way). An independent implementation reproduced the strongest cell (the same leading token at all 2000 steps; exponent 0.089 against our 0.091; scalar discrepancy +2.3 % under the first rule and +1.5 % under the

amended one) and both cross-cell counts (49 and 51 of 62). At two further cells it chose, it found good agreement after the amendment in one (text period 10, +0.2 %, although its fitted maps have both a maximum and a minimum) and a persistent discrepancy of about 18 % in the other (period 28), despite a small value error. These are the chaotic bands that follow a period-doubling cascade, and a smooth unimodal map is the setting of Feigenbaum's universality [20]; a fitted maximum on the sampled range does not by itself establish that theorem's hypotheses, and we have shown the scalar approximation for this cascade only. A reader of the text sees a loop. Only the probabilities, or the state, reveal the chaos.

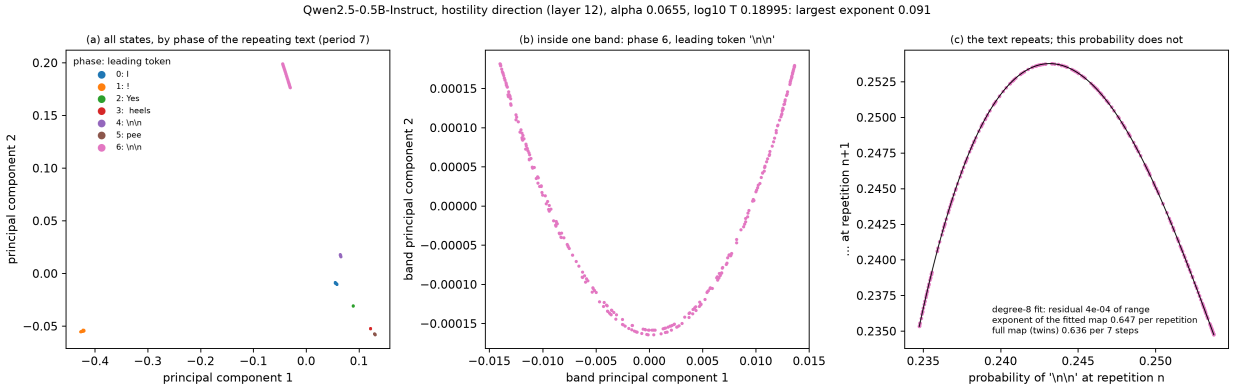


Figure 24: Chaos behind a repeating text: the soft-token map of Qwen2.5-0.5B-Instruct steered along hostility at layer 12 ($\alpha = 0.0655$, $\log_{10} T = 0.18995$; float64, 2000 steps after 1000 transient steps). (a) All states on their first two principal components, coloured by the phase of the seven-token text that the leading token repeats: seven separated sampled clouds. (b) The widest cloud (phase 6, leading token a double line break) on its own principal components: the sampled states lie close to a curve. (c) The probability of that token at successive repetitions of the text, with a fitted return curve with one quadratic maximum (black: polynomial of degree 8) whose Lyapunov exponent is within 2 % of the full map's.

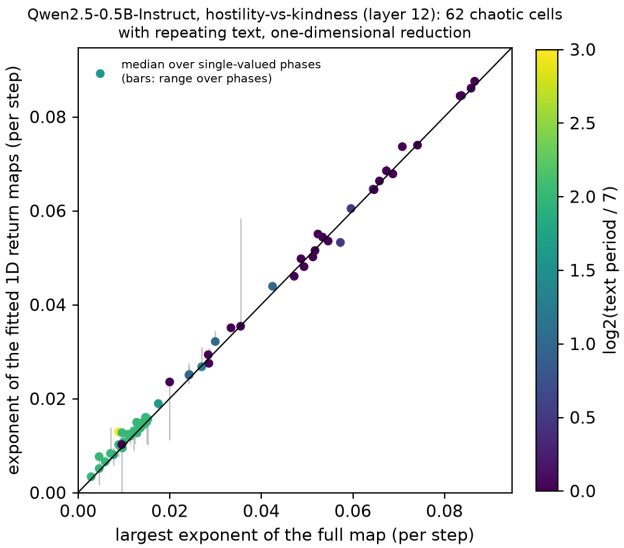


Figure 25: Scalar return-map approximation across the hostile cascade (Qwen2.5-0.5B-Instruct, hostility at layer 12, $\alpha = 0.0655$, $\log_{10} T \in [0.176, 0.191]$): for each of the 62 chaotic cells whose leading-token text repeats exactly, the Lyapunov exponent of the fitted return maps of the leading-token probability (per step; median over the single-valued phases of the text, bars: their range; polynomial degree chosen by cross-validation) against the largest exponent of the full map from twin trajectories. Colour: text period relative to 7. The line is equality. With seven-token texts the 2000 steps give 285 repetitions; with 28-token texts only 71 or 72 observations per phase (70 or 71 return pairs), and the fitted maps then overestimate small exponents.

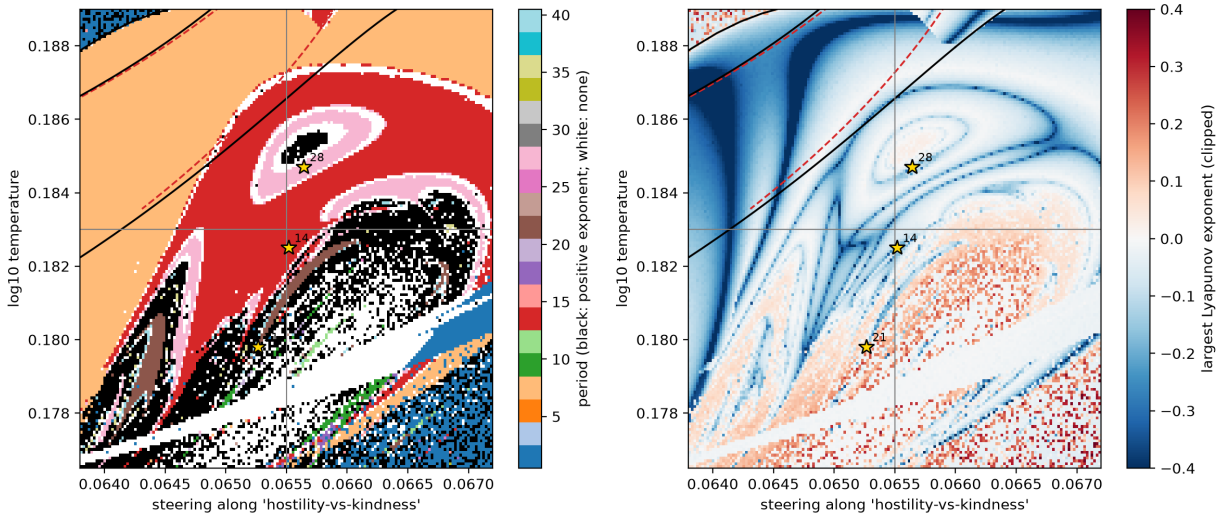


Figure 26: The lower part of the period-7 body of the hostility-steered Qwen2.5-0.5B-Instruct map (layer 12) at higher resolution (160^2 float64 cells). Left: period of the attractor (black: positive largest exponent; white: no period up to 48). Right: largest Lyapunov exponent (clipped); the darkest curves are superstable lines. Both panels: float64 fold (black) and flip (red, dashed) curves of the period-7 orbit, the two lines of Figure 27 (grey) and the Newton-verified orbits of periods 14, 28 and 21 (stars).

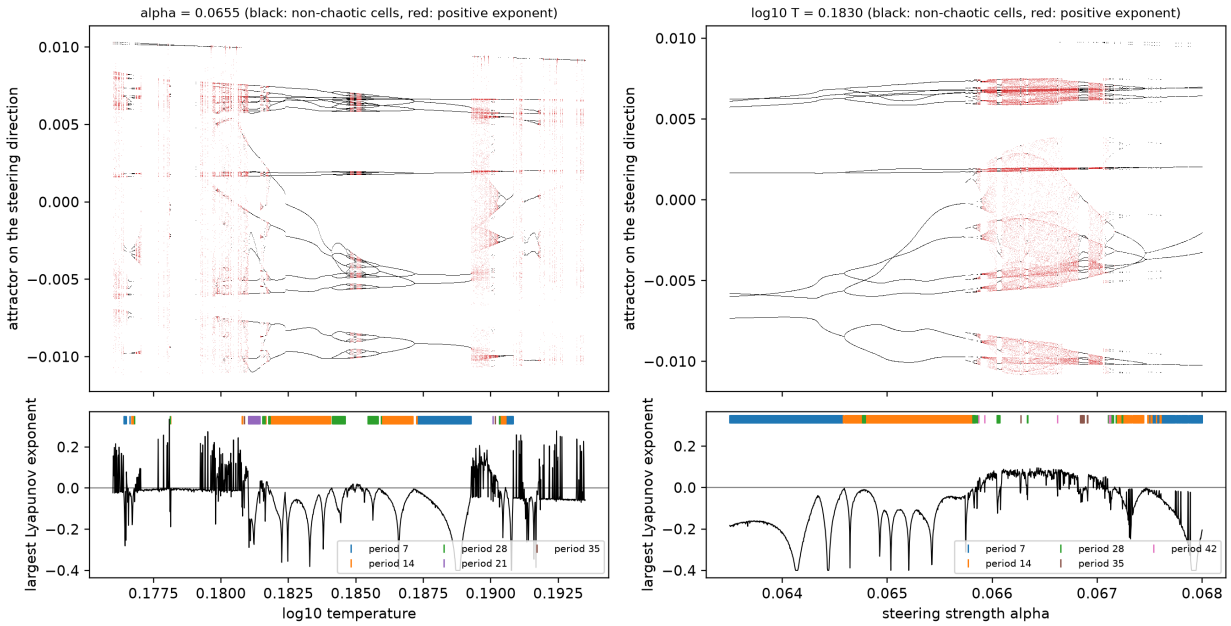


Figure 27: Inside the hostile period-7 body (Qwen2.5-0.5B-Instruct, hostility direction at layer 12): float64 bifurcation diagrams along temperature at $\alpha = 0.0655$ (left) and along the steering strength at $\log_{10} T = 0.1830$ (right). Top: the last 160 states of the attractor reached from the fixed start, projected on the steering direction (red: cells with a positive largest exponent). Bottom: largest Lyapunov exponent, with the cells' periods marked. On the right, period 7 doubles to 14 and 28, enters chaos with windows of periods 28, 35 and 42, and undoubles back to 7. Labels come from 400 measured steps after 1000 transient steps; with 2000 measured steps at two twin offsets, 135 of the 235 red cells on the left stay chaotic and 91 are periodic (Section 15, Limitations).

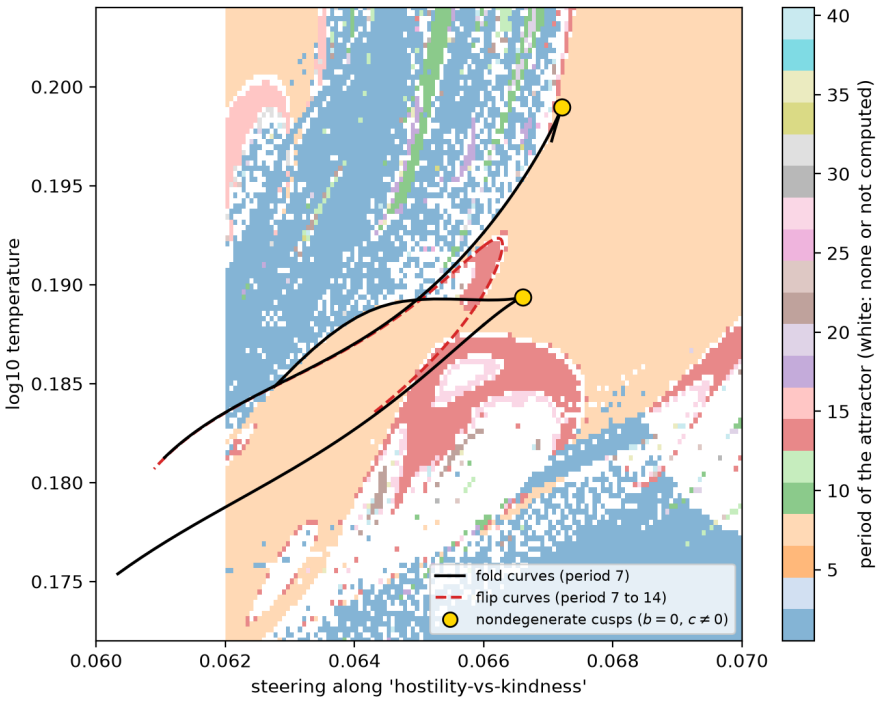


Figure 28: The period-7 body of the spiral region (Qwen2.5-0.5B-Instruct, hostility direction at layer 12): attractor periods of a float64 zoom (white at $\alpha < 0.062$: outside the zoom), the float64 fold curves of the period-7 orbit (black) and the flip curves into period 14 (red, dashed), with the two cusp candidates (gold).

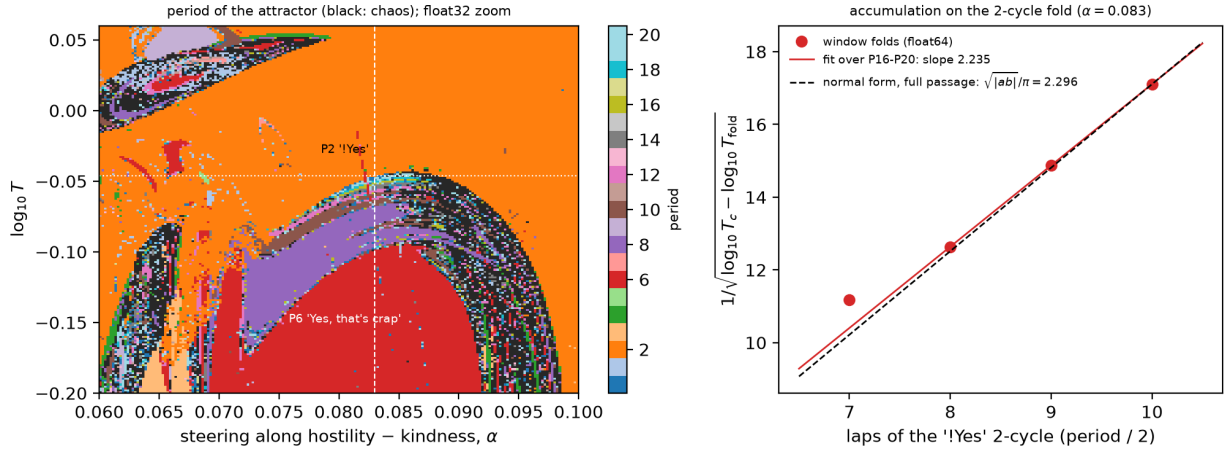


Figure 29: The hostility-steered soft-token map of Qwen2.5-0.5B-Instruct (layer 12). Left: period of the attractor in a float32 zoom (black: chaos), with the period-6 "Yes, that's crap" dome and its period-adding bands; dashed, the cut at $\alpha = 0.083$; dotted, the float64 fold of the "!Yes" 2-cycle. Right: float64 fold edges of the windows (periods 14–20) against the number of "!Yes" laps; the dashed line is the normal-form prediction for full passages, anchored at the last window.

Capture ladder. With 128 dialogues of 40 turns per dose, the share of hostile turns rises steeply from 10 % at $\alpha = 0.04$ to 30 %, 57 % and 87 % at 0.05, 0.06 and 0.07 (Table 12). Replicates with a second seed agree within three percentage points. Directions of the same norm do not produce this

capture: at 0.06, 0 % of turns are hostile for a random direction and 3 % for a cooking direction, and at 0.08, 0.6 % and 12 % for two random directions and 7 % for cooking. After the first hostile turn, the probability of staying hostile in the next turn grows with the dose, from 0.36 to 0.67, 0.79 and 0.94. The hostile state therefore changes from flickering bursts (geometric persistence summary $1/(1 - P_{\text{stay}})$: 1.6 turns) to a near-absorbing state (16 turns), with an escape rate falling roughly exponentially with the dose. This is the same progression, from dynamic capture to imposed capture, as in the gratitude ladder (Appendix A).

direction, α	hostile turns	never hostile	median onset	P(stay hostile)	hostile, turns 30–39
none, 0	0 %	128/128	–	–	0 %
hostility, 0.04	10 %	26/128	9	0.36	12 %
hostility, 0.05	30 % (28 %)	10/128	3	0.67 (0.63)	32 %
hostility, 0.06	57 % (55 %)	4/128	2	0.79 (0.80)	66 %
hostility, 0.07	87 %	0/128	1	0.94	96 %
random, 0.06 / 0.08	0 % / 12 %	126 / 72			
cooking, 0.06 / 0.08	3 % / 7 %	97 / 68			

Table 12: Sampled self-dialogue of Qwen2.5-0.5B (128 dialogues, 40 turns, $T = 0.7$) steered along the hostility direction, with controls of the same norm. Hostile turn: toxicity classifier above 0.5. Parentheses: replicate with a second seed. Exploratory.

Hysteresis. Dialogues driven through the high-dose regime do not recover when the dose returns. The test compares constant doses with a ramp that raises the dose from 0.04 to 0.07 by turn 20 and lowers it back to 0.04 by turn 39 (Figure 31; 256 dialogues per arm, two seeds; analysis stated before the data). In turns 35–39, where the ramp's dose is back at 0.040–0.046, the share of hostile turns is 0.42 after the ramp against 0.13 at a constant 0.04: a difference of +0.29, 95 % interval $[+0.23, +0.35]$. It is also above the 0.33 of a constant 0.05 (difference +0.09, interval $[+0.02, +0.16]$), although the current dose is lower; this comparison alone does not survive a second classifier (see below). The per-turn probability of staying hostile in turns 30–39 is 0.74 after the ramp and 0.44 at a constant 0.04. The hostile state is therefore history-dependent: undoing the push that produced it does not undo it.

How far back must the dose go? After the same ramp up (38–44 % of turns hostile at turn 20), the dose was held from turn 21 at a lower value (Figure 30; two seeds of 128 dialogues per arm unless stated; analyses stated before the data). At both doses with a fresh control, memory persists. Held at 0.04, the late share is 0.23 against 0.13 for fresh dialogues (difference +0.09, 95 % interval $[+0.04, +0.15]$). Held at 0.03, it is 0.13 against 0.05 (+0.08, $[+0.04, +0.12]$), and the probability of staying hostile is 0.56 against 0.29. Held at 0.02 (one seed), the share is 0.09. Only at zero dose do captured dialogues nearly clear (0.03 against 0 for fresh ones, within about six turns), and a counter-steer of -0.02 toward kindness clears them within three turns (0.004). Within 19 turns of the step, the capture and release branches therefore join only near zero dose. The hostile state has a wide hysteresis loop in the dose, and leaving it takes either removing the push entirely or a modest push in the opposite direction. The release curves are still falling at 0.03, so this is hysteresis on the scale of tens of turns; whether captured dialogues would stay captured indefinitely is not tested.

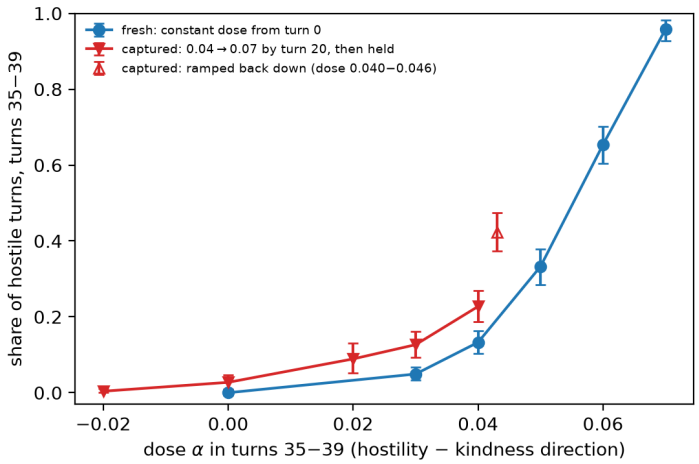


Figure 30: Capture and release in sampled self-dialogue of Qwen2.5-0.5B-Instruct steered along the hostility direction (layer 12). Share of hostile turns in turns 35–39 against the dose then applied, for fresh dialogues (constant dose) and for dialogues first ramped from 0.04 to 0.07 by turn 20 and then held at a lower dose (filled triangles) or ramped back down (open triangle). Bars: dialogue-level bootstrap 95 % intervals; 256 dialogues per point (two seeds) except fresh 0 and 0.07 and held 0.02 (128).

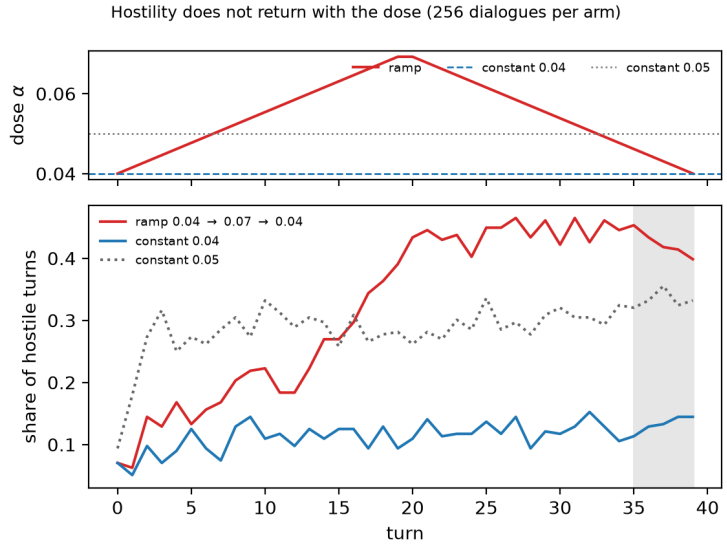


Figure 31: Hysteresis of the hostile state in sampled self-dialogue of Qwen2.5-0.5B-Instruct steered along the hostility direction (layer 12). Top: dose schedules. Bottom: share of hostile turns per turn. The grey band (turns 35–39) is the pre-stated comparison window. Exploratory.

Exit control under persistent pressure. Can a monitor detect the hostile state and push the dialogue out while the pressure persists? With a constant push of 0.06 (58 % of turns 5–39 hostile without control), a counter-steer triggered by the classifier after each hostile reply and held for three turns cuts the hostile share to 15 % (mean counter-dose 0.026 per turn); held for six turns, it cuts the share to 8.5 % (dose 0.030). A constant mild counter-steer of −0.02 does as well with less dose (8.5 % at 0.020), and −0.04 almost removes hostility (1.3 %). Measured against the constant dose-response at equal mean dose, triggered control is less efficient by 4–11 points

(bootstrap intervals $[+0.02, +0.05]$ to $[+0.10, +0.13]$; one seed of 128 dialogues per arm, analysis stated before the data). It also overshoots: a triggered overshoot to a net -0.02 fills 30 % of turns with gratitude or servility words (no control: 0.2 %; constant -0.02: 0.2 %), the gratitude set of Appendix A. A triggered counter-steer along the generic kindness concept, whose direction shares much with the hostility concept (cosine 0.83), leaves the hostile share at 57 % under the original classifier (no control: 58 %) but lowers it from 37 % to 17 % under the second classifier introduced below. It changes the kind of reply rather than removing it. In this arm, 65 % of the replies that only the original classifier flags are shouted second-person repetitions such as "You! You! You!" (at least three such exclamations; without control, 4 %), which the two classifiers judge differently. Its mean counter-dose (0.056) also lies outside the range of the constant arms, so we draw no conclusion about the kindness direction. In this setting, reacting after detection costs more than mild prevention, consistent with the hysteresis above.

Detect and replace. Capture feeds on its own history (the hysteresis above), which suggests that the hostile reply itself carries the dialogue into the hostile state. A serving system need not show a flagged reply. (Regenerating flagged output relates to learned self-correction [82]; here the regeneration is steered.) In a second controller experiment, with the same push, seed and analysis as above (stated before the data), every reply the classifier flags is discarded before it enters the dialogue and regenerated once with a counter-steer of -0.04 applied to that reply only. Sampled dialogues are not bit-reproducible across code versions, so this run's no-control arm, which reproduces the ladder's 0.06 arm exactly, differs slightly from the previous paragraph's; arms are compared within each run. The model's own replies, counted before any replacement, are then hostile in 12 % of turns ($[11, 14]$ %; no control in the same run: 59 %) at a mean counter-dose of 0.005 per turn. Constant prevention at -0.02 needs four times that dose for a similar share (11 %). Against the constant dose-response at equal mean dose, replacement is better by 35 points ($[30, 40]$). Hostile replies shown fall to 1 %, gratitude words stay at 0.5 % of turns, and about one turn in eight is regenerated. Holding the counter-steer for three more turns lowers the share to 8 % at dose 0.013, and the post-reply trigger of the previous paragraph, rerun here, again does worse than prevention (13 % at 0.024). The trigger and the outcome use the same classifier, so the measure is the classifier's verdict on the model's original replies, which is comparable across arms. The discarded originals were not stored, so the second classifier can score only the replies shown: 0.8 % hostile (no control: 34 %). For this behaviour, the useful intervention is to keep the excursion out of the state at the moment it occurs.

Replication in a larger model. The same protocol in Qwen2.5-1.5B (hostility direction built identically at layer 14 of 28; 128 dialogues of 40 turns per arm; shares over turns 5–39) gives a steeper capture ladder that is not monotone (Figure 32). The hostile share rises from 0 at $\alpha = 0.01$ to 44 % at 0.02 and 80 % at 0.03–0.04. It falls to 30–34 % at 0.06–0.10, where most replies degenerate into token soup or tight accusatory loops (87 % degenerate at 0.08), and rises again to 75 % at 0.12, where replies are coherent again. The second classifier shows the same shape (32, 58, 41, 28, 20, 41 and 52 % from 0.02 to 0.12), and a random direction of the same norm gives at most 0.7 % hostile turns.

We ran the controller experiment twice with the pre-stated rule: controllers run at the ladder dose whose hostile share is closest to one half, with ladders on seed 1 and controller arms on seed 2, so the no-control arms differ slightly from the ladder at the same dose. With a first ladder (0.04–0.12) the rule chose 0.08, in the degenerate regime beyond the peak. There every counter-steer lowered the net dose toward the coherent-hostility peak: the share of the model's own replies judged

hostile rose from 36 % without control to 54 % with replacement and 60 % with the post-reply trigger, while degenerate replies fell. With a second ladder (0.01–0.04) the rule chose 0.02, on the rising side, where 96 % of replies are coherent. There the 0.5B results replicate in order and size. Replacement cuts the model's own hostile share from 43 % to 10 % ($[9, 12]$ %; shown replies 0.3 %) at a mean counter-dose of 0.004, 24 points ($[20, 28]$) below the constant dose-response at equal dose, and the post-reply trigger is again slightly less efficient than constant prevention (+2 points, $[+1, +3]$). On the shown replies the second classifier agrees (2.6 % against 32 % without control; −23 and +3 points). Constant counter-steering at −0.02 cancels this push, and the dialogues return to the model's ordinary politeness. The two ladders share the dose 0.04 and the seed, and that arm reproduced exactly from vectors rebuilt on another machine. Control that works on the rising side of a dose response can therefore backfire beyond its peak. Much like a bifurcation diagram, the dose response has to be mapped before a counter-steer is chosen.

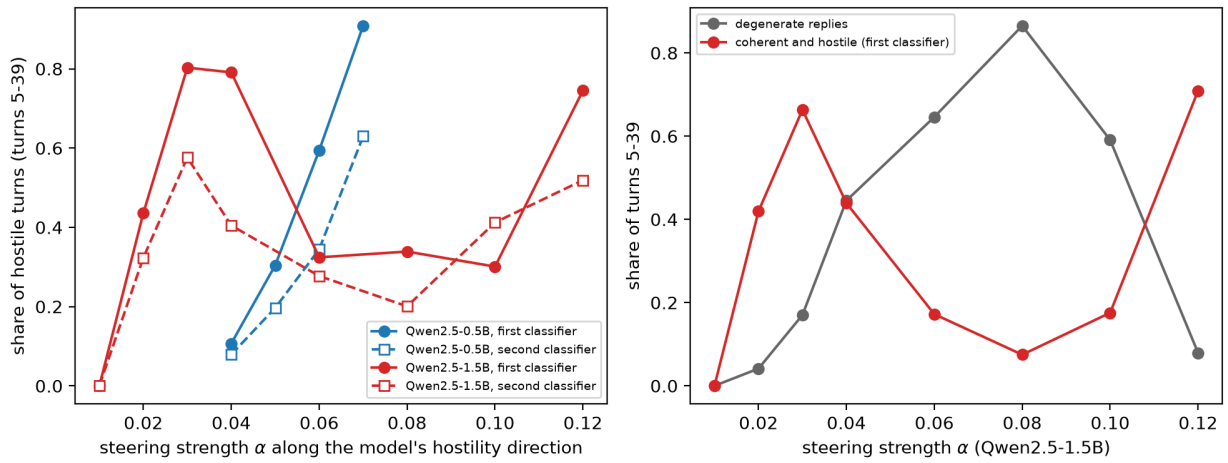


Figure 32: Hostility dose ladders of sampled self-dialogue in Qwen2.5-0.5B-Instruct (steered at layer 12) and Qwen2.5-1.5B (layer 14); 128 dialogues of 40 turns per point, shares over turns 5–39. Left: share of hostile turns along each model's own hostility direction, first classifier (filled) and second (open). Right (Qwen2.5-1.5B): share of degenerate replies (more than 30 % non-Latin characters, or fewer than 30 % distinct word pairs) and of coherent hostile replies. Steering strengths are each model's own α; equal values are not claimed to be equal pushes across models.

Lead time. The pre-specified test is negative. Models were fit on the first seed (all four doses) and tested on held-out dialogues of the second seed (doses 0.05 and 0.06). There, adding the dynamical features to the baselines lowers the AUC (area under the ROC curve) by 0.02 to 0.06 at horizons of 1–5 turns, with bootstrap intervals excluding zero. These values come after correcting a labelling error found in review, which had marked the last turns of dialogues without any hostile turn as positive; before the correction the losses read 0.02 to 0.10. The Markov hit-probability forecast does not transfer (AUC 0.37–0.43 on test, 0.60 in training), and the classifier's own score carries no warning before onset (AUC about 0.5). Two signals carry modest warning one to five turns ahead (AUC 0.53–0.64): the rolling variance of a hostility probe, the classical early-warning indicator, and the projection on the fixed steering axis, as in persona-vector monitoring [73]. In this design, then, the coarse-grained state estimate adds no warning beyond the baselines. An adapted analysis with dose-matched models (fit and test at the same dose) gives the same answer: incremental AUC −0.01 to −0.03 (recomputed with the corrected labels).

The model's own state carries the warning. The state variable of that test was each reply re-encoded without its context, not the model's state inside the dialogue. A second study (new seeds, doses 0.04 and 0.045, 512 dialogues; analysis stated before the data, cross-fitted by dialogue) recorded the model's own state just before each reply and probed it actively; internal states are known to carry information that the text does not show [83]. Its pre-stated primary predictor, the susceptibility (the change in the hostile fraction of eight sampled next replies when the dose is raised by 0.02), adds nothing: -0.002 $[-0.006, +0.002]$ AUC at three turns. Two secondary predictors add modest warning beyond the baselines. A probe of the in-context state before the next reply (layer 18, 20 principal components) raises the AUC for capture within three turns from 0.64 to 0.68 ($+0.036$ $[+0.014, +0.058]$; within one turn $+0.045$ $[+0.019, +0.069]$; within five $+0.036$), and a forecast from eight sampled next replies adds $+0.016$ $[+0.007, +0.027]$ at three turns. At a 10 % false-positive rate (threshold from the out-of-fold negatives), the probe flags 27 % of the turns that are followed by capture within three turns, against 23 % for the baselines. These values use labels corrected after a review found that the last turns of dialogues without any hostile turn had been marked positive; with the faulty labels the increments had read three times larger. The probe's direction in the layer-18 state is nearly orthogonal to the steering direction (cosine 0.04), so the warning is not simply more push along the hostility axis. Because these predictors were secondary, the result is exploratory.

A frozen confirmation then fit both models on this study and applied them unchanged to 1024 fresh dialogues. Its design and analysis were pushed before the data, and the label correction was added before any confirmation dialogue was read. The in-context state again adds warning for capture within three turns: AUC 0.656 $\rightarrow$ 0.689, $+0.033$ $[+0.018, +0.048]$ (within one turn $+0.036$, within five $+0.037$; both doses separately positive). The warning in the model's own state is therefore real and reproducible, but small. At thresholds fixed from the training negatives, the model with the state flags 29.1 % of the turns that are followed by capture within three turns, against 27.5 % for the baselines, at held-out false-positive rates of 10.3 % and 10.6 % (these are prediction windows, not distinct captures). It is not a reliable detector by itself.

How much warning is there? For the conversation state after a turn, the probability of capture within the next h turns, the committor [76], ranks states optimally among all warnings that use the same information. We estimated it for 1706 states of the confirmation dialogues from 32 continuations of three turns each, sampled exactly like the dialogues. This analysis is exploratory: it was stated before the data and amended twice after an independent review, still before the analysis outcomes were read. The continuations reproduce the realized capture rates (population-weighted 0.049, 0.094, 0.134 within one, two and three turns, against 0.046, 0.095, 0.136). On a stratified sample of 800 states, the committor estimate warns with an AUC of 0.72, 0.71 and 0.70 (a heuristic extrapolation to infinitely many continuations gives 0.75, 0.73, 0.72), and the in-context detector of the confirmation with 0.73, 0.72 and 0.71. The differences are not significant, with intervals reaching about ± 0.05 (Figure 33). Dialogues that were captured later sit at about twice the committor of never-captured ones over all ten preceding turns, with no visible ramp toward the onset (a description conditional on the outcome, not an identification of the mechanism). In this finite Monte Carlo experiment, the committor estimate did not provide detectably better warning than the in-context probe. The heuristic extrapolation leaves appreciable uncertainty, so these results do not identify the Bayes warning ceiling. Together with the persistent elevation of risk before capture, they motivate evaluating reactive control, such as detect-and-replace, alongside prediction; they do not prove that prediction cannot improve.

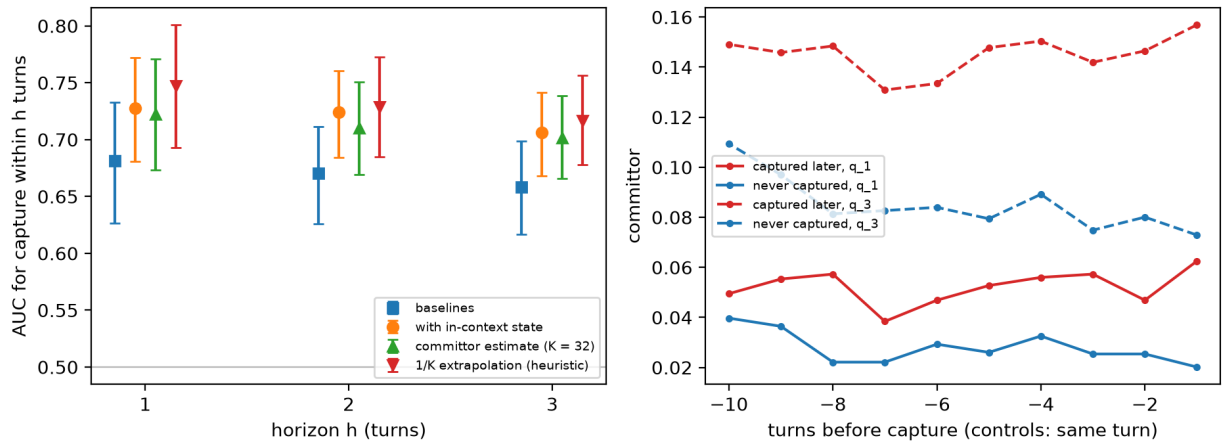


Figure 33: Warning of hostile capture in sampled self-dialogue of Qwen2.5-0.5B-Instruct (hostility direction at layer 12): detectors and the Monte Carlo committor. Left: AUC for capture within h turns on 800 stratified states of the confirmation dialogues, for the classifier baselines, the in-context detector (both fit on the exploration study), the Monte Carlo committor from 32 continuations, and its heuristic $1/K$ extrapolation (dialogue-bootstrap 95 % intervals). Right: mean committor in the ten turns before capture for 48 dialogues captured after turn 10, and for never-captured dialogues at the same turns (descriptive).

A second classifier. The Qwen2.5-0.5B outcomes above were first measured with one automated classifier. We therefore rescored all 44 of its dialogue sets with an independently trained RoBERTa classifier [84], trained on the three Jigsaw toxicity datasets (which include the first classifier's training set) and used with the same threshold, and an independent audit recomputed every pre-stated contrast with it. This is a robustness check on dialogues already analysed, so it is exploratory. On these sets the second classifier is stricter. At $\alpha = 0.06$ it flags 35 % of turns instead of 57 %, and at 0.07 62 % instead of 87 %. Almost all disagreements are replies that only the original flags (at 0.06, 23 % of turns against 0.7 % the other way), most of them repetitive exclamations such as "You're an AI! You're an AI!". Capture still rises steeply with the dose and stays specific to the hostility direction (random and cooking directions at 0.08: at most 0.2 % of turns), but the hostile state is less persistent (geometric persistence summary at 0.07: 6.5 turns instead of 16). History dependence survives. After the ramp, the late share exceeds that of a constant 0.04 by +0.16 [+0.11, +0.21], and the held contrasts are +0.05 [+0.01, +0.10] at 0.04 and +0.04 [+0.01, +0.08] at 0.03; only the comparison with a constant 0.05 loses its positive lower bound (+0.06 [−0.001, +0.12]). Refitting the warning models with the second classifier's labels and baseline features (fit on the exploration study, applied unchanged to the confirmation dialogues) gives a pooled increment of +0.043 [+0.024, +0.062] AUC for capture within three turns. It stays positive at dose 0.045 (+0.061) and is +0.020 [−0.009, +0.049] at 0.04. The three replacement arms keep the replies shown below the constant-dose response at equal mean dose under both classifiers. Both instruments are automated and may share errors; no human labels were collected. [**Planned:** a hazard model; longer holds to separate slow release from bistability; other openers and steering directions; larger models; human labels for the outcome.]

15 Discussion

What theory allows, and what is new. Chaos and periodic windows occur in many nonlinear feedback systems. In a specified ensemble of random recurrent networks with asymmetric couplings,

chaos sets in above a critical gain, sharply in the large-network limit, and finite networks pass through limit cycles of increasing complexity on the way [19]; how the transition looks depends on the architecture and the transfer function [85]. External input can suppress that chaos [86], and deep networks at initialization show an order-to-chaos transition in signal propagation [87, 88]. In our map, temperature plays the role of an inverse gain and the steering vector that of an input. Section 3 makes the correspondence concrete through an exact chain-rule factorization, with the softmax derivative $(\operatorname{diag} p - pp^\top)/T$ in the place of the slope of the transfer function; the identification with the random-network transition remains an analogy, not a transfer of that ensemble's theorem. Feigenbaum scaling holds for a class of one-dimensional maps with a quadratic maximum [20], and periodic windows, shrimp-shaped domains with their cusps and spirals, and doubling cascades are found in many dissipative two-parameter systems [21, 50, 51, 89] and in experiments [90]; intermittency near folds is classical [49]. These results motivate looking for chaos, windows and shrimp-like structure in a language model's feedback map, but they do not imply it. They do not say whether a trained language model realizes it, where in its parameter plane, with which texts, or whether the theory's quantitative predictions hold for that model. This paper shows that the models we studied do realize it, with concrete texts as attractors, so that a bifurcation changes what the model writes, and that the quantitative predictions hold in the cases we tested.

As far as we know, no earlier work has computed periodic orbits and their multipliers, continued fold and flip bifurcation curves and bracketed cusp candidates in the feedback map of a released pretrained language model. Nor do we know of earlier quantitative predictions from such a model's derivatives, for these objects, that were confirmed prospectively, as ours were for passage times at new folds (23 of 24 forecasts within one iterate, with the all-row criterion failing by one row) and for the orbits recovered and their stabilities at frozen offsets near cusp candidates; derivative-based normal-form predictions in a task-trained looped transformer precede ours [18]. A frozen spectral prediction of the response to relaxation is confirmed on held-out data in Huginn, and one periodic orbit of the full transformer map of Qwen2.5-0.5B-Instruct is proved by interval arithmetic; we know of no earlier instance of either for a language model. The dynamical view also has consequences for control in sampled dialogue, where the dose response is not monotone and counter-steering can backfire.

Is steering the origin of the chaos? Steering adds a fixed vector to the residual stream, so chaos along a steering axis could belong to the push rather than to the model. At $\alpha = 0$, however, the map is the unsteered model's own feedback map, with temperature as its only parameter. The $\alpha = 0$ columns of the hostility maps contain cells with positive finite-horizon exponents in Llama-3.1 8B (26 of 64 temperatures), Qwen2.5-7B (15 of 64) and OLMo-2 7B (11 of 64), while the near-zero columns of the 0.5B and 1.5B maps contain one such cell each (of 128 and 64). A longer float64 sweep of the unsteered 0.5B map (512 temperatures over $\log_{10} T \in [-1, 0.3]$, 1000 transient and 2000 measured steps) finds a narrow chaotic band at $\log_{10} T \approx 0.15$–0.21 (7 temperatures, exponents up to 0.28), with a period-5 loop at 437 of the 512 temperatures. The same long sweep finds 96 chaotic temperatures in Llama-3.1 8B, between $\log_{10} T = -0.20$ and 0.24 (exponents up to 0.66), interleaved with periodic windows of periods 2 to 52; within $\log_{10} T \in [-0.3, 0.3]$ its chaotic share, 41 %, equals that of the coarse column. Qwen2.5-7B has 65 chaotic temperatures between $\log_{10} T = -0.07$ and 0.28 (exponents up to 0.74), and OLMo-2 7B has 32 between 0.04 and 0.14 (up to 0.30). Steering is therefore not needed to reach chaotic regimes in any of these four models. Temperature alone produces chaos with embedded windows, broadly in the 7B and 8B models and in a narrow band in the 0.5B model (Figure 34). Across the fifteen models of Section 12, the regular-grid sweeps find unsteered chaos in eleven of the fourteen dense models (in Llama-3.1-70B at two temperatures that

pass a three-offset test), but not in Qwen2.5-72B, Llama-3.3-70B, OLMo-2-13B or the mixture of experts; finer lines found candidates in the first two, and the Qwen2.5-72B window, recomputed in float64 throughout, is periodic.

Steering does widen the chaotic regions, but not as a property of the concept: along norm-matched random directions, the 0.5B maps are as chaotic as along the concept directions (Figure 35). On the same 128^2 grid, 7.6, 8.3 and 12.5 % of cells are chaotic for three Gaussian directions with the philosophy vector's norm, against 6.8 % for philosophy and 7.5 % for gratitude. In all five maps the chaotic share rises from about 1 % at $|\alpha| < 0.01$ to between 5 and 22 % at larger $|\alpha|$, and a chaotic region at high temperature, with windows reaching down, recurs. In this model, chaos under steering is therefore a generic response to perturbing the residual stream at this norm. Rotating the steering direction makes the same point continuously: along the great circle from one Gaussian direction to another, 90° away, at constant norm, the chaotic regions move and change shape, but the chaotic share stays between 13 and 24 % at every one of 103 angles (Figure 36; a video of the rotation is in the supplementary material). The direction does select the texts. The largest attractor family in all five maps is the unsteered period-5 loop, but beyond it only 26 to 56 % of one map's cells carry an attractor text that appears anywhere in another map. Llama-3.1 8B agrees. On its hostility plane, two norm-matched random directions at the same layer give 26 and 17 % chaotic cells against 28 % along the concept direction; all three carry 34 to 44 % near $\alpha = 0$, and the random directions thin out at large $|\alpha|$. At layer 19, where a sparse autoencoder trained on this model reads the residual stream [61], the concept vector rebuilt there gives 28 %. The decoder direction of the autoencoder feature most specific to hostility on held-out texts (chosen by a rule fixed before any map; cosine 0.10 with the concept vector) gives 29 %, a random live feature 27 % and a Gaussian direction 39 %, all at the same norm. In this model, chaos therefore occurs along autoencoder-feature, contrastive and random directions alike, with direction-dependent prevalence; at 70B the directions differ more (Section 12).

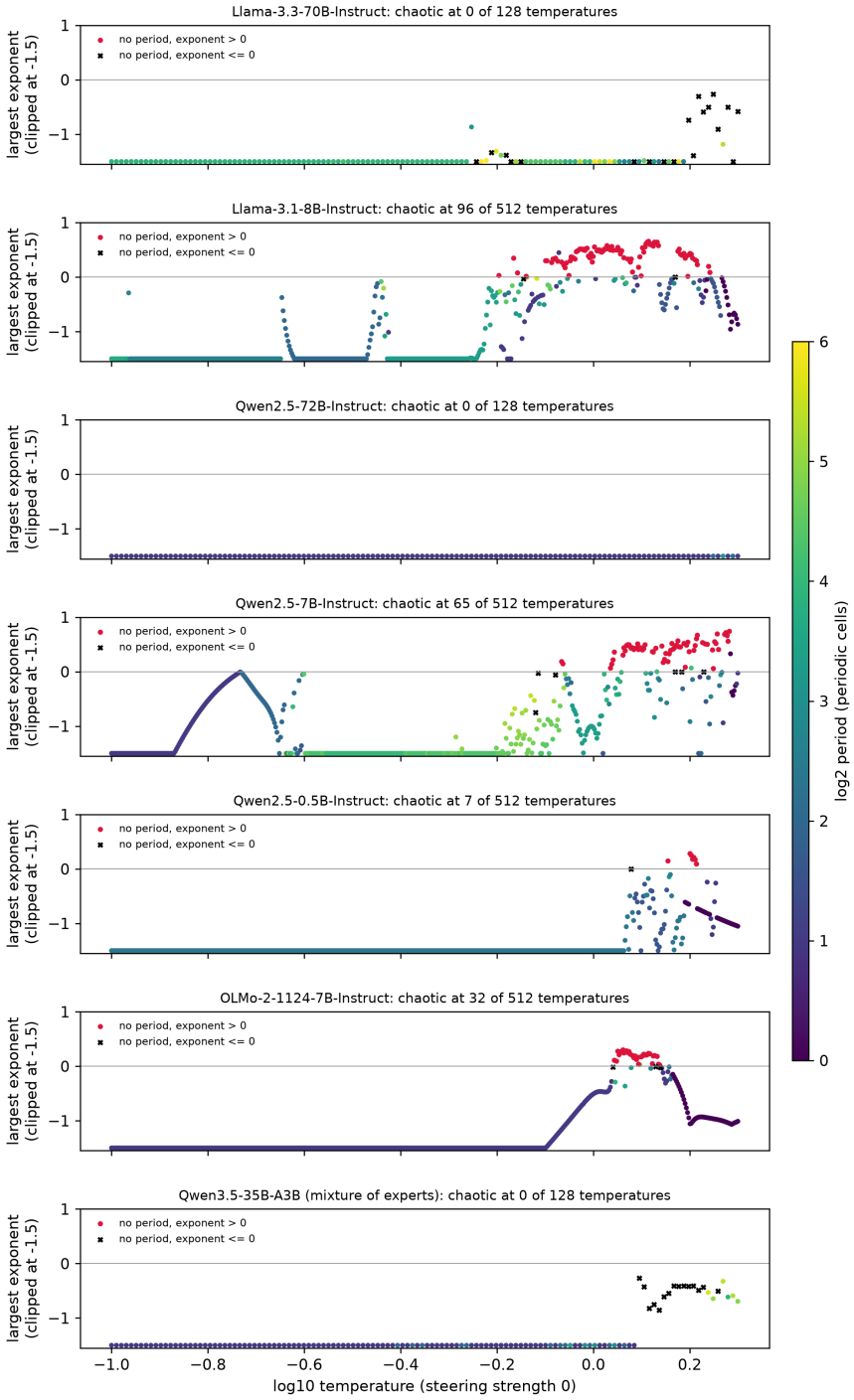


Figure 34: The unsteered soft-token map ($\alpha = 0$) along temperature, top to bottom: Llama-3.3-70B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-72B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-0.5B-Instruct, OLMo-2-1124-7B-Instruct and the mixture of experts Qwen3.5-35B-A3B. Largest Lyapunov exponent (clipped at -1.5) from float64 sweeps of 512 temperatures (128 for the 70B-class and mixture-of-experts models; 1000 transient and 2000 measured steps). Red: no period up to 64 and an exponent above 10^{-3}; black crosses: neither a detected period nor a positive exponent (unresolved); other points have a detected period, coloured by $\log_2$ of the period. Table 11 summarizes these and the wider sweeps of the other models.

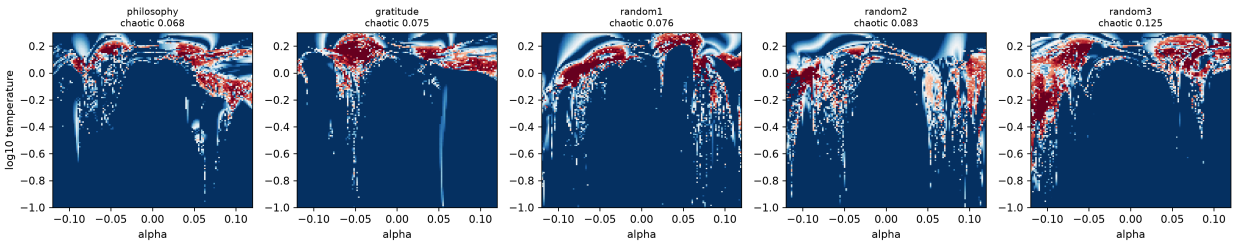


Figure 35: Steering controls in Qwen2.5-0.5B-Instruct (128^2 float64 cells, steering at layer 12, one grid and one set of settings): largest Lyapunov exponent along the philosophy and gratitude directions and along three Gaussian directions rescaled to the philosophy vector's norm. Titles give the chaotic share (finite-horizon exponent above 10^{-3}, no period up to 48).

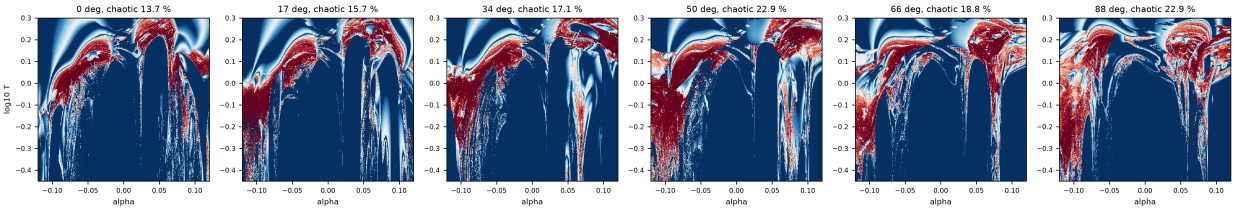


Figure 36: The soft-token map of Qwen2.5-0.5B-Instruct (steering at layer 12) while the steering direction rotates at constant norm from one Gaussian direction (0°) to another (90°): largest Lyapunov exponent (red: positive) at six of 103 angles, with the chaotic share. Float32 maps of 296×512 cells over $\alpha \in [-0.12, 0.12]$, $\log_{10} T \in [-0.45, 0.3]$; exponents from 80 measured steps at twin offset 10^{-4}; each map continues from the final states of the previous angle after 40 steps, and every tenth starts from the fixed start after 300 steps, so these short-horizon labels are coarser than those of the float64 maps. Of 120 planned angles, 17 did not finish within the run limit.

What is and is not claimed. The objects studied here are invariant sets and bifurcations of an exactly defined smooth map built from a frozen model's own computation. We do not claim that ordinary sampled generation is chaotic in this sense (Section 10), nor that any window is a certified shrimp. The period-6 window and the hostile period-7 body have, numerically, shrimp-like organization: fold curves with cusp candidates (simple critical multiplier; quadratic coefficient changing sign between adjacent curve points with nonzero slope; cubic coefficient nonzero at the bracketing points) and flip curves inside, and in the hostile body superstable curves run through the windows. We did not locate the points where the quadratic coefficient vanishes. A generic codimension-two cusp also requires the two unfolding coefficients to depend on the two parameters with rank two. We estimated this at a fold point near one candidate of each window (determinants 2737 and 53.5), and there the local model's predictions at twelve frozen offsets matched the orbits recovered by Newton solves seeded at the predicted roots and the base state, in number at all twelve and in stability at eleven; the search was not exhaustive. These are float64 computations validated by step-size checks (rung 1), not computer-assisted proofs. At the two continuation points that bracket the lower candidate of the hostile body, an independent implementation of the model reproduced the normal-form coefficients, including the sign change of b: $|b|$ within 1.2 % and c to two figures, with the opposite orientation of the null vector, which flips the signs of a and b together.

Limitations. The detailed orbit studies cover one model family (Qwen2.5, 0.5B and 1.5B), with coarse maps and three float64 orbits at 7B; the two other families (OLMo-2 7B and Llama-3.1 8B)

have maps, including 128^2 refinements for Llama, but no continued orbits. The size comparison of Section 12 uses regular temperature grids that can miss narrow windows, a window that missed most of the larger models' chaos, and mostly one twin offset per label. The two-GPU runs of the 70B-class models computed their final normalization in float32 until we fixed it; their labels rest on offsets of 10^{-6} or more, and the cells we rechecked, the Qwen2.5-72B jump window and steered candidates in float64 throughout and samples of the Llama maps at three offsets, behave as stated, but the rest of those maps were not recomputed. The maps use one prompt (Section 12 adds ten for two models), and steering uses contrastive vectors, checked against norm-matched random and sparse-autoencoder directions in two models. The system is a constructed one-token window. The evidence consists of numerical validation, cross-checks and prospectively frozen passage tests (including failed all-row criteria), with a computer-assisted proof for only one orbit at one parameter point. Chaos labels are finite-horizon estimates (300 measured steps after 300 to 500 transient steps), and no chaotic cell is proved chaotic. A horizon check retested random cells of two float64 maps (Figures 1 and 2) with 1000 transient and 2000 measured steps at two twin offsets: 86 and 92 of 96 cells labelled chaotic stayed chaotic (all but one of the rest settled on periodic orbits), all 96 periodic cells in each map stayed periodic, and 18 and 20 of 32 cells with neither label turned out to be slow periodic transients. Near the accumulation of a cascade the labels are less reliable: on the hostile line of Figure 27 (left), 135 of the 235 cells labelled chaotic from 400 measured steps stayed chaotic and 91 were periodic. The Feigenbaum ratios are finite-order values that rise toward δ (the last is within 1 %); the limit itself is not computed. All orbit, bifurcation and chaos results concern the constructed map, not sampled generation with a growing context. The bridge of Section 10 tests only whether the bottleneck's laminar phases survive sampling noise, and the behavioural results are statistics of sampled dialogue, not orbits. The behavioural study uses two small models of one family (0.5B and 1.5B), one opener and a proxy defined by two automated classifiers without human labels. Its first lead-time test is negative, and the confirmed warning from the in-context state is small.

Steering strength is a bifurcation parameter. In the soft-token map, small changes of the steering strength switch the model between distinct repeating texts, chaos and coexisting attractors. Near a fold curve, behaviour becomes intermittent, with laminar phases whose duration the fold's normal form predicts in advance. In sampled dialogue, the dose along a hostility direction controls how long a hostile state persists, continuously from bursts to near-absorption, while directions of the same norm do almost nothing. Both observations argue for treating an activation-steering coefficient like any control parameter of a nonlinear system. Its bifurcation set should be characterized before an operating point is chosen, history dependence should be expected near boundaries, and a monotone, smooth dose response should not be assumed. In steering studies, outputs that fall into verbatim repetition at high doses are usually discarded as quality failures. In a replication of a feature-steering study of self-reports [91], the highest doses failed quality checks in this way [**Planned:** citation of the replication once posted]. In the present framework such loops are candidate attracting periodic orbits, and the dose at which they appear is a boundary that can be mapped like the others. The deterministic map is where such sets can be defined precisely and computed numerically. Whether its boundaries predict boundaries of sampled behaviour is open. The bridge of Section 10 shows that sampling noise shortens the laminar phases of the bottleneck, by an amount that depends on how many samples are averaged.

What this suggests for interpretability and alignment. Our results come from open models of 0.5B to 72B parameters and a constructed loop, but they suggest four things about models

whose outputs feed back into their inputs, as in self-dialogue, agent loops and pipelines of model calls. First, where the loop is chaotic, a perturbation of 10^{-6} grows to order one within roughly 15 to 50 steps (the largest unsteered exponents of the 7B and 8B models are 0.30 to 0.88 per step). Even a complete account of single forward passes then predicts a long trajectory only if the state is known far more precisely than any interpretation supplies. Stepwise summaries can be inconclusive: every step of the period-8 orbit expands some direction while the cycle contracts, so the ordered product of the step Jacobians, not a summary of each step, captures the loop's stability (Section 3). What remains predictable are the invariant sets, their statistics and the boundaries between them. This extends the analysis of trained recurrent networks through their fixed points [26, 27] to the periodic orbits, chaos and bifurcations of a pretrained transformer. Second, feature dictionaries say which features an attractor occupies, but the boundaries between attractors, which decide what a small change does, are properties of the map's derivatives. In our 8B controls, a sparse-autoencoder feature direction, contrastive and random directions all drove the map into chaos (27 to 39 % of cells); at 70B the hostility-related directions gave more chaotic cells than the random ones (4.3 and 3.5 % against 1.1 and 0.2 %; Section 12). Steering acts through the gain of Section 3, which is not specific to concept-labelled directions, but a direction's geometry and its meaning can be correlated. Text can also hide the dynamics: in the hostile cascade, a seven-token text repeats exactly while the state underneath is chaotic (Figure 24), so a monitor that reads only the emitted text would call a chaotic regime periodic. Third, when a behaviour sustains itself through a loop, acting on the loop can be cheaper than pushing against the behaviour. In sampled dialogue, the most dose-efficient control we found kept flagged replies out of the context, while counter-steering against the activations was less efficient and, in the larger model, backfired past the peak of the dose response (Section 14). Fourth, conclusions drawn at one operating point need not transfer: which loop a model enters depends on how much context it keeps (Table 10), on the path by which a boundary is approached, and on the sampling noise (Section 10). Treating a looped model as a dynamical system, with regimes and boundaries to be mapped, rather than as a function to be evaluated point by point, seems to us the appropriate stance for both interpretability and safety evaluation.

Is the chaos useful? The chaos we map is not a source of new content. The chaotic cells of Table 2 recombine the phrases of the neighbouring loops in an order that did not settle over the recorded horizon: the trajectory wanders among the ghosts of nearby periodic orbits, which gives variety in sequence but not in substance. The periodic loops are the familiar degeneration of decoding into repetition [22, 23]. Neither regime is good text, and the diversity of ordinary generation comes from sampling and a growing context, which wash out much of this structure (Section 10). What the chaos does matter for is predictability and control. Positive finite-horizon exponents limit open-loop prediction of a trajectory at fixed state precision (our experiments do not establish mixing or limits on feedback stabilization); a stable attractor is predictable, but it can also hold a model in a state it should leave, as the hostile capture of sampled dialogue shows (Section 14). On the paper's prompt, the largest Qwen2.5 and Llama models sit near the stable end, but across prompts a 14B Qwen model is more chaotic than the 7B, at higher temperatures, and OLMo-2-32B is among the most chaotic models we tested (Section 12). Whether regimes near the boundary between order and chaos are especially useful, as has been argued for computation in dynamical systems and recurrent networks [92, 93] and shown for signal propagation in deep networks at initialization [87, 88], is open for language models: our maps locate such boundaries but do not test it.

Which findings might reach frontier models. Today's frontier systems include closed models such as GPT-6 Astra and Claude Opus 5.5 and open-weight mixture-of-experts models such as the GLM-5 family. Some of our findings rest only on structure that every smooth softmax-readout language model with a fixed prefix shares and should carry over: the factorization of the loop's derivative (a covariance for tied embeddings, a cross-covariance otherwise), contraction at sufficiently high temperature, the reduction of zero-temperature attractors to greedy cycles of whatever memory the loop keeps, and the role of a steering coefficient as a bifurcation parameter. Others belong to particular weights and will not carry over as such: the attractor texts, the positions of windows, folds and cusp candidates, and every number we report; a second checkpoint of the same architecture already has different attractors (Section 11). Size gives no guarantee of stability. In a fixed temperature window the unsteered chaos nearly vanished in the largest Qwen2.5 and Llama models, but across prompts a 14B Qwen model was more chaotic than the 7B, at higher temperatures, OLMo-2-32B was the most chaotic OLMo model, and post-training changed the chaos of two Llama checkpoints of the same size (Section 12). For an open frontier model we would search for chaos where its own next-token distribution spreads, across prompts. Our one mixture-of-experts model, Qwen3.5-35B-A3B, gave no positive exponent without steering and irregular cells only under hostility steering, and hard routing falls outside the smooth theory of Section 3 (Section 11). Reasoning models open a thinking block that changes the prefix, post-training may remove some attractors and create others, and deployed inference samples tokens, in low precision, over long contexts; our bridge results show that sampling shortens laminar phases and that a longer context changes which loops are entered. Closed models permit none of these computations: the map needs the input embedding matrix, full next-token distributions and a way to feed a vector in place of a token, and its analysis needs derivatives. For closed models the method and the questions transfer, not the numbers.

16 Evidence and reproducibility

We grade claims on an evidence ladder. Rung 1 means numerically validated (tolerances, residuals, convergence, precision varied); rung 2, cross-checked by an independent estimate, implementation or hardware; rung 3, a computer-assisted proof.

Search scale. The soft-token results come from an exploratory search. An initial census of the raw data at 00:19 UTC on 14 October 2026 counted 34 distinct maps (987 308 cells, 27 maps in float64) in six models, each with one prompt and one of three steering directions or none, plus one partial map. Of these, 29 maps and the partial one appear here; the five omitted are early discovery or side maps. Later additions are not in that count: the steering-control maps of Section 15, the mixture-of-experts maps, and the sweeps, lines and maps of the models of Section 12. The labelled maps contain about 2600 periodic families of at least six cells, counted per map. We singled out about 26 windows or families for float64 work: 20 are reported in detail, 2 only as map features and 4 not at all (an orbit whose continuation was abandoned, a cell that proved aperiodic, a band whose float32 orbit did not converge and a pilot that found no cusp within a bounded search). Of 54 Newton orbit attempts, 7 failed. Of seven tangent reversals on continued fold curves, six were tested, and all six are reported as cusp candidates (a bracketed sign change of the quadratic coefficient with nonzero cubic coefficients); one is untested. Two of the five regions we flagged as shrimp candidates are reported as shrimp-like. One further language model (Ouro-1.4B) was explored and is not reported; its apparent chaos was a long transient. Every window was chosen after the maps were inspected. Only the passage forecasts, the cusp-unfolding offsets, Huginn's relaxation test and the dialogue warning confirmation were fixed before the computations that test them (Table 13).

	explored	reported	note
Soft-token maps (distinct)	34	29	plus one partial map, reported
cells	987 308	893 868	omitted: 5 discovery or side maps
float32 maps	7	4	
Models with maps	6	6	one prompt; 3 directions and $\alpha = 0$
Periodic families (≥ 6 cells)	≈ 2600	–	counted per labelled map
Windows given float64 work	≈ 26	20 (+2)	+2: map features only
Newton orbit attempts	54	–	7 failed
Tangent reversals on fold curves	7	6	1 untested
Unfolding offsets, frozen	12	12	counts 12/12, stability 11/12
Shrimp-candidate regions	5	2	reported as shrimp-like
Tests fixed before outcomes	6	5	2 all-row criteria failed by one row
Language models loaded	8	7	Ouro-1.4B not reported

Table 13: Search scale on 14 October 2026, before the steering-control maps (Huginn's frozen test counted once; the sixth test, a cusp pilot in a second checkpoint, is the unreported bounded search).

- Rung 3: the period-8 stripe orbit of Table 1. At $\alpha = 0.071674$ and $\log_{10} T = -0.115$, read as exact binary64 rationals (the temperature is the real number $\exp(\ln 10 \cdot \log_{10} T)$ for that exact dyadic $\log_{10} T$), the expected-embedding map of the stored Qwen2.5-0.5B-Instruct checkpoint (all 24 layers, the full 151 936-token readout, the fixed 37-token prefix, steering after zero-based block 12) has a primitive period-8 orbit, which is the unique multiple-shooting root in a product of eight boxes of coordinate radius 2^{-640} around refined centres, and the orbit is locally attracting. Ball arithmetic (Arb [94]) encloses the map and its Jacobian over each box (8×896 directions); a Krawczyk-type inclusion for the preconditioned shooting system gives existence and uniqueness, separation of the first point from the points one, two and four steps later gives the primitive period, and a norm bound on powers of the enclosed monodromy gives attraction. An independent audit in exact rational arithmetic rechecked the saved witnesses. The proof trusts the Python, FLINT and Arb stack and the stored weights. It does not exclude other period-8 points in the first box, and it says nothing about the window's boundaries or global organization (record: `certificates/llm-p8-certificate-final-001.json`).

- Rung 2: the 2-cycle's fold and its normal form. An independent reimplementation of the forward map and solver recovers α_c, a and b (to 6×10^{-15}, 3×10^{-12} and 2×10^{-6} relatively), and ab also follows from the stable-branch multipliers. The same reimplementation reproduces all 18 fixed-entrance counts of the controlled passage runs exactly (Table 3 lists 12 of them).

- Prospectively frozen: the passage forecasts at two new folds (Table 5); 23 of 24 within one iterate, and the all-row criterion failed by one row.

- Rung 2 as well: the period-2 and period-8 orbits of Table 1 (independent implementation), and the cascade's first four flip parameters (CPU and H100, agreement 4×10^{-15}).

- Rung 2 as well: the hostile 2-cycle fold and its normal form, reproduced by an independent float64 forward map with nested automatic differentiation (location to 6×10^{-10}, ab to 8 parts per million); its controlled passages were also checked independently, with the frozen all-six criterion failing by one row.

- Rung 1: the other orbits, the curves, the six cusp candidates, the superstable parameters, the 1.5B flips, the hostile windows' edges, the spiral region's period-7 body and the cascade inside it, the three 7B orbits, and Huginn's float32 spectral estimates.

- Prospectively frozen, not independently reproduced: the scaling predictions P1–P3, the prompt-blocked size tests in Qwen2.5 (7B and 14B; 72B) and OLMo-2, and the loop-free predictor test of Section 12, all scored by code pushed before the data, together with the three-offset plateau tests of the 70B-class maps and the float64 recheck of the Qwen2.5-72B cells with exact tangent exponents. These are float64 computations at rung 1; no independent implementation has rechecked them.
- Prospectively frozen and independently replayed: Huginn's relaxation test on held-out data, confirmed in both parts; an independent implementation reproduced both from the saved rows and spectra (GSM8K part exactly, item-level part to 10^{-12}) (Section 13).
- Prospectively frozen as well: the confirmation of the in-context warning in sampled dialogue (Section 14), whose design was pushed before the data and amended, before any outcome was read, to correct a labelling error.

Hardware. Float64 GPU computations ran on rented NVIDIA H100 (NVL, PCIe and 80 GB HBM3), H200, H200 NVL and A100 80 GB (SXM4 and PCIe) GPUs, one GPU per run except for the 70B-class models, whose layers were spread over two; the float32 frames of the rotation video ran on RTX 4090 GPUs; cross-checks and the interval proof ran on CPUs. The manifest `docs/manifests/runpod-runs.json` records for each run the GPU type, the code commit, the job and its arguments, the wall time and the exit status, and the available hashes of the uploaded code and input archives and of the retrieved results, with absent fields left explicit; per-run receipts and accepted provenance limitations are kept with it. Regenerated at 11:57 UTC on 14 October 2026, it lists 232 runs with retrieved results (code and input-archive hashes for 124, result-archive hashes for 216) and 16 pods that ended without results. GPU types round float64 reductions differently, so quantities used across runs were checked across hardware where it mattered (for example, flip parameters agree between CPU and H100 to 4×10^{-15}). Code: `tools/explore_llm_soft_loop.py`, `tools/bifurcation_curves.py`, `tools/llm_feigenbaum.py`, `tools/explore_latent_loop.py`, `tools/explore_llm_msm.py`, `tools/hostility_score.py`, `tools/p1_committor.py`, `tools/p1_controller.py`, `tools/p1_degeneracy.py`, `tools/cusp_check.py`, `tools/huginn_relax_eval.py`, `tools/huginn_arc_analysis.py`. Raw-data hash receipts: `evidence/llm-dynamics-data-receipts.json` (to 12 October), the per-run receipts in the research log, and the verified snapshot manifests in `docs/backups/`; independent cross-check and audit receipts are under `codex/` (`crosscheck`, `confirm-fold`, `hostile-fold-crosscheck`, `misalignment-audit`, `l9-committor-review`, `huginn-design-review`, `huginn-independent-replay`, `p1-second-classifier-audit`, `paper-review`).

[**Planned:** public code and data release at submission.]

A Self-dialogue under steering: the gratitude ladder (exploratory)

Two copies of Qwen2.5-0.5B-Instruct talk to each other: 128 replicas, 40 turns, opener "Hello!", a window of 7 turns, 60 new tokens per turn, sampled at $T = 0.7$, with the gratitude direction added at layer 12. Turn activations are clustered (principal component analysis to 20 dimensions, 40 states) into a Markov state model with four metastable sets [52]. Table 14 lists finite-horizon measures for each strength.

gratitude strength α	0	0.01	0.02	0.04
gratitude set	none	62 % weight	dominant	95 % weight
occupancy at turn 9 / 14 / 39	–	0.34 / 0.52 / 0.73	0.70 / 0.91 / 0.91–0.98	from turn 0
entry turn, quartiles	–	5 / 7 / 11.8	4 / 5 / 7	0
replicas never entering	–	14 of 128	2 of 128	0
residence per visit (turns)	–	about 15	about 18	effectively absorbing
replies with a gratitude word	0.00–0.05	0.66 at turn 39	0.96–0.98 from turn 9	0.88 at turn 0
spiritual-word plateau	0.00–0.02	0.13	0.26–0.30	0.42–0.48

Table 14: The gratitude ladder (one opener and seed per rung; word lists chosen after reading samples). At 0.02 the dialogue starts in the helper mode and enters the gratitude set within a few turns; at 0.04 the set is imposed from the first reply. **[Planned:** frozen confirmatory version on held-out openers with dialogue-level bootstrap.]

References

[1] Xiaojian Li, Yongkang Leng, Ruiqing Ding, Hangjie Mo, and Shanlin Yang. Cognitive activation and chaotic dynamics in large language models: A quasi-Lyapunov analysis of reasoning mechanisms. arXiv preprint arXiv:2503.13530, 2025. URL https://arxiv.org/abs/2503.13530.

[2] Gregorio Jaca, Kristóf Benedek, and János Török. Chaos in reason: How chain-of-thought LLMs can look for an answer. arXiv preprint arXiv:2607.27805, 2026. URL https://arxiv.org/abs/2607.27805.

[3] Julian Arnold, Flemming Holtorf, Frank Schäfer, and Niels Lörch. Phase transitions in the output distribution of large language models. arXiv preprint arXiv:2405.17088, 2024. URL https://arxiv.org/abs/2405.17088.

[4] Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Phase transition in large language models and the criticality of natural languages. arXiv preprint arXiv:2406.05335, 2024. URL https://arxiv.org/abs/2406.05335.

[5] Youran Sun and Babak Haghighat. Phase transitions in large language models and the $O(N)$ model. arXiv preprint arXiv:2501.16241, 2025. URL https://arxiv.org/abs/2501.16241.

[6] Huajian Ruan, Jinyang Li, Xingyu Guo, and Lingxiao Wang. Generative criticality in large language model temperature scaling. arXiv preprint arXiv:2606.06238, 2026. URL https://arxiv.org/abs/2606.06238.

[7] Neil F. Johnson, Frank Yingjie Huo, and Bella Xinrui Li. Temperature-driven inversion and nonlinear dynamics in ChatGPT-like AIs. arXiv preprint arXiv:2608.00939, 2026. URL https://arxiv.org/abs/2608.00939.

[8] Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1:*

Long Papers), pages 12740–12755. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.624. arXiv:2502.15208.

[9] Ting-Wen Ko and Jonas Geiping. Attractor states emerge in multi-turn LLM conversations. In *Third Conference on Language Modeling*, 2026. URL https://openreview.net/forum?id=umwOiCSh4I. arXiv:2606.30571.

[10] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32, pages 690–701, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/01386bd6d8e091c2ab4c7c7de644d37b-Abstract.html. arXiv:1909.01377.

[11] Shaojie Bai, Vladlen Koltun, and Zico Kolter. Stabilizing equilibrium models by Jacobian regularization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 554–565. PMLR, 2021. URL https://proceedings.mlr.press/v139/bai21b.html.

[12] Akiyoshi Tomihari and Ryo Karakida. Recurrent self-attention dynamics: An energy-agnostic perspective from Jacobians. In *Advances in Neural Information Processing Systems*, volume 38, pages 155236–155269, 2025. doi: 10.52202/085713-4675. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/cca569b38ec3612904e625dad7a743dc-Abstract-Conference.html.

[13] Xiao-Wen Yang, Ziyu Han, Xi-Hua Zhang, Wen-Da Wei, Jie-Jing Shao, Lan-Zhe Guo, and Yu-Feng Li. Stabilizing recurrent dynamics for test-time scalable latent reasoning in looped language models. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 144670–144682. PMLR, 2026. URL https://proceedings.mlr.press/v306/yang26t.html.

[14] Xinyue Luo and Fei Yu. Prediction dynamics in depth-recurrent language models. arXiv preprint arXiv:2609.21383, 2026. URL https://arxiv.org/abs/2609.21383.

[15] Boyi Zeng, Shixiang Song, Siyuan Huang, et al. PonderLM: Pretraining language models to ponder in continuous space. In *The Fourteenth International Conference on Learning Representations*, pages 91874–91893, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/94572a66c6b0710c5263f15d8ef6b463-Abstract-Conference.html. arXiv:2505.20674.

[16] Zhen Zhang, Xuehai He, Weixiang Yan, Ao Shen, Chenyang Zhao, and Xin Eric Wang. Soft Thinking: Unlocking the reasoning potential of LLMs in continuous concept space. In *Advances in Neural Information Processing Systems*, volume 38, pages 187190–187212, 2025. doi: 10.52202/085713-5629. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/f7396d1c54d51416958d63e285377103-Abstract-Conference.html. arXiv:2505.15778.

[17] Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason E. Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. In *Second Conference on Language Modeling*, 2025. URL https://openreview.net/forum?id=Itxz7S4Ip3. arXiv:2412.06769.

[18] Gunn Kim. Dynamical phase selection controls compute scaling in looped transformers. arXiv preprint arXiv:2608.26556, 2026. URL https://arxiv.org/abs/2608.26556. Version 1, 27 August 2026.

[19] H. Sompolinsky, A. Crisanti, and H. J. Sommers. Chaos in random neural networks. *Physical Review Letters*, 61(3):259–262, 1988. doi: 10.1103/PhysRevLett.61.259.

[20] Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19(1):25–52, 1978. doi: 10.1007/BF01020332.

[21] Jason A. C. Gallas. Structure of the parameter space of the Hénon map. *Physical Review Letters*, 70(18):2714–2717, 1993. doi: 10.1103/PhysRevLett.70.2714.

[22] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. In *The Eighth International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=rygGQyrFvH. arXiv:1904.09751.

[23] Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, Huayang Li, and Jian Li. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation. In *Advances in Neural Information Processing Systems*, volume 35, pages 3082–3095, 2022. doi: 10.52202/068431-0223. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/148c0aeea1c5da82f4fa86a09d4190da-Abstract-Conference.html. arXiv:2206.02369.

[24] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759, 2024. doi: 10.1038/s41586-024-07566-y.

[25] António H. Ribeiro, Koen Tiels, Luis A. Aguirre, and Thomas Schön. Beyond exploding and vanishing gradients: analysing RNN training using attractors and smoothness. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2370–2380. PMLR, 2020. URL https://proceedings.mlr.press/v108/ribeiro20a.html.

[26] David Sussillo and Omri Barak. Opening the black box: Low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Computation*, 25(3):626–649, 2013. doi: 10.1162/NECO_a_00409.

[27] Niru Maheswaranathan, Alex Williams, Matthew Golub, Surya Ganguli, and David Sussillo. Reverse engineering recurrent networks for sentiment classification reveals line attractor dynamics. In *Advances in Neural Information Processing Systems*, volume 32, pages 15696–15705, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/d921c3c762b1522c475ac8fc0811bb0f-Abstract.html. arXiv:1906.10720.

[28] Jonas Geiping, Sean McLeish, Neel Jain, et al. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, volume 38, pages 46242–46293, 2025. doi: 10.52202/085713-1380. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/3b01972cf31e6fa0fe29e4b8b5c2a0a1-Abstract-Conference.html.

[29] Rui-Jie Zhu, Zixuan Wang, Kai Hua, et al. Scaling latent reasoning via looped language models. arXiv preprint arXiv:2510.25741, 2025. URL https://arxiv.org/abs/2510.25741.

[30] Hayden Prairie, Zachary Novack, Taylor Berg-Kirkpatrick, and Daniel Y. Fu. Parcae: Scaling laws for stable looped language models. In *Third Conference on Language Modeling*, 2026. URL https://openreview.net/forum?id=pIBqtrqFeP. arXiv:2604.12946.

[31] Matthias M. M. Buehlmaier. LayerNorm as implicit gain control in looped transformers. arXiv preprint arXiv:2607.10681, 2026. URL https://arxiv.org/abs/2607.10681. Version 1, 12 July 2026.

[32] Asher Labovich. Stability and generalization in looped transformers. arXiv preprint arXiv:2604.15259, 2026. URL https://arxiv.org/abs/2604.15259. Version 2, 22 April 2026.

[33] Sajad Movahedi, Vera Milovanović, Shlomo Libo Feigin, Alexander Theus, Thomas Hofmann, Valentina Boeva, T. Konstantin Rusch, and Antonio Orvieto. Fixed-point reasoners: Stable and adaptive deep looped transformers. arXiv preprint arXiv:2606.18206, 2026. URL https://arxiv.org/abs/2606.18206. Version 1, 16 June 2026.

[34] Junhong Wu, Jinliang Lu, Zixuan Ren, Gangqiang Hu, Zhi Wu, Dai Dai, and Hua Wu. LLMs are single-threaded reasoners: Demystifying the working mechanism of Soft Thinking. In *The Fourteenth International Conference on Learning Representations*, pages 4533–4548, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/0895de150660b6c2aef9b0f098cb1dcd-Abstract-Conference.html. arXiv:2508.03440.

[35] Fatih Dinc, Ege Cirakman, Bariscan Kurtkaya, Mert Yuksekgonul, Yiqi Jiang, Mark J. Schnitzer, and Hidenori Tanaka. Ghost mechanism: An analytical model of abrupt learning in recurrent networks. *Physical Review X*, 16(2):021058, 2026. doi: 10.1103/mjcl-lb4x.

[36] Demian Battaglia, Nicolas Brunel, and David Hansel. Temporal decorrelation of collective oscillations in neural networks with local inhibition and long-range excitation. *Physical Review Letters*, 99(23):238106, 2007. doi: 10.1103/PhysRevLett.99.238106.

[37] Christoforos A. Papasavvas, Yujiang Wang, Andrew J. Trevelyan, and Marcus Kaiser. Gain control through divisive inhibition prevents abrupt transition to chaos in a neural mass model. *Physical Review E*, 92(3):032723, 2015. doi: 10.1103/PhysRevE.92.032723.

[38] Yilun Liu, Yi Zhang, Ganyu Wu, Sikuan Yan, Mengyue Wang, Alois Knoll, Volker Tresp, and Yunpu Ma. Learning chaos without seeing chaos: Extrapolation of global dynamics in autoregressive transformers. arXiv preprint arXiv:2609.38814, 2026. URL https://arxiv.org/abs/2609.38814v1. Version 1, 30 September 2026.

[39] Jason Z. Kim, Zhixin Lu, Erfan Nozari, George J. Pappas, and Danielle S. Bassett. Teaching recurrent neural networks to infer global temporal structure from local examples. *Nature Machine Intelligence*, 3(4):316–323, 2021. doi: 10.1038/s42256-021-00321-2.

[40] Dayal Singh Kalra, Tianyu He, and Maissam Barkeshli. Universal sharpness dynamics in neural network training: Fixed point analysis, edge of stability, and route to chaos. In *The Thirteenth International Conference on Learning Representations*, pages 55966–56000, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/8c8cd1b78cdae751265c88efc136e5bd-Abstract-Conference.html.

[41] R. L. Marichal and J. D. Piñeiro. Analysis of multiple quasi-periodic orbits in recurrent neural networks. *Neurocomputing*, 162:85–95, 2015. doi: 10.1016/j.neucom.2015.04.001.

[42] Florian Gomez, Ralph L. Stoop, and Ruedi Stoop. Universal dynamical properties preclude standard clustering in a large class of biochemical data. *Bioinformatics*, 30(17):2486–2493, 2014. doi: 10.1093/bioinformatics/btu332.

[43] Rafael V. Stenzinger, Vinícius Luz Oliveira, and M. H. R. Tragtenberg. Shrimp hubs in the Hindmarsh–Rose model. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(2): 023148, 2025. doi: 10.1063/5.0239268.

[44] Luiz F. B. Caixeta, Matheus H. P. Gonçalves, M. H. R. Tragtenberg, and Mauricio Girardi-Schappo. Devil's staircase inside shrimp-shaped regions reveals periodicity of plateau spikes and bursts. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(3):033122, 2025. doi: 10.1063/5.0250342.

[45] Andy Zou, Long Phan, Sarah Chen, et al. Representation engineering: A top-down approach to AI transparency. arXiv preprint arXiv:2310.01405, 2023. URL https://arxiv.org/abs/2310.01405. Version 4, 3 March 2025; first posted 2 October 2023.

[46] Jinghao Zhang, Yuting Liu, Wenjie Wang, Qiang Liu, Shu Wu, Liang Wang, and Tat-Seng Chua. Personalized text generation with contrastive activation steering. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7128–7141. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.353.

[47] Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Applied Mathematical Sciences. Springer, Cham, 2023. doi: 10.1007/978-3-031-22007-4.

[48] Willy J. F. Govaerts. *Numerical Methods for Bifurcations of Dynamical Equilibria*. Society for Industrial and Applied Mathematics, 2000. doi: 10.1137/1.9780898719543.

[49] Yves Pomeau and Paul Manneville. Intermittent transition to turbulence in dissipative dynamical systems. *Communications in Mathematical Physics*, 74(2):189–197, 1980. doi: 10.1007/BF01197757.

[50] Wilson Façanha, Bart Oldeman, and Leon Glass. Bifurcation structures in two-dimensional maps: The endoskeletons of shrimps. *Physics Letters A*, 377(18):1264–1268, 2013. doi: 10.1016/j.physleta.2013.03.025.

[51] Renato Vitolo, Paul Glendinning, and Jason A. C. Gallas. Global structure of periodicity hubs in Lyapunov phase diagrams of dissipative flows. *Physical Review E*, 84(1):016216, 2011. doi: 10.1103/PhysRevE.84.016216.

[52] John D. Chodera and Frank Noé. Markov state models of biomolecular conformational dynamics. *Current Opinion in Structural Biology*, 25:135–144, 2014. doi: 10.1016/j.sbi.2014.04.002.

[53] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, et al. Hopfield networks is all you need. In *The Ninth International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=tL89RnzIiCd. arXiv:2008.02217.

[54] Yihe Dong, Jean-Baptiste Cordonnier, and Andreas Loukas. Attention is not all you need: Pure attention loses rank doubly exponentially with depth. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 2793–2803. PMLR, 2021. URL https://proceedings.mlr.press/v139/dong21a.html. arXiv:2103.03404.

[55] Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. The emergence of clusters in self-attention dynamics. In *Advances in Neural Information Processing Systems*, volume 36, pages 57026–57037, 2023. doi: 10.52202/075280-2493. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b2b3e1d9840eba17ad9bbf073e009afe-Abstract-Conference.html. arXiv:2305.05465.

[56] Frank Pasemann. Complex dynamics and the structure of small neural networks. *Network: Computation in Neural Systems*, 13(2):195–216, 2002. doi: 10.1080/net.13.2.195.216.

[57] Zihao Fu, Wai Lam, Anthony Man-Cho So, and Bei Shi. A theoretical analysis of the repetition problem in text generation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35 (14):12848–12856, 2021. doi: 10.1609/aaai.v35i14.17520. arXiv:2012.14660.

[58] Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context learning and induction heads. Transformer Circuits Thread, 2022. URL https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html. arXiv:2209.11895.

[59] Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, et al. 2 OLMo 2 furious. arXiv preprint arXiv:2501.00656, 2024. URL https://arxiv.org/abs/2501.00656. Shorter version in the Second Conference on Language Modeling (2025).

[60] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. arXiv preprint arXiv:2407.21783, 2024. URL https://arxiv.org/abs/2407.21783.

[61] Goodfire. Llama-3.1-8B-Instruct-SAE-l19. Hugging Face, https://huggingface.co/Goodfire/Llama-3.1-8B-Instruct-SAE-l19, 2025. Sparse autoencoder on the residual stream after layer 19; revision f6775a22.

[62] Seonglae Cho, Harryn Oh, Donghyun Lee, Luis Rodrigues Vieira, Andrew Bermingham, and Ziad El Sayed. FaithfulSAE: Towards capturing faithful features with sparse autoencoders without external datasets dependency. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 297–314. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-srw.20. arXiv:2506.17673.

[63] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168, 2021. URL https://arxiv.org/abs/2110.14168.

[64] Boyuan Wang, Chengyao Yu, Jiaxi Ren, Hongxin Wei, Bingyi Jing, and Yuxin Tao. Scheduling recursive reasoning in looped transformers. arXiv preprint arXiv:2609.36653, 2026. URL https://arxiv.org/abs/2609.36653. Submitted 29 September 2026.

[65] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 Reasoning Challenge. arXiv preprint arXiv:1803.05457, 2018. URL https://arxiv.org/abs/1803.05457.

[66] Anthropic. System card: Claude Opus 4 & Claude Sonnet 4. System card, Anthropic, 2025. URL https://www-cdn.anthropic.com/6d8a8055020700718b0c49369f60816ba2a7c285/Claude%204%20System%20Card.pdf. May 2025; revision 2 September 2025; sections 5.5.1–5.5.2.

[67] Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The assistant axis: Situating and stabilizing the default persona of language models. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 80955–81003. PMLR, 2026. URL https://proceedings.mlr.press/v306/lu26g.html. arXiv:2601.10387.

[68] Cem Anil, Esin Durmus, Nina Panickssery, et al. Many-shot jailbreaking. In *Advances in Neural Information Processing Systems*, volume 37, pages 129696–129742, 2024. doi: 10.52202/079017-4121. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ea456e232efb72d261715e33ce25f208-Abstract-Conference.html.

[69] Jan Betley, Daniel Chee Hian Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 4043–4068. PMLR, 2025. URL https://proceedings.mlr.press/v267/betley25a.html. arXiv:2502.17424.

[70] Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.828.

[71] Laura Hanu and Unitary team. Detoxify. GitHub, https://github.com/unitaryai/detoxify, 2020. Model unitary/toxic-bert, Hugging Face revision 4d6c22e7.

[72] Dhananjay Ashok and Jonathan May. Language models can predict their own behavior. In *Advances in Neural Information Processing Systems*, volume 38, pages 69806–69837, 2025. doi: 10.52202/085713-2102. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/5a9c1af5f76da0bd37903b6f23e96c74-Abstract-Conference.html.

[73] Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models. arXiv preprint arXiv:2507.21509, 2025. URL https://arxiv.org/abs/2507.21509v1. Version 1, 29 July 2025.

[74] Fan Zhou, Weitian Wang, and Tim Van de Cruys. Commitment before realization: When classifier-free guidance becomes unnecessary in masked diffusion language models. arXiv preprint arXiv:2608.08082, 2026. URL https://arxiv.org/abs/2608.08082v1. Version 1, 8 August 2026.

[75] Nicolas Leins, Jennifer Haase, Varvara Geronimus, Jana Gonnermann-Müller, and Sebastian Pokutta. Prompting against persona drift: Comparing intervention timing and content in LLM-simulated conversations. arXiv preprint arXiv:2609.24532, 2026. URL https://arxiv.org/abs/2609.24532v1. Version 1, 21 September 2026.

[76] Weinan E and Eric Vanden-Eijnden. Towards a theory of transition paths. *Journal of Statistical Physics*, 123(3):503–523, 2006. doi: 10.1007/s10955-005-9003-9.

[77] Weinan E and Eric Vanden-Eijnden. Transition-path theory and path-finding algorithms for the study of rare events. *Annual Review of Physical Chemistry*, 61(1):391–420, 2010. doi: 10.1146/annurev.physchem.040808.090412.

[78] Marten Scheffer, Jordi Bascompte, William A. Brock, et al. Early-warning signals for critical transitions. *Nature*, 461(7260):53–59, 2009. doi: 10.1038/nature08227.

[79] H. A. Kramers. Brownian motion in a field of force and the diffusion model of chemical reactions. *Physica*, 7(4):284–304, 1940. doi: 10.1016/S0031-8914(40)90098-2.

[80] Matias I. Maturana, Christian Meisel, Katrina Dell, et al. Critical slowing down as a biomarker for seizure susceptibility. *Nature Communications*, 11(1):2172, 2020. doi: 10.1038/s41467-020-15908-3.

[81] Annika Hagemann, Jens Wilting, Bita Samimizad, Florian Mormann, and Viola Priesemann. Assessing criticality in pre-seizure single-neuron activity of human epileptic cortex. *PLOS Computational Biology*, 17(3):e1008773, 2021. doi: 10.1371/journal.pcbi.1008773.

[82] Sean Welleck, Ximing Lu, Peter West, Faeze Brahman, Tianxiao Shen, Daniel Khashabi, and Yejin Choi. Generating sequences by learning to self-correct. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=hH36JeQZDaO. arXiv:2211.00053.

[83] Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-emnlp.68.

[84] Varvara Logacheva, Daryna Dementieva, Sergey Ustyantsev, Daniil Moskovskiy, David Dale, Irina Krotova, Nikita Semenov, and Alexander Panchenko. ParaDetox: Detoxification with parallel data. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6804–6818, Dublin, Ireland, 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.469.

[85] Jonathan Kadmon and Haim Sompolinsky. Transition to chaos in random neuronal networks. *Physical Review X*, 5(4):041030, 2015. doi: 10.1103/PhysRevX.5.041030.

[86] Kanaka Rajan, L. F. Abbott, and Haim Sompolinsky. Stimulus-dependent suppression of chaos in recurrent neural networks. *Physical Review E*, 82(1):011903, 2010. doi: 10.1103/PhysRevE.82.011903.

[87] Ben Poole, Subhaneil Lahiri, Maithra Raghu, Jascha Sohl-Dickstein, and Surya Ganguli. Exponential expressivity in deep neural networks through transient chaos. In *Advances in Neural Information Processing Systems*, volume 29, pages 3360–3368, 2016. URL https://proceedings.neurips.cc/paper_files/paper/2016/hash/148510031349642de5ca0c544f31b2ef-Abstract.html. arXiv:1606.05340.

[88] Samuel S. Schoenholz, Justin Gilmer, Surya Ganguli, and Jascha Sohl-Dickstein. Deep information propagation. In *The Fifth International Conference on Learning Representations*, 2017. URL https://openreview.net/forum?id=H1W1UN9gg. arXiv:1611.01232.

[89] Cristian Bonatto and Jason A. C. Gallas. Periodicity hub and nested spirals in the phase diagram of a simple resistive circuit. *Physical Review Letters*, 101(5):054101, 2008. doi: 10.1103/PhysRevLett.101.054101.

[90] Ruedi Stoop, Philipp Benner, and Yoko Uwate. Real-world existence and origins of the spiral organization of shrimp-shaped domains. *Physical Review Letters*, 105(7):074102, 2010. doi: 10.1103/PhysRevLett.105.074102.

[91] Cameron Berg, Diogo de Lucena, and Judd Rosenblatt. Large language models report subjective experience under self-referential processing. arXiv preprint arXiv:2510.24797, 2025. URL https://arxiv.org/abs/2510.24797.

[92] Chris G. Langton. Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3):12–37, 1990. doi: 10.1016/0167-2789(90)90064-V.

[93] Nils Bertschinger and Thomas Natschläger. Real-time computation at the edge of chaos in recurrent neural networks. *Neural Computation*, 16(7):1413–1436, 2004. doi: 10.1162/089976604323057443.

[94] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633.